M2 Quantum Engineering Internship Report

2025–26

Derivation of Chern-Simons Theory from the Toric Code

by

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Abstract

This report studies the derivation of Chern-Simons theory from the Toric Code by comparing their structures as topological quantum field theories. The central methodology adopted is not a direct microscopic-to-continuum limit of the lattice Hamiltonian, but a topological-data matching procedure: first, Chern-Simons theory and the Toric Code are each formulated as continuum $2+1$-dimensional TQFTs on a torus; then their Abelian anyon behaviour, observable loop operators, and ground-state structures are compared. In the Chern-Simons description, the relevant observables are Wilson loop operators associated with non-contractible cycles of the torus, while in the Toric Code the corresponding observables appear as non-contractible string operators acting on the lattice ground-state space. By studying how Abelian anyons move, braid, fuse, and act on the torus Hilbert space in both models, the report shows how topology controls the physical degrees of freedom in each theory. This comparison also clarifies the role of the Toric Code as a topological quantum memory, where quantum information is protected by global topological features rather than local microscopic details. Through this analysis, the report builds a bridge between the discrete stabilizer structure of the Toric Code and the continuum field-theoretic structure of Chern-Simons theory, preparing the final proof that both models describe the same underlying Abelian topological phase.

Contents

Chapter 1

What is Quantum Field Theory?

Quantum Field Theory (QFT) is the mathematical framework in which the fundamental objects of nature are not individual particles placed inside an empty universe, but fields defined over spacetime. In ordinary quantum mechanics, one usually begins with a fixed number of particles and studies their wavefunctions. In QFT, the field is taken to be the primary object, and particles appear as localized excitations, ripples, or quanta of that field. For example, an electron is interpreted as an excitation of the electron field, a photon as an excitation of the electromagnetic field, and the Higgs boson as an excitation of the Higgs field. Thus, QFT replaces the picture of the universe as an empty box containing particles with the picture of the universe as a collection of interacting quantum fields.

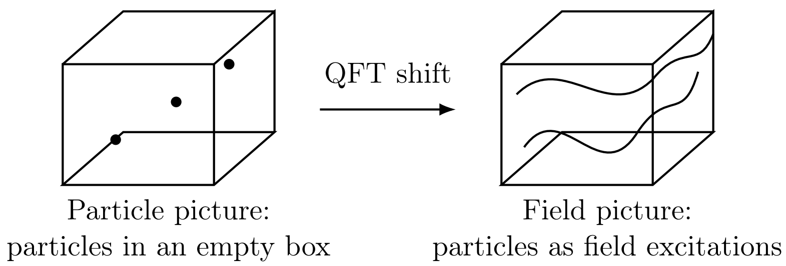


Figure 1.1: The conceptual shift from ordinary particle mechanics to QFT. QFT treats fields as fundamental, while particles arise as localized excitations of those fields.

1.1 Why Quantum Field Theory is Needed

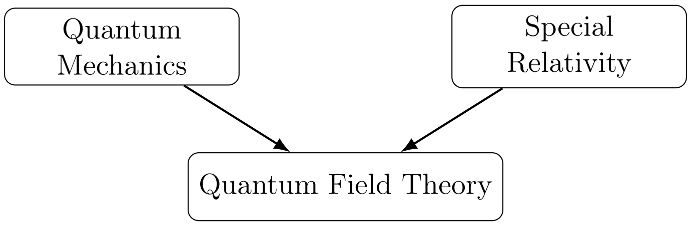
Quantum Field Theory was developed because ordinary quantum mechanics and special relativity cannot be consistently combined while keeping the number of particles fixed. In non-relativistic quantum mechanics, the number of particles is usually assumed to be fixed, and observables such as position and momentum are described by operators,

$$\hat{x}, \qquad \hat{p}. \tag{1.1}$$

However, special relativity allows energy to be converted into mass through

$$E = mc^2. \tag{1.2}$$

Therefore, at sufficiently high energies, particles can be created and annihilated. A theory with a fixed particle number is no longer sufficient. QFT resolves this by replacing single-particle wavefunctions with quantum fields whose excitations can appear and disappear dynamically. In this sense, QFT is the mathematical marriage of quantum mechanics and special relativity.



Allows fluctuating particle number through creation and annihilation of field excitations.

Figure 1.2: QFT combines the quantum-mechanical operator framework with the relativistic possibility of particle creation and annihilation.

A useful physical picture is to imagine a tuning fork. If one tuning fork vibrates, it can cause another nearby tuning fork to vibrate. Similarly, in QFT, a field can fluctuate, interact with other fields, and transfer energy into excitations. These excitations are what we observe as particles. Thus, particle creation and annihilation are not mysterious additions to the theory; they arise naturally from the quantum dynamics of fields.

1.2 Fields as the Fundamental Objects

A field assigns a mathematical value to every point of spacetime. If the spacetime coordinate is denoted by

$$x^\mu = (t, x, y, z), \tag{1.3}$$

then a scalar field is written as

$$\phi(x^\mu). \tag{1.4}$$

This means that at every spacetime point x^μ, the field has a value. Different physical particles correspond to different kinds of fields. For example,

$$\phi(x) \quad \text{may represent a scalar field,} \tag{1.5}$$

$$\psi_\alpha(x) \quad \text{may represent a spinor field,} \tag{1.6}$$

and

$$A_\mu(x) \quad \text{may represent a vector or gauge field.} \tag{1.7}$$

The type of field determines the type of particle excitation. Scalar fields describe spin-0 particles, spinor fields describe spin-1/2 particles such as fermions, and vector fields describe spin-1 gauge particles such as photons.

In standard quantum mechanics, one quantizes the position and momentum of a particle. In QFT, one quantizes the field itself. The field becomes an operator-valued object:

$$\phi(x) \longrightarrow \hat{\phi}(x). \tag{1.8}$$

This is the conceptual shift from quantum mechanics to quantum field theory.

1.3 Parameters Needed to Define a QFT

A QFT is not specified merely by saying that a field exists. To define a QFT, one must specify several mathematical and physical ingredients.

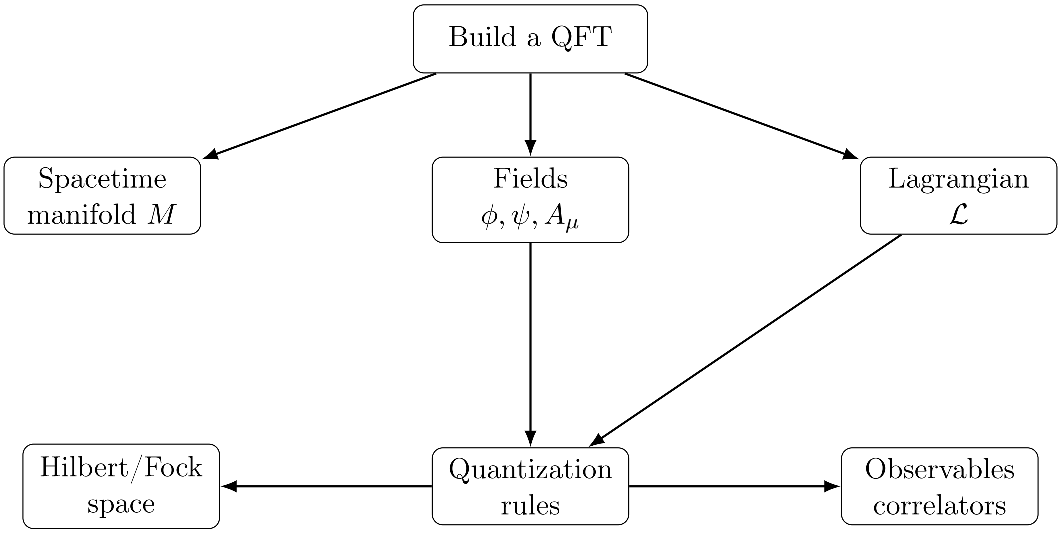


Figure 1.3: Basic data needed to define a quantum field theory. The spacetime manifold specifies where the fields live, the Lagrangian specifies their dynamics, and quantization turns the classical fields into quantum operators.

1.3.1 The Spacetime Manifold

First, one specifies the spacetime manifold on which the fields live. In ordinary relativistic QFT, this is often Minkowski spacetime,

$$\mathbb{R}^{1,3}, \tag{1.9}$$

with one time direction and three spatial directions. In this report, however, we are particularly interested in $2+1$-dimensional field theories, where spacetime has one time direction and two spatial directions:

$$M^{2+1} = \mathbb{R} \times \Sigma. \tag{1.10}$$

Later, when we place a theory on a torus, the spatial surface will be

$$\Sigma = T^2. \tag{1.11}$$

1.3.2 The Quantum Fields

Second, one specifies the fields of the theory. These fields are the fundamental dynamical objects. For example, a scalar field theory may contain

$$\phi(x), \tag{1.12}$$

whereas a gauge theory may contain a gauge field

$$A_\mu(x). \tag{1.13}$$

In the case of Chern-Simons theory, the fundamental field is a gauge connection one-form,

$$a = a_\mu dx^\mu. \tag{1.14}$$

1.3.3 The Lagrangian Density and Action

Third, one specifies the Lagrangian density,

$$\mathcal{L}. \tag{1.15}$$

The Lagrangian density contains the physical information of the theory, such as kinetic terms, mass terms, and interaction terms. The action is then defined by integrating the Lagrangian density over spacetime:

$$S = \int d^4x\,\mathcal{L} \tag{1.16}$$

in $3+1$ dimensions, or more generally,

$$S = \int_M \mathcal{L}. \tag{1.17}$$

For example, a scalar field theory may contain a Lagrangian density of the form

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - V(\phi), \tag{1.18}$$

where m is the mass parameter and $V(\phi)$ contains interaction terms. In this way, the numerical constants appearing inside $\mathcal{L}$, such as masses and coupling constants, determine the physical behaviour of the theory.

1.3.4 Conjugate Momentum and Canonical Quantization

Fourth, one defines the conjugate momentum of the field. For a field $\phi(x)$, the conjugate momentum is

$$\pi(x) = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi)}. \tag{1.19}$$

Canonical quantization is then imposed by replacing classical fields with operators and requiring equal-time commutation relations:

$$[\hat{\phi}(\mathbf{x},t), \hat{\pi}(\mathbf{y},t)] = i\hbar\delta^3(\mathbf{x}-\mathbf{y}). \tag{1.20}$$

This is the field-theoretic analogue of the ordinary quantum-mechanical commutation relation

$$[\hat{x},\hat{p}] = i\hbar. \tag{1.21}$$

It states that the field and its conjugate momentum cannot both be sharply specified at the same spatial point.

For fermionic fields, the corresponding quantization uses anti-commutators rather than commutators:

$$\{\hat{\psi}_a(\mathbf{x},t), \hat{\psi}_b^\dagger(\mathbf{y},t)\} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y}). \tag{1.22}$$

1.3.5 Fock Space and the Vacuum

The Hilbert space of QFT is usually a Fock space rather than a fixed-particle Hilbert space. The Fock space contains sectors with different particle numbers:

$$\mathcal{F} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \cdots . \tag{1.23}$$

Here,

$$\mathcal{H}_0 \tag{1.24}$$

is the vacuum sector,

$$\mathcal{H}_1 \tag{1.25}$$

is the one-particle sector, and so on.

The vacuum state is denoted by

$$|0\rangle. \tag{1.26}$$

It is the lowest-energy state of the theory. Particle states are created by acting on the vacuum with creation operators:

$$a_p^\dagger|0\rangle = |p\rangle. \tag{1.27}$$

Annihilation operators remove excitations:

$$a_p|p\rangle = |0\rangle. \tag{1.28}$$

Thus, in QFT, particles are created and annihilated as excitations of the underlying field.

1.4 Correlation Functions and Observables

The physical observables of a QFT are often encoded in correlation functions. For example, an n-point correlation function is written as

$$\langle 0|\phi(x_1)\phi(x_2)\cdots\phi(x_n)|0\rangle. \tag{1.29}$$

This object measures how field excitations at different spacetime points are related.

In ordinary quantum mechanics, one may ask for the probability of finding a particle at a point. In QFT, one instead asks for the amplitude that an excitation created at one point propagates and is detected at another point. This is encoded in two-point functions such as

$$\langle 0|\phi(x)\phi(y)|0\rangle. \tag{1.30}$$

More generally, correlation functions are the main objects from which scattering amplitudes, response functions, and physical observables are computed.

In the path-integral formulation, correlation functions are written as

$$\langle \mathcal{O} \rangle = \frac{\int \mathcal{D}\phi\, \mathcal{O}[\phi] e^{iS[\phi]}}{\int \mathcal{D}\phi\, e^{iS[\phi]}}. \tag{1.31}$$

This formula says that the observable $\mathcal{O}$ is evaluated by summing over all field configurations, weighted by the phase $e^{iS[\phi]}$.

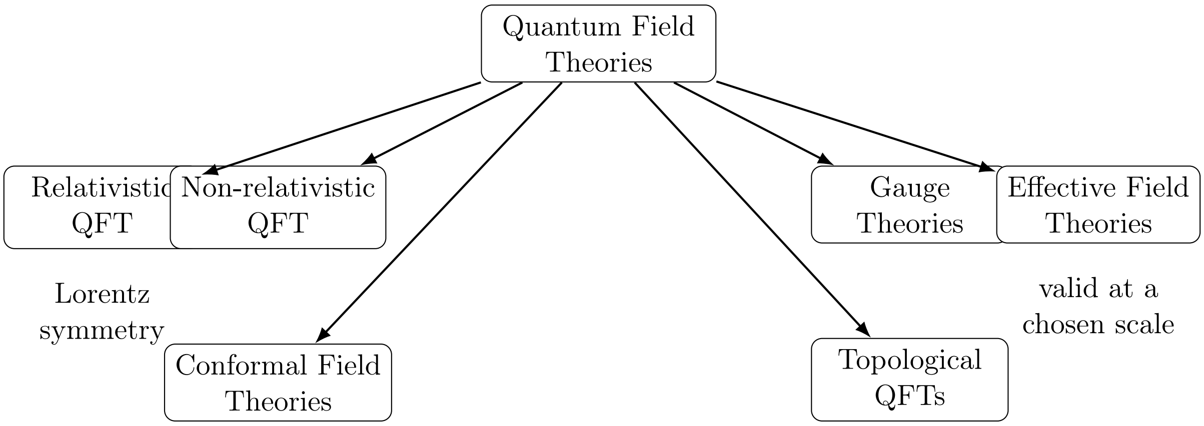


Figure 1.4: A broad classification of quantum field theories according to their symmetries, degrees of freedom, and physical regimes. This report focuses on topological quantum field theories.

1.5 Types of Quantum Field Theories

There are many different types of QFTs, classified by their symmetries, fields, and physical regimes. Some important examples are shown in Fig. 1.4.

- **Relativistic QFTs**, which are invariant under Lorentz transformations and combine quantum mechanics with special relativity.
- **Non-relativistic QFTs**, which are used in condensed matter physics and many-body quantum systems.
- **Gauge theories**, where fields have local gauge redundancies. Quantum electrodynamics is an Abelian gauge theory, while Yang-Mills theory is a non-Abelian gauge theory.
- **Effective field theories**, which describe physics at a particular energy scale without requiring complete knowledge of the microscopic theory.
- **Conformal field theories**, which are invariant under conformal transformations and often appear at critical points.
- **Topological quantum field theories**, whose observables depend on topological rather than geometric data.

This report is mainly concerned with the final class: topological quantum field theories. Chern-Simons theory is a central example of a $2+1$-dimensional continuum TQFT, while the Toric Code is a lattice model whose low-energy topological sector can be described by such a continuum TQFT.

1.6 Working Definition

For the purposes of this report, we use the following working definition:

A QFT is a mathematical framework that combines quantum mechanics with fields, allowing particle number to fluctuate through the creation and annihilation of field excitations.

Equivalently, a QFT is specified by:

$$\boxed{(M,\ \text{fields},\ \mathcal{L},\ S,\ \text{quantization rules},\ \mathcal{H},\ \text{observables})\,.} \tag{1.32}$$

Here M is the spacetime manifold, the fields are the fundamental objects, $\mathcal{L}$ and S determine their dynamics, the quantization rules turn classical fields into quantum operators, $\mathcal{H}$ is the Hilbert or Fock space of states, and the observables are computed through operators or correlation functions.

This definition prepares the transition to the next chapter. A topological quantum field theory is a special kind of QFT in which the physical quantities do not depend on local metric geometry, such as lengths and angles, but only on global topological data.

Chapter 2

What is a Topological Quantum Field Theory?

In the previous chapter, a quantum field theory was introduced as a framework in which fields live on spacetime and particles arise as quantum excitations of these fields. A Topological Quantum Field Theory (TQFT) is a special kind of QFT in which the physical observables do not depend on local geometric measurements such as distance, angle, curvature, or time duration. Instead, the observables depend only on global topological information, such as the number of holes of a space, the linking of loops, the braiding of worldlines, and the global structure of the underlying manifold. This is the central idea that will allow us to compare Chern-Simons theory and the Toric Code.

2.1 From Geometry to Topology

Ordinary physics often depends on geometry. For example, if two electrons are separated by a distance r, the Coulomb interaction depends on that distance. Similarly, in relativistic field theory, the spacetime metric determines distances, time intervals, and causal structure. In a conventional field theory, local geometric information is therefore physically meaningful.

A TQFT behaves differently. It is insensitive to continuous deformations of the spacetime geometry. Stretching, bending, or smoothly deforming the manifold does not change the topological observables, provided the topology is unchanged. Therefore, a TQFT does not ask how long a path is or how much time a process takes. It asks whether a loop winds around a hole, whether two worldlines braid, or whether a surface has non-contractible cycles.

For this report, the most important consequence is the following:

> A TQFT is a quantum field theory whose physically meaningful observables are invariant under smooth deformations of the underlying spacetime manifold.

2.2 The Metric Independence Criterion

A practical way to recognize a TQFT is to inspect whether its action depends on the spacetime metric. In many ordinary field theories, the metric tensor $g_{\mu\nu}$ appears explicitly

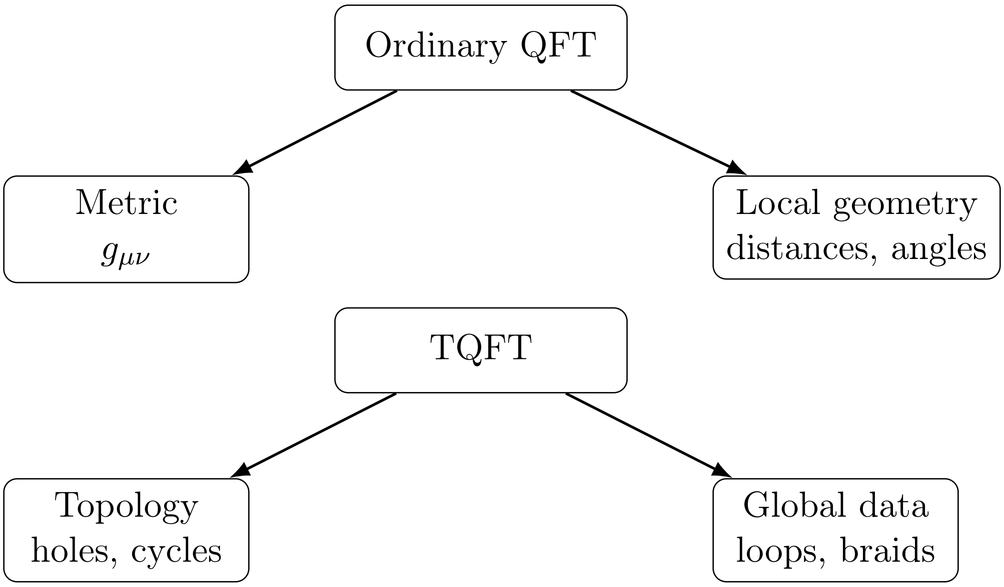


Figure 2.1: Ordinary QFTs generally depend on local geometric data, while TQFTs depend only on global topological data.

in the action. For example, a Yang-Mills-type theory contains contractions such as

$$F_{\mu\nu}F^{\mu\nu}, \tag{2.1}$$

where raising indices requires the inverse metric $g^{\mu\nu}$. Such a theory depends on the geometry of spacetime.

A topological action, by contrast, can often be written without any reference to the metric. The Chern-Simons action is the central example for this report:

$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi}\int_M a \wedge da. \tag{2.2}$$

The expression $a \wedge da$ is a differential form integrated over a three-dimensional manifold. No metric tensor appears. Thus,

$$\frac{\delta S_{\mathrm{CS}}}{\delta g_{\mu\nu}} = 0. \tag{2.3}$$

This is the mathematical reason Chern-Simons theory is topological. Witten emphasizes that Chern-Simons theory achieves general covariance not by making the metric dynamical, as in general relativity, but by starting with a gauge-invariant Lagrangian that does not contain a metric [1]. Simon also treats Chern-Simons theory as a central example of a TQFT because of this metric independence [5].

2.3 Physical and Mathematical Viewpoints

There are two complementary ways to understand a TQFT: the physical viewpoint and the mathematical viewpoint.

From the physical viewpoint, a TQFT is described through an action or Hamiltonian whose low-energy observables are topological. In this language, we study fields, Wilson loops, anyons, braiding phases, and ground-state degeneracy. This is the viewpoint most useful for comparing Chern-Simons theory with the Toric Code.

From the mathematical viewpoint, a TQFT can be described axiomatically as a functor. In the Atiyah-Segal style of thinking, a TQFT assigns algebraic data to manifolds. A closed d-dimensional spatial manifold Σ is assigned a Hilbert space or vector space,

$$\Sigma \longmapsto V(\Sigma), \tag{2.4}$$

and a $(d+1)$-dimensional spacetime cobordism M between spatial manifolds is assigned a linear map,

$$M : \Sigma_{\text{in}} \to \Sigma_{\text{out}} \quad \longmapsto \quad Z(M) : V(\Sigma_{\text{in}}) \to V(\Sigma_{\text{out}}). \tag{2.5}$$

This means that the spacetime manifold acts like an evolution operator between topological state spaces.

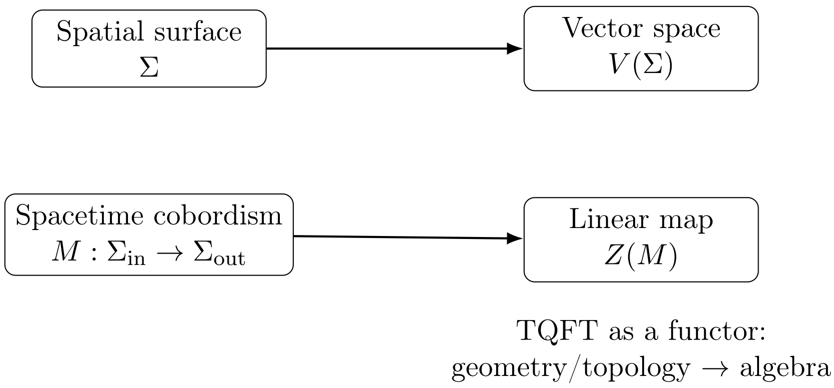


Figure 2.2: Mathematical viewpoint of a TQFT: spatial manifolds are assigned vector spaces, and spacetime manifolds are assigned linear maps.

In this report, we primarily use the physical viewpoint, but the functorial picture is useful because it explains why a TQFT on a torus should assign a finite-dimensional Hilbert space to T^2.

2.4 Pure TQFT and the Absence of Local Dynamics

In a pure TQFT, the Hamiltonian is often trivial or constraint-like in the topological sector:

$$H = 0 \tag{2.6}$$

or, more physically,

$$H|\psi\rangle = E_0|\psi\rangle \tag{2.7}$$

for all states in the degenerate topological ground-state space. This does not mean that the theory is physically meaningless. It means that the theory does not describe local propagating waves in the usual sense. Instead, the nontrivial content is global.

The important data are:

$$\text{loops}, \qquad \text{holes}, \qquad \text{worldlines}, \qquad \text{braids}, \qquad \text{ground-state degeneracy}. \tag{2.8}$$

This is why TQFTs are natural for describing topological phases of matter. In such phases, local perturbations do not easily destroy the stored information, because the information is encoded globally rather than locally. This idea will later become central when discussing the Toric Code as a topological quantum memory.

2.5 When Are Two TQFTs Equivalent?

To compare two TQFTs, one should not compare their microscopic Hamiltonians or wavefunctions directly. Two theories may look very different microscopically and still describe the same topological phase. Instead, one compares their universal topological data.

For Abelian anyon theories, the essential data include:

- the set of anyon types or superselection sectors;
- the fusion rules;
- the topological spins, often encoded in the T-matrix;
- the mutual braiding phases, often encoded in the S-matrix;
- the ground-state degeneracy on closed spatial manifolds such as the torus.

Thus, two topological theories are considered equivalent at the level relevant to this report if they possess the same anyon content, the same fusion algebra, the same braiding statistics, the same topological spins, and the same ground-state degeneracy on the torus.

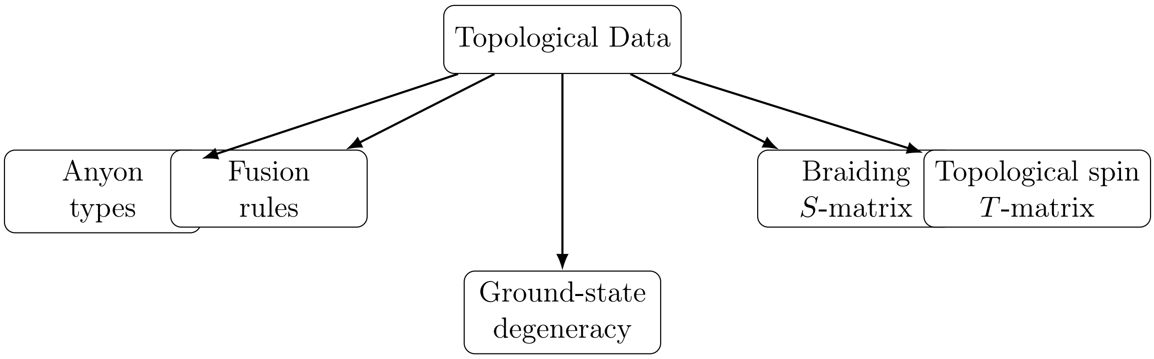


Figure 2.3: Topological data used to compare two TQFTs. In this report, this data will be extracted from both Chern-Simons theory and the Toric Code.

This is the methodology used throughout the report:

$$\boxed{\text{Extract topological data from the discrete Toric Code} \quad \Longleftrightarrow \quad \text{extract topological data from the con}} \tag{2.9}$$

If the two sets of data match, then the two theories describe the same topological phase.

2.6 Toric Code as a Motivating Example of a TQFT

The Toric Code is a lattice model, but its low-energy sector behaves like a TQFT. It has four anyon sectors,

$$1, \qquad e, \qquad m, \qquad \varepsilon = e \times m. \tag{2.10}$$

These sectors are Abelian because each fusion process has a unique output. The fusion rules are

$$e \times e = 1, \qquad m \times m = 1, \qquad e \times m = \varepsilon, \qquad \varepsilon \times \varepsilon = 1. \tag{2.11}$$

Thus, the fusion algebra is

$$\mathbb{Z}_2 \times \mathbb{Z}_2. \tag{2.12}$$

The topological spins are

$$\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1. \tag{2.13}$$

Therefore, e and m are bosonic Abelian anyons, while their composite $\varepsilon = e \times m$ is fermionic. In the ordered basis

$$(1, e, m, \varepsilon), \tag{2.14}$$

the T-matrix is

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{2.15}$$

The mutual braiding information is encoded in the S-matrix. For the Toric Code,

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{2.16}$$

The entry expressing the mutual braiding of e and m contains the phase -1, meaning that taking e around m produces a nontrivial Abelian braiding phase. This is exactly the kind of topological data that must be reproduced by the continuum field theory.

2.7 Towards the Continuum Chern-Simons Description

The purpose of this report is not merely to say that the Toric Code and Chern-Simons theory are both topological. That would be too weak. Many different TQFTs exist, and not all of them describe the same topological phase.

Instead, the goal is to show that the relevant continuum Chern-Simons theory reproduces the Toric Code topological data. The continuum theory that will appear later is the doubled Abelian Chern-Simons theory, also written as a mutual BF theory. In K-matrix form, the relevant matrix is

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{2.17}$$

Using the standard Abelian Chern-Simons K-matrix formulas, this continuum theory reproduces the same four anyon sectors, the same $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion algebra, the same mutual braiding phase -1, and the same torus ground-state degeneracy as the Toric Code.

The central strategy of this report is therefore to prove equivalence by matching topological data, not by directly equating a microscopic lattice Hamiltonian with a continuum action.

2.8 Working Definition

For the purposes of this report, we use the following working definition:

> A Topological Quantum Field Theory is a quantum field theory whose observables are invariant under smooth deformations of spacetime and depend only on global topological data such as cycles, braids, fusion rules, and ground-state degeneracy.

This definition prepares the next chapter, where we specialize to 2+1-dimensional continuum TQFTs. This is the natural dimension for both anyon physics and Chern-Simons theory, because anyonic braiding is fundamentally a $2+1$-dimensional phenomenon.

Chapter 3

$2+1$-Dimensional Continuum Topological Quantum Field Theory

The previous chapter introduced a Topological Quantum Field Theory as a quantum field theory whose observables depend only on global topological data. In this chapter, we specialize to the case that is most important for this report: a $2+1$-dimensional continuum TQFT. This is the natural setting for Chern-Simons theory, Abelian anyons, and the low-energy topological sector of the Toric Code. The phrase $2+1$-dimensional means that the theory has two spatial dimensions and one time dimension. The word continuum means that the fields are defined on a smooth manifold rather than on a discrete lattice.

3.1 Why $2+1$ Dimensions are Special

A $2+1$-dimensional spacetime consists of a two-dimensional spatial surface evolving in time. If the spatial surface is denoted by Σ, then the spacetime manifold is written as

$$M = \mathbb{R}_t \times \Sigma. \tag{3.1}$$

Here $\mathbb{R}_t$ represents the time direction, while Σ represents the two-dimensional space on which the physical system lives.

This dimension is special because anyonic statistics can occur only in two spatial dimensions. In three spatial dimensions, exchanging two identical point particles twice can be continuously deformed into doing nothing, which restricts ordinary particles to bosonic or fermionic statistics. In two spatial dimensions, however, particle worldlines form braids in spacetime. These braids cannot always be untangled without crossing worldlines. Therefore, an exchange can produce a nontrivial phase or, in the non-Abelian case, a nontrivial matrix action on the Hilbert space. This is why $2+1$-dimensional TQFTs naturally describe anyons and topological phases such as those appearing in the fractional quantum Hall effect and the Toric Code [2, 4, 5].

3.2 Continuum Versus Discrete Descriptions

The word continuum refers to the mathematical structure of the space on which the theory is defined. In a continuum theory, spacetime is a smooth manifold. Locally, it resembles

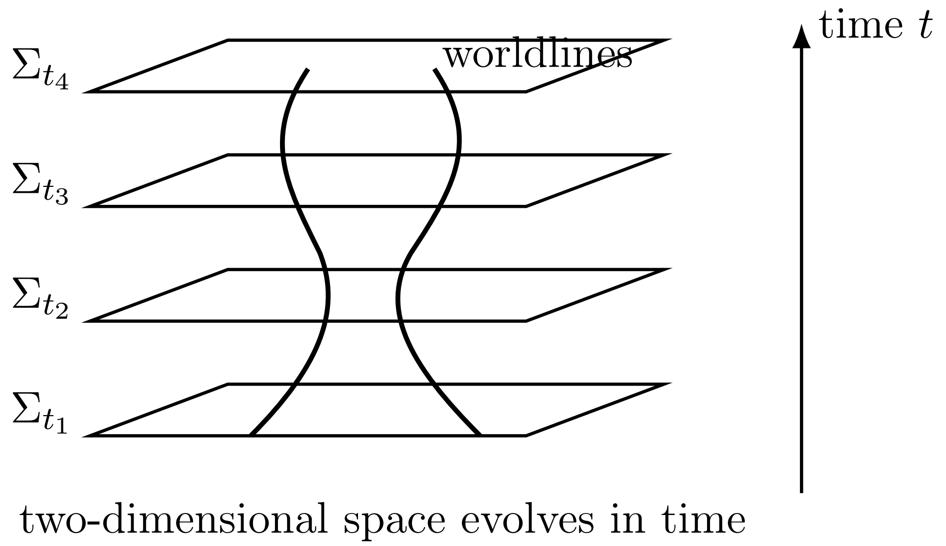


Figure 3.1: A 2 + 1-dimensional spacetime can be viewed as a two-dimensional spatial surface Σ evolving in time. Particle histories become worldlines in the three-dimensional spacetime $M = \mathbb{R}_t \times \Sigma$.

Euclidean space, and coordinates can take continuously many values. For example, on ordinary flat space one may write

$$x^\mu = (t, x, y), \tag{3.2}$$

where x and y vary continuously.

A field in a continuum QFT is therefore a function or differential-geometric object defined at every point of spacetime. For example, a scalar field is

$$\phi : M \to \mathbb{R}, \tag{3.3}$$

while a gauge field may be written as a one-form

$$a = a_\mu dx^\mu. \tag{3.4}$$

Calculus, differential forms, derivatives, and integrals can then be used to describe how fields vary across spacetime.

This is different from a discrete lattice model. In a lattice theory, space is divided into sites, links, plaquettes, or other discrete cells. The Toric Code is originally defined in this way: qubits live on the edges of a lattice, and stabilizer operators act on stars and plaquettes. There are no points between neighbouring lattice sites in the microscopic definition. A continuum TQFT, by contrast, describes the long-distance topological behaviour after microscopic lattice details have been ignored.

3.3 Definition of a $2 + 1$D Continuum TQFT

A 2 + 1-dimensional continuum TQFT is a quantum field theory defined on a smooth three-dimensional spacetime manifold

$$M = \mathbb{R}_t \times \Sigma, \tag{3.5}$$

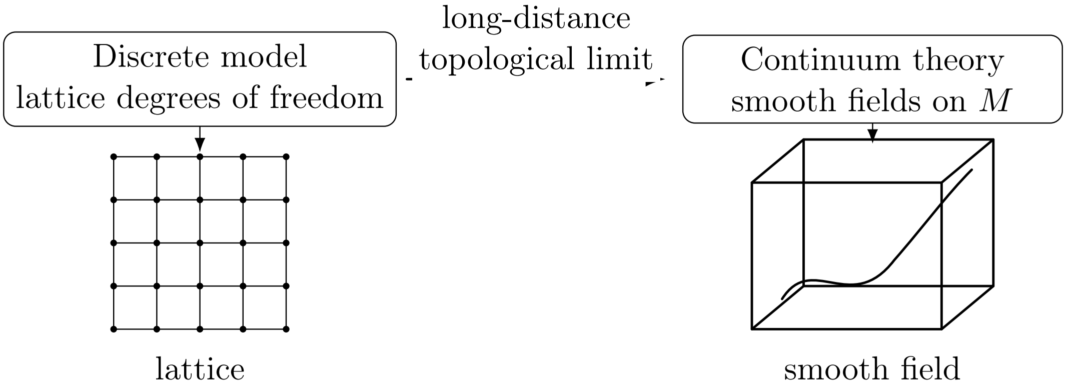


Figure 3.2: A discrete lattice model has finitely separated microscopic degrees of freedom, while a continuum field theory uses smooth fields defined over a differentiable manifold.

where Σ is a two-dimensional spatial surface. The theory is topological if its observables do not depend on the local metric geometry of M. In other words, its physical quantities are invariant under smooth deformations of the manifold.

One way to express this condition is through the stress-energy tensor. For a field theory with action S, the stress-energy tensor is schematically obtained by varying the action with respect to the metric:

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta S}{\delta g^{\mu\nu}}. \tag{3.6}$$

If the action does not depend on the metric, then

$$\frac{\delta S}{\delta g^{\mu\nu}} = 0, \tag{3.7}$$

and hence

$$T_{\mu\nu} = 0. \tag{3.8}$$

This expresses the absence of local metric-dependent energy flow in a pure topological theory. The theory does not care how far apart objects are, how fast they move, or what local angles are drawn on the manifold. It only cares about global topological questions: Did a worldline braid around another worldline? Did a loop wind around a hole? Is a cycle contractible or non-contractible?

For this reason, a 2 + 1-dimensional continuum TQFT assigns a Hilbert space to a two-dimensional spatial surface,

$$\Sigma \longmapsto V(\Sigma), \tag{3.9}$$

while a three-dimensional spacetime cobordism M assigns a linear map,

$$M : \Sigma_{\text{in}} \to \Sigma_{\text{out}} \quad \longmapsto \quad Z(M) : V(\Sigma_{\text{in}}) \to V(\Sigma_{\text{out}}). \tag{3.10}$$

This is the topological version of time evolution. It is not controlled by local geometry, but by the topology of the spacetime history.

A 2 + 1-dimensional continuum TQFT is a quantum field theory on a smooth three-dimensional spacetime $M = \mathbb{R}_t \times \Sigma$, whose physical observables are independent of the metric and depend only on global topological data of M and Σ.

3.4 Why Actions are Natural for TQFTs

In ordinary quantum mechanics, the action is a functional of a particle path. If a particle follows a path $q(t)$, the action is

$$S[q] = \int dt\, L(q, \dot{q}, t). \tag{3.11}$$

It assigns a number to an entire history of the particle.

In field theory, the configuration is not a particle path but a field history. If the field is $\phi(x, t)$, then the action is

$$S[\phi] = \int d^{d+1}x\, \mathcal{L}(\phi, \partial_\mu \phi). \tag{3.12}$$

Thus, the action assigns a number to an entire spacetime field configuration.

The path integral then sums over all possible field histories:

$$Z = \int \mathcal{D}\phi\, e^{iS[\phi]}. \tag{3.13}$$

This is why actions are particularly natural in TQFT. A TQFT is concerned with whole spacetime histories, worldlines, braids, loops, and manifolds. The action formalism directly assigns quantum amplitudes to these histories.

A Hamiltonian description is also possible in many systems, especially when a preferred time direction is chosen. For example, the Toric Code is naturally defined by a Hamiltonian,

$$H_{\text{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p. \tag{3.14}$$

This Hamiltonian description is extremely useful for quantum memory and stabilizer-code physics. However, for continuum TQFTs such as Chern-Simons theory, the action formulation is often more natural because it does not require separating spacetime into local time evolution plus local spatial dynamics. Instead, it treats the full spacetime manifold as the object of interest.

3.5 Chern-Simons Theory as the Prototype

The central continuum TQFT used in this report is Chern-Simons theory. The fundamental variable is a gauge connection one-form. For an Abelian $U(1)$ theory, this is written as

$$a = a_\mu dx^\mu. \tag{3.15}$$

The Abelian Chern-Simons action is

$$S_{\text{CS}}[a] = \frac{k}{4\pi} \int_M a \wedge da. \tag{3.16}$$

Here M is a three-dimensional spacetime manifold. The field a is a one-form, da is a two-form, and therefore $a \wedge da$ is a three-form. This is why Chern-Simons theory naturally lives in $2 + 1$ dimensions. Witten emphasizes that precisely in three dimensions one can

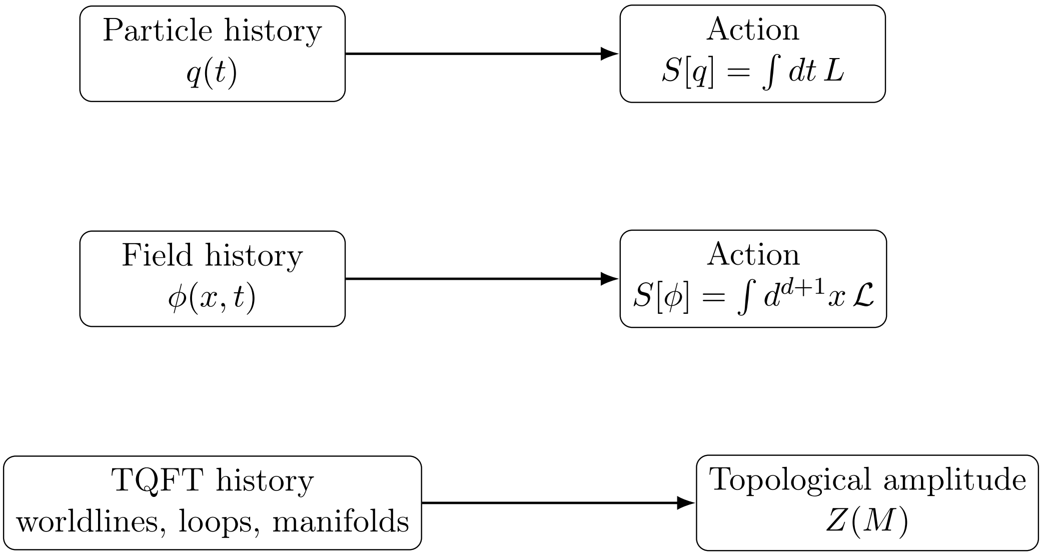


Figure 3.3: The action formalism assigns a number or phase to an entire history. This is why it is natural for TQFTs, where the relevant objects are spacetime histories, loops, and braids.

write the Chern-Simons three-form as a metric-independent action, making it a central example of a topological quantum field theory [1].

The geometric meaning is that a gauge connection can be used to build curvature,

$$F = da \tag{3.17}$$

for Abelian gauge theory, or

$$F = dA + A \wedge A \tag{3.18}$$

for non-Abelian gauge theory. Chern-Simons theory builds a topological functional from this gauge-geometric data. When used as the action of a $2+1$-dimensional quantum field theory, its observables become Wilson loops and link invariants. This is the reason Chern-Simons theory is the natural continuum language for studying anyons and topological phases.

3.6 Working Definition

For this report, the working definition is:

> A $2 + 1$-dimensional continuum TQFT is a metric-independent quantum field theory defined on a smooth three-dimensional spacetime $M = \mathbb{R}_t \times \Sigma$, where Σ is a two-dimensional spatial surface. Its nontrivial physics is encoded in global topological data such as Wilson loops, braiding, non-contractible cycles, and finite-dimensional Hilbert spaces assigned to spatial surfaces.

This chapter prepares the next step of the report. We will now study the most important example of such a theory: Chern-Simons theory as a $2 + 1$-dimensional continuum TQFT.

Chapter 4

Chern-Simons Theory as a $2+1$D Continuum TQFT

The aim of this chapter is to show that Chern-Simons theory satisfies the defining properties of a $2+1$-dimensional continuum topological quantum field theory. Starting from the Abelian Chern-Simons action, we prove that the theory is naturally three-dimensional, gauge invariant on closed manifolds, metric independent, locally flat, and governed by global Wilson-loop observables rather than local propagating bulk degrees of freedom. This is the continuum field theory that will later be placed on a torus and compared with the Toric Code.

In this report, we mainly use the Abelian $U(1)$ Chern-Simons theory because the Toric Code anyons that we eventually compare with are Abelian. The non-Abelian theory is important historically and mathematically, especially in Witten's construction of knot and three-manifold invariants, but the Abelian theory is the correct starting point for the Toric Code/BF comparison [1, 4, 5].

4.1 Gauge Connections and Curvature

Chern-Simons theory is a gauge theory. The fundamental field is not a scalar field or a particle coordinate, but a gauge connection. In the Abelian case, this connection is a one-form

$$a = a_\mu dx^\mu. \tag{4.1}$$

The field strength or curvature of this connection is the two-form

$$f = da. \tag{4.2}$$

In component notation,

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu. \tag{4.3}$$

Thus, the connection a is analogous to the electromagnetic vector potential, while $f = da$ is analogous to the physical electric and magnetic field strength.

The geometric meaning of a connection is that it tells us how to compare phases at different spacetime points. A charged quantum state transported along a path accumulates a phase determined by the gauge connection. In this sense, the connection records how local phase conventions are glued together across spacetime.

$$
\begin{tikzcd}
\boxed{\text{Local phasepoint } x} \arrow[r, "\substack{\text{parallel transport}\\ \text{by connection } a}"] & \boxed{\text{Local phasepoint } y}
\end{tikzcd}
$$

$$
\begin{tikzcd}
\boxed{\text{Connection} a = a_\mu dx^\mu} \arrow[r, "\text{exterior derivative}"'] & \boxed{\text{Curvature} f = da}
\end{tikzcd}
$$

Figure 4.1: A gauge connection determines how phases are compared along paths. Its curvature measures the failure of parallel transport around a small loop to return to the original phase.

For Abelian $U(1)$ gauge theory, a gauge transformation is

$$a \longmapsto a + d\lambda, \tag{4.4}$$

where λ is a scalar function on spacetime. Since

$$d(a + d\lambda) = da + d^2\lambda = da, \tag{4.5}$$

using $d^2 = 0$, the field strength is unchanged. Therefore, a and $a + d\lambda$ describe the same physical gauge configuration. Gauge theory is therefore a theory with redundancy in its mathematical description: different gauge potentials can represent the same physical field.

4.2 Definition of the Abelian Chern-Simons Action

Let M be a three-dimensional spacetime manifold. The Abelian Chern-Simons action is

$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi} \int_M a \wedge da. \tag{4.6}$$

Here,

$$a = a_\mu dx^\mu \tag{4.7}$$

is a gauge connection one-form, and

$$da \tag{4.8}$$

is a field-strength two-form. Therefore,

$$a \wedge da \tag{4.9}$$

is a three-form. Since a three-form can be integrated naturally over a three-dimensional manifold, the Chern-Simons action is naturally defined in $2+1$ spacetime dimensions. In component notation, it is equivalent to

$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi} \int_M d^3x \, \epsilon^{\mu\nu\rho} a_\mu \partial_\nu a_\rho, \tag{4.10}$$

up to conventional normalizations.

The Chern-Simons action is naturally $2 + 1$-dimensional because $a \wedge da$ is a three-form, and a three-form is integrated over a three-dimensional spacetime manifold.

4.3 Gauge Invariance of the Abelian Chern-Simons Action

We now show that the Abelian Chern-Simons action is gauge invariant on a closed spacetime manifold. Consider the gauge transformation

$$a \longmapsto a' = a + d\lambda. \tag{4.11}$$

Then

$$S_{\text{CS}}[a + d\lambda] = \frac{k}{4\pi} \int_M (a + d\lambda) \wedge d(a + d\lambda). \tag{4.12}$$

Since $d^2\lambda = 0$,

$$d(a + d\lambda) = da. \tag{4.13}$$

Therefore,

$$S_{\text{CS}}[a + d\lambda] = \frac{k}{4\pi} \int_M (a + d\lambda) \wedge da \tag{4.14}$$

$$= \frac{k}{4\pi} \int_M a \wedge da + \frac{k}{4\pi} \int_M d\lambda \wedge da. \tag{4.15}$$

The first term is the original action:

$$\frac{k}{4\pi} \int_M a \wedge da = S_{\text{CS}}[a]. \tag{4.16}$$

For the second term, use the exterior derivative product rule:

$$d(\lambda da) = d\lambda \wedge da + \lambda d(da). \tag{4.17}$$

Since $d(da) = d^2 a = 0$, we have

$$d(\lambda da) = d\lambda \wedge da. \tag{4.18}$$

Thus,

$$S_{\text{CS}}[a + d\lambda] = S_{\text{CS}}[a] + \frac{k}{4\pi} \int_M d(\lambda da). \tag{4.19}$$

By Stokes' theorem,

$$\int_M d(\lambda da) = \int_{\partial M} \lambda da. \tag{4.20}$$

Therefore,

$$S_{\text{CS}}[a + d\lambda] = S_{\text{CS}}[a] + \frac{k}{4\pi} \int_{\partial M} \lambda da. \tag{4.21}$$

If the spacetime manifold has no boundary,

$$\partial M = 0, \tag{4.22}$$

then the boundary term vanishes, and hence

$$S_{\text{CS}}[a + d\lambda] = S_{\text{CS}}[a]. \tag{4.23}$$

Thus the Abelian Chern-Simons action is gauge invariant on a closed manifold.

Gauge invariance means that the action depends only on the physical gauge configuration, not on the arbitrary representative a chosen from the gauge-equivalence class $a \sim a + d\lambda$.

4.4 Metric Independence

The defining topological property of Chern-Simons theory is metric independence. The Abelian Chern-Simons action

$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi} \int_M a \wedge da \tag{4.24}$$

contains no spacetime metric $g_{\mu\nu}$. It uses only differential forms and the orientation of the three-dimensional manifold. Therefore,

$$\frac{\delta S_{\mathrm{CS}}}{\delta g_{\mu\nu}} = 0. \tag{4.25}$$

Equivalently, the stress-energy tensor vanishes in the topological sector:

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta S}{\delta g^{\mu\nu}} = 0. \tag{4.26}$$

This expresses the fact that the theory does not care about local distances, angles, areas, or curvature measured by a metric. It only depends on global topological information such as loops, linking, braiding, and holonomy. Simon explains this metric independence by writing the action as the integral of a differential three-form, and Witten emphasizes that Chern-Simons theory achieves general covariance by starting with a Lagrangian that does not contain a metric [1, 5].

4.5 Equations of Motion: Flatness

We now derive the classical equation of motion. Start with

$$S_{\mathrm{CS}}[a] = \frac{k}{4\pi} \int_M a \wedge da. \tag{4.27}$$

Vary the field:

$$a \longmapsto a + \delta a. \tag{4.28}$$

Then, to first order in δa,

$$\delta S_{\mathrm{CS}} = \frac{k}{4\pi} \int_M \left(\delta a \wedge da + a \wedge d(\delta a) \right). \tag{4.29}$$

Use the identity

$$d(a \wedge \delta a) = da \wedge \delta a - a \wedge d(\delta a), \tag{4.30}$$

because a is a one-form. Rearranging gives

$$a \wedge d(\delta a) = da \wedge \delta a - d(a \wedge \delta a). \tag{4.31}$$

Since da is a two-form and δa is a one-form,

$$da \wedge \delta a = \delta a \wedge da. \tag{4.32}$$

Therefore,

$$\delta S_{\mathrm{CS}} = \frac{k}{4\pi} \int_M \left(2\delta a \wedge da - d(a \wedge \delta a) \right). \tag{4.33}$$

On a closed manifold, the boundary term vanishes by Stokes' theorem:

$$\int_M d(a \wedge \delta a) = \int_{\partial M} a \wedge \delta a = 0. \tag{4.34}$$

Thus,

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M \delta a \wedge da. \tag{4.35}$$

For the action to be stationary for arbitrary variations δa, we require

$$da = 0. \tag{4.36}$$

Hence,

$$f = da = 0. \tag{4.37}$$

The gauge connection is therefore locally flat. Witten identifies the stationary points of the Chern-Simons action with flat connections, and this flatness condition is one of the central reasons the theory has no ordinary local bulk dynamics [1].

4.6 No Local Propagating Bulk Degrees of Freedom

The equation of motion

$$f = da = 0 \tag{4.38}$$

means that the gauge field has no local curvature in the bulk. Locally, a flat Abelian connection is pure gauge:

$$a = d\lambda. \tag{4.39}$$

Therefore local fluctuations can be removed by a gauge transformation. Unlike Maxwell theory, pure Chern-Simons theory does not contain local propagating photon-like bulk modes. Its nontrivial information is global rather than local.

This does not make the theory empty. It means that the physical content is stored in global quantities such as

$$\oint_C a, \tag{4.40}$$

where C is a closed curve. If C wraps a non-contractible cycle of the spatial manifold, this integral can be nontrivial even though $da = 0$ locally.

4.7 Wilson Loops as Topological Observables

Since local field values are not gauge-invariant observables, the natural observables of Chern-Simons theory are Wilson loops. For a closed curve $C \subset M$, define

$$W_q(C) = \exp\left(iq \oint_C a \right). \tag{4.41}$$

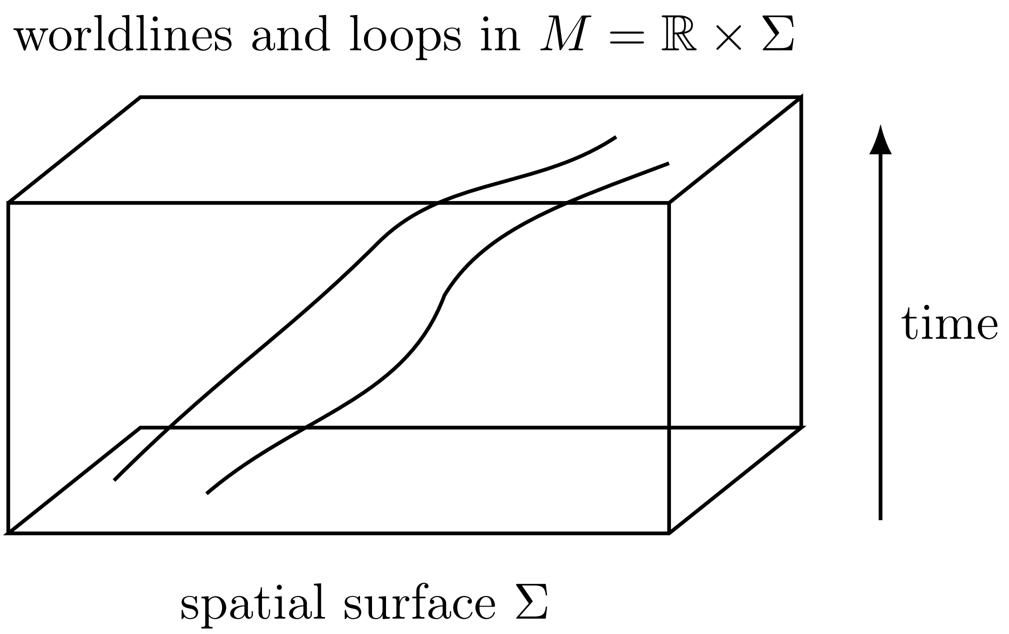


Figure 4.2: Pure Chern-Simons theory has no local propagating bulk modes. Its physical information is carried by global line and loop observables in three-dimensional spacetime.

Here q is the charge coupled to the gauge field. The integral

$$\oint_C a \tag{4.42}$$

measures the holonomy of the gauge connection around the loop C.

Under the gauge transformation $a \mapsto a + d\lambda$,

$$\oint_C a \longmapsto \oint_C (a + d\lambda) \tag{4.43}$$

$$= \oint_C a + \oint_C d\lambda. \tag{4.44}$$

For a closed curve C,

$$\oint_C d\lambda = 0, \tag{4.45}$$

assuming λ is single-valued. Therefore,

$$W_q(C) \tag{4.46}$$

is gauge invariant.

Wilson loops are topological observables because they are associated with closed curves in spacetime. Their expectation values can depend on how curves link or braid, but not on local geometric details such as the exact length of the curve. Simon explains that Wilson-loop insertions in the Chern-Simons path integral produce link and knot invariants, precisely because the action itself is metric independent [5].

4.8 Path Integral and Topological Partition Function

The quantum theory is formally defined by the path integral

$$Z(M) = \int \mathcal{D}a\, e^{iS_{\mathrm{CS}}[a]}. \tag{4.47}$$

Substituting the Abelian Chern-Simons action gives

$$Z(M) = \int \mathcal{D}a \, \exp\left[i\frac{k}{4\pi} \int_M a \wedge da \right]. \tag{4.48}$$

Since S_{CS} is metric independent, $Z(M)$ is expected to depend only on the topology of the three-manifold M, not on its metric geometry. Simon writes this Chern-Simons functional integral explicitly and states that it produces a manifold invariant when the level is properly quantized [5].

With Wilson-loop insertions, the normalized expectation value is

$$\langle W_{q_1}(C_1) \cdots W_{q_n}(C_n) \rangle = \frac{1}{Z(M)} \int \mathcal{D}a\, W_{q_1}(C_1) \cdots W_{q_n}(C_n) e^{iS_{\mathrm{CS}}[a]}. \tag{4.49}$$

These expectation values depend on the topology of the embedded curves, such as linking and braiding. Thus, the Chern-Simons path integral assigns topological amplitudes to line configurations inside the three-manifold.

4.9 Level Quantization

The coefficient k in the Chern-Simons action is called the level. For the quantum path integral to be well-defined under large gauge transformations, the level must be quantized:

$$k \in \mathbb{Z}. \tag{4.50}$$

The reason is that quantum mechanics uses the phase

$$e^{iS_{\mathrm{CS}}}, \tag{4.51}$$

not merely the classical action itself. If a large gauge transformation shifts the action by an integer multiple of 2π, then the quantum phase remains unchanged:

$$e^{i(S_{\mathrm{CS}}+2\pi n)} = e^{iS_{\mathrm{CS}}}, \qquad n \in \mathbb{Z}. \tag{4.52}$$

Witten and Simon both discuss this quantization condition in terms of gauge transformations with nontrivial winding number, while Tong discusses level quantization in Abelian Chern-Simons theory in the quantum Hall context [1, 4, 5].

4.10 Proof Summary

We can now summarize the proof that Abelian Chern-Simons theory is a 2+1-dimensional continuum TQFT.

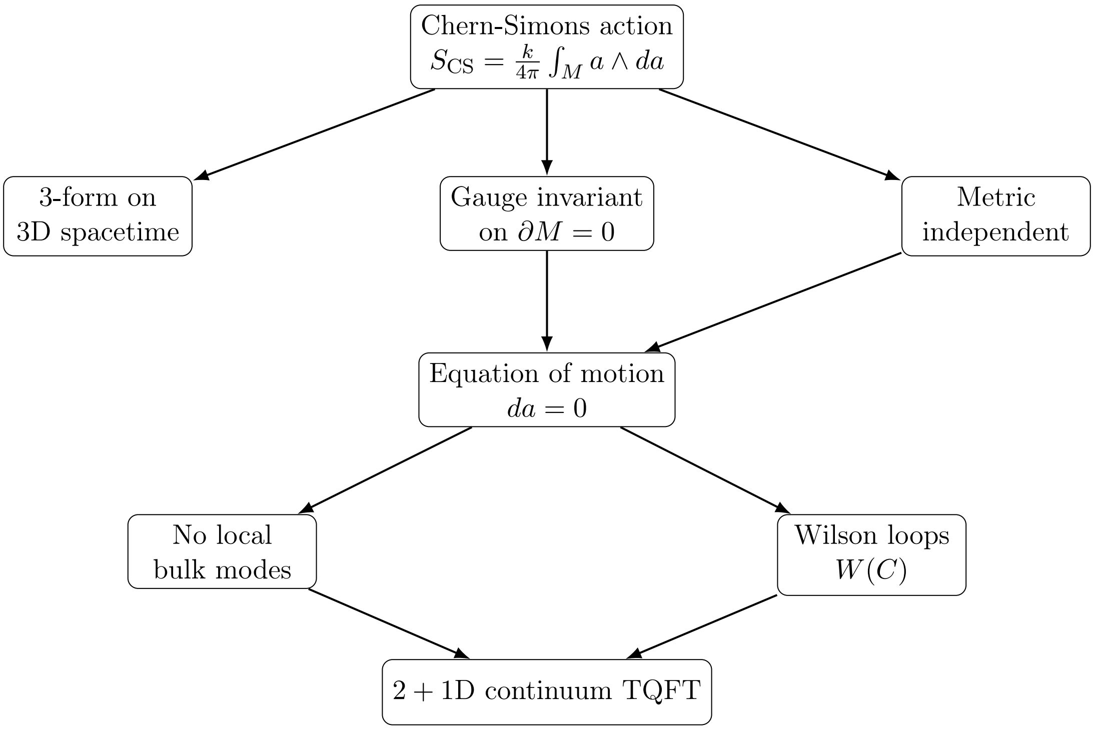


Figure 4.3: Proof structure showing why pure Abelian Chern-Simons theory defines a $2 + 1$-dimensional continuum TQFT.

The action is naturally $2 + 1$-dimensional because it integrates a three-form over a three-dimensional spacetime manifold. It is gauge invariant on closed manifolds, independent of the spacetime metric, and its equation of motion imposes flatness:

$$da = 0. \tag{4.53}$$

Consequently, pure Chern-Simons theory has no local propagating bulk degrees of freedom. Its natural physical observables are Wilson loops, whose expectation values depend on global topological data such as linking and braiding. Finally, the quantum theory requires a quantized level $k \in \mathbb{Z}$, ensuring that the path-integral phase is well-defined.

Pure Abelian Chern-Simons theory is a $2+1$-dimensional continuum TQFT because its action is gauge invariant, metric independent, imposes local flatness, has no local propagating bulk modes, and has Wilson loops as its natural topological observables.

This completes the general proof. In the next chapter, we place this continuum TQFT on a torus and study the global Wilson-loop degrees of freedom that arise from the two non-contractible cycles of T^2.

Chapter 5

Chern-Simons Theory as a $2 + 1$D Continuum TQFT on a Torus

In the previous chapter, we proved that pure Abelian Chern-Simons theory is a $2 + 1$-dimensional continuum TQFT because its action is gauge invariant, metric independent, locally flat, and described by Wilson-loop observables. In this chapter, we place the theory on a specific spatial topology: the torus. This is the first point where the global topology of the spatial manifold becomes physically visible. Although the local equation of motion still imposes flatness, the torus has two non-contractible cycles, and these cycles support nontrivial global holonomies. Quantizing these holonomies produces a finite-dimensional topological Hilbert space.

The guiding idea of this chapter is:

$$\boxed{\text{Chern-Simons theory on } M = \mathbb{R}_t \times T^2 \text{ has no local bulk modes, but it has global Wilson-loop degrees}} \tag{5.1}$$

We use Witten's general TQFT framework for Chern-Simons theory on $M = \mathbb{R} \times \Sigma$, and Tong's explicit Abelian torus calculation for the Wilson-loop algebra and torus ground-state degeneracy [1,4,5].

5.1 The Continuum Torus

The spatial manifold is chosen to be the two-dimensional torus

$$\Sigma = T^2 = S^1 \times S^1. \tag{5.2}$$

This means that the torus is the product of two independent circles. Let the angular coordinates on the two circles be

$$x^1, \qquad x^2. \tag{5.3}$$

The periodic identifications are

$$x^1 \sim x^1 + 2\pi R_1, \qquad x^2 \sim x^2 + 2\pi R_2. \tag{5.4}$$

Thus, the torus is a continuum surface: the coordinates vary continuously, but they are periodically identified. Equivalently, the torus may be represented as a square whose opposite edges are glued together.

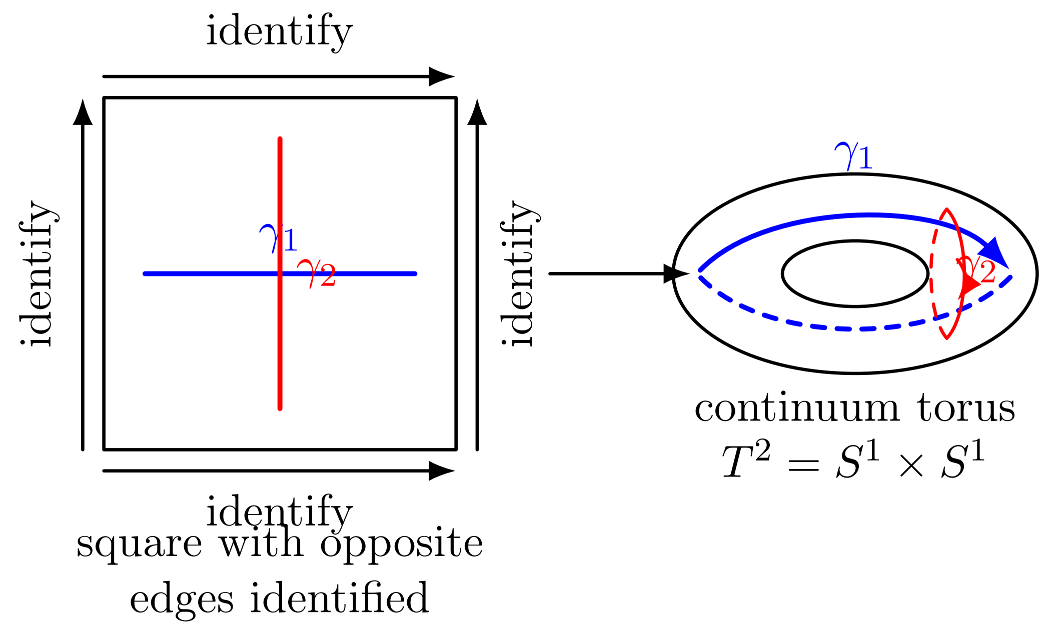


Figure 5.1: The continuum torus T^2 can be represented as a square with opposite edges identified. The two independent non-contractible cycles are γ_1, which runs around the main circumference of the torus, and γ_2, which runs around the tube cross-section.

The fundamental topological feature of the torus is that it contains two independent non-contractible cycles,

$$\gamma_1, \qquad \gamma_2. \tag{5.5}$$

A loop around either of these cycles cannot be continuously shrunk to a point. This is what makes Chern-Simons theory on a torus richer than Chern-Simons theory on the plane or sphere.

5.2 The Spacetime Manifold $M = \mathbb{R}_t \times T^2$

To define a 2 + 1-dimensional field theory, we evolve the two-dimensional torus in time. The spacetime manifold is therefore

$$M = \mathbb{R}_t \times T^2. \tag{5.6}$$

Here $\mathbb{R}_t$ is the time direction and T^2 is the spatial surface. At every time slice, the spatial system is a torus:

$$\Sigma_t = T^2. \tag{5.7}$$

Since T^2 is two-dimensional and $\mathbb{R}_t$ contributes one time dimension, the total spacetime is three-dimensional. This is why the theory is still a 2 + 1-dimensional continuum TQFT.

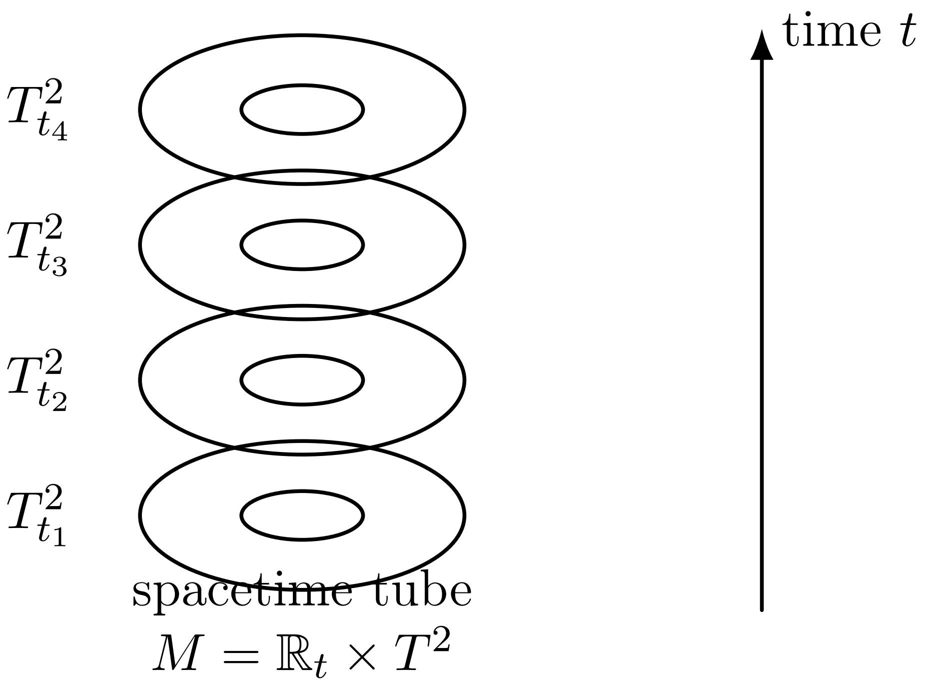


Figure 5.2: Chern-Simons theory on a torus is defined on the three-dimensional spacetime manifold $M = \mathbb{R}_t \times T^2$. Each time slice is a continuum torus.

5.3 Witten's Framework Specialized to the Abelian Torus Case

Witten's Chern-Simons framework considers a three-dimensional spacetime of the form

$$M = \mathbb{R} \times \Sigma, \tag{5.8}$$

where Σ is a closed spatial surface. Canonical quantization of the Chern-Simons theory assigns a Hilbert space to Σ:

$$\Sigma \longmapsto \mathcal{H}_\Sigma. \tag{5.9}$$

In Witten's general non-Abelian framework, this Hilbert space is related to conformal blocks of a corresponding two-dimensional conformal field theory. In this report, we specialize this idea to the Abelian case because the Toric Code anyons that will be compared later are Abelian.

For the Abelian theory, we take

$$G = U(1), \tag{5.10}$$

and the gauge field is a one-form

$$a = a_\mu dx^\mu. \tag{5.11}$$

The corresponding spatial surface is

$$\Sigma = T^2. \tag{5.12}$$

Thus, the TQFT assignment becomes

$$T^2 \longmapsto \mathcal{H}_{T^2}. \tag{5.13}$$

The goal of this chapter is to compute the topological structure of this Hilbert space using the Abelian Wilson loops around the two cycles of the torus.

5.4 Abelian Chern-Simons Action on $\mathbb{R}_t \times T^2$

Following Tong's Abelian torus calculation, the dynamical Chern-Simons action can be written as

$$S_{\mathrm{CS}} = \frac{e^2}{\hbar} \int d^3x \, \frac{m}{4\pi} \epsilon^{\mu\nu\rho} a_\mu \partial_\nu a_\rho. \tag{5.14}$$

Here $m \in \mathbb{Z}$ is the Abelian Chern-Simons level. Suppressing the conventional constants e and $\hbar$, the same action may be written in differential-form notation as

$$S_{\mathrm{CS}}[a] = \frac{m}{4\pi} \int_M a \wedge da, \qquad M = \mathbb{R}_t \times T^2. \tag{5.15}$$

The gauge field decomposes into temporal and spatial components:

$$a = a_0 dt + a_1 dx^1 + a_2 dx^2. \tag{5.16}$$

The spatial field strength is

$$f_{12} = \partial_1 a_2 - \partial_2 a_1. \tag{5.17}$$

5.5 Flatness Constraint and Global Holonomies

In Chern-Simons theory, the temporal component a_0 acts as a Lagrange multiplier. Varying the action with respect to a_0 imposes the constraint

$$f_{12} = 0. \tag{5.18}$$

This means that the spatial gauge connection is flat. Locally, a flat Abelian connection is pure gauge:

$$a = d\lambda. \tag{5.19}$$

However, on a torus, local flatness does not imply global triviality. The torus has two non-contractible cycles, and a flat connection may have nonzero holonomy around them.

Define the holonomies

$$w_i = \oint_{\gamma_i} a_j dx^j, \qquad i = 1, 2. \tag{5.20}$$

These quantities measure the gauge connection around the two independent torus cycles. They are global variables: they cannot be detected in a small local patch, but they are visible to Wilson loops wrapping around the full torus.

This is the central reason the torus is topologically important:

> The equation $f_{12} = 0$ removes local curvature, but the torus still permits nontrivial global holonomies around its two non-contractible cycles.

5.6 Wilson Loops as Gauge-Invariant Torus Observables

The holonomies w_i themselves can shift under large gauge transformations. Therefore, the gauge-invariant observables are the exponentiated Wilson loops

$$W_i = \exp\left(i\frac{e}{\hbar} \oint_{\gamma_i} a_j dx^j \right), \qquad i = 1, 2. \tag{5.21}$$

Equivalently,

$$W_i = \exp\left(i\frac{e}{\hbar}w_i\right). \tag{5.22}$$

Thus, the two fundamental topological observables of Abelian Chern-Simons theory on the torus are

$$W_1, \qquad W_2. \tag{5.23}$$

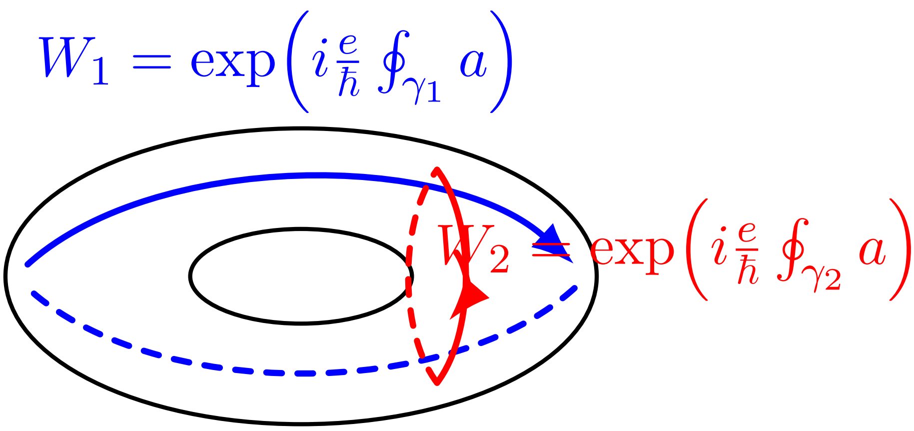


Figure 5.3: The two basic Wilson-loop observables on the torus wrap around the two non-contractible cycles γ_1 and γ_2.

These Wilson loops are topological observables. Their values do not depend on the precise shape or length of the loops, but only on which non-contractible cycle they wrap.

5.7 Canonical Commutation Relation of the Holonomies

Chern-Simons theory is first order in time derivatives. As a result, the two spatial components of the gauge field become conjugate variables. Tong gives the canonical commutation relation in the Abelian torus calculation as

$$[a_1(x), a_2(x')] = \frac{2\pi i}{m}\frac{\hbar^2}{e^2}\delta^2(x - x'), \tag{5.24}$$

up to orientation convention. Integrating this relation around the two independent cycles gives the holonomy commutator

$$[w_1, w_2] = \frac{2\pi i}{m}\frac{\hbar^2}{e^2}, \tag{5.25}$$

again up to the sign determined by the orientation of γ_1 and γ_2. The important physical point is that the two holonomies do not become independent commuting quantum numbers. They become conjugate topological variables.

Thus,

$$\boxed{w_1 \text{ and } w_2 \text{ are canonically conjugate global degrees of freedom.}} \tag{5.26}$$

This noncommutativity is purely topological. It comes from the intersection of the two fundamental cycles of the torus, not from any local propagating dynamics.

5.8 Wilson-Loop Algebra

The Wilson loops are exponentials of the holonomies:

$$W_1 = e^{iew_1/\hbar}, \qquad W_2 = e^{iew_2/\hbar}. \tag{5.27}$$

Using the Baker-Campbell-Hausdorff formula for exponentials of operators with a central commutator, Tong obtains the torus Wilson-loop algebra

$$\boxed{W_1 W_2 = e^{2\pi i/m} W_2 W_1.} \tag{5.28}$$

This is the central result of Abelian Chern-Simons theory on the torus.

Equation (5.28) says that the two Wilson loops around the independent cycles do not commute. Their noncommutativity is a phase, and therefore the algebra is Abelian in the anyonic sense: braiding acts by a scalar phase rather than by a non-Abelian matrix on a fusion space.

Geometrically, the reason is that γ_1 and γ_2 intersect once on the torus. In $2+1$-dimensional spacetime, this intersection becomes a linking of histories. The phase in (5.28) is the topological memory of this linking.

5.9 Ground-State Degeneracy on the Torus

The algebra

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1 \tag{5.29}$$

has a smallest irreducible representation of dimension m. Therefore, Abelian $U(1)_m$ Chern-Simons theory on a torus assigns an m-dimensional Hilbert space to T^2:

$$\boxed{\dim \mathcal{H}_{T^2} = m.} \tag{5.30}$$

More generally, for a closed genus-g surface, the corresponding Abelian Chern-Simons ground-state degeneracy is

$$\dim \mathcal{H}_{\Sigma_g} = m^g. \tag{5.31}$$

For the torus, $g = 1$, and hence

$$\dim \mathcal{H}_{T^2} = m. \tag{5.32}$$

One explicit representation is the clock-shift representation. Let

$$|n\rangle, \qquad n = 0, 1, \ldots, m-1, \tag{5.33}$$

be a basis of $\mathcal{H}_{T^2}$. Then the Wilson loops may act as

$$W_1 |n\rangle = e^{2\pi i n/m} |n\rangle, \tag{5.34}$$

and

$$W_2 |n\rangle = |n+1 \bmod m\rangle. \tag{5.35}$$

These operators satisfy

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1. \tag{5.36}$$

This explicitly shows how the topology of the torus produces a finite-dimensional quantum state space.

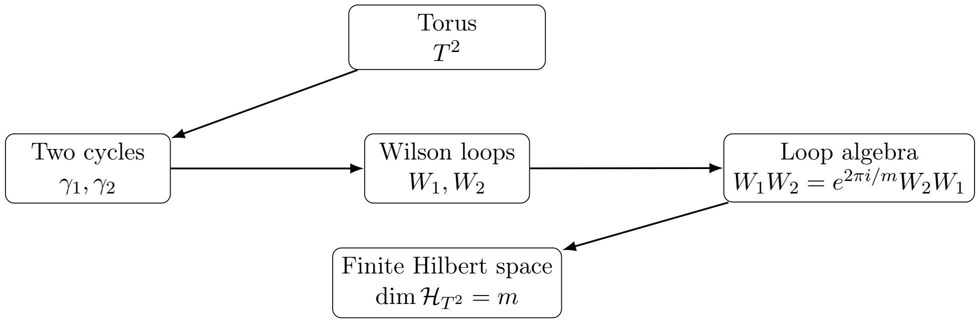


Figure 5.4: The topology of the torus produces two non-contractible cycles. The associated Wilson loops obey a noncommuting algebra, whose smallest irreducible representation gives the torus ground-state degeneracy.

5.10 Why This is a 2+1D Continuum TQFT on a Torus

We can now summarize the proof. The theory is $2 + 1$-dimensional because it is defined on

$$M = \mathbb{R}_t \times T^2, \tag{5.37}$$

where T^2 is two-dimensional space and $\mathbb{R}_t$ is time. It is a continuum theory because the gauge field

$$a = a_\mu dx^\mu \tag{5.38}$$

is a smooth differential form on the spacetime manifold. It is topological because the Chern-Simons action is metric independent and the local equation of motion imposes flatness:

$$f_{12} = 0. \tag{5.39}$$

Nevertheless, the torus has two non-contractible cycles, so flat connections may still carry global holonomies. The gauge-invariant Wilson loops around these cycles obey the nontrivial algebra

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1, \tag{5.40}$$

which gives a finite-dimensional topological Hilbert space

$$\dim \mathcal{H}_{T^2} = m. \tag{5.41}$$

> Chern-Simons theory on a continuum torus is a 2+1-dimensional TQFT because its local dynamics impose flatness, while its nontrivial physics is carried by global Wilson loops around the two non-contractible cycles of T^2.

This chapter establishes the first model in our comparison:

$$\mathrm{TQFT}_1 = \text{Chern-Simons theory on } \mathbb{R}_t \times T^2. \tag{5.42}$$

Later, we will construct the second model,

$$\mathrm{TQFT}_2 = \text{Toric Code as a } 2 + 1\text{D continuum TQFT on } T^2, \tag{5.43}$$

then compare their anyon sectors, Wilson/string operators, braiding phases, and torus ground-state degeneracies.

Chapter 6

Abelian Anyons in $2+1$D Chern-Simons Theory as a TQFT on a Torus

In the previous chapter, Chern-Simons theory was placed on the spacetime manifold

$$M = \mathbb{R}_t \times T^2, \tag{6.1}$$

where the spatial surface is a continuum torus. We saw that the torus has two independent non-contractible cycles, and that Chern-Simons theory on this space has global Wilson-loop degrees of freedom. In this chapter, we study the Abelian anyons that arise in this model. The aim is to understand how anyons appear as Wilson-line excitations, how they braid, how they fuse, and how their motion around the torus acts on the finite-dimensional topological Hilbert space.

The important point is that in a TQFT, the word "dynamics" does not mean local Newtonian motion or propagating field waves. Instead, the meaningful processes are topological processes: creation of anyon-antianyon pairs, braiding of worldlines, transport around non-contractible cycles, and annihilation. These processes are encoded by Wilson lines in $2+1$-dimensional spacetime.

6.1 Abelian Chern-Simons Theory on $M = \mathbb{R}_t \times T^2$

We begin with Abelian $U(1)_m$ Chern-Simons theory on the spacetime manifold

$$M = \mathbb{R}_t \times T^2. \tag{6.2}$$

The Abelian Chern-Simons action is

$$S_{\text{CS}}[a] = \frac{m}{4\pi} \int_M a \wedge da, \tag{6.3}$$

where

$$a = a_\mu dx^\mu \tag{6.4}$$

is a compact $U(1)$ gauge field, and

$$m \in \mathbb{Z} \tag{6.5}$$

is the Chern-Simons level.

Tong writes the same theory in component notation as

$$S_{\text{CS}} = \frac{e^2}{\hbar} \int d^3x \, \frac{m}{4\pi} \epsilon^{\mu\nu\rho} a_\mu \partial_\nu a_\rho. \tag{6.6}$$

For most of this chapter, we set

$$e = \hbar = 1 \tag{6.7}$$

to simplify notation. These constants can always be restored by dimensional analysis. The essential topological content is controlled by the integer level m.

6.2 Anyons as Wilson-Line Excitations

In Chern-Simons theory, an anyon is represented by a Wilson line. Let

$$C \subset M \tag{6.8}$$

be the spacetime worldline of a particle. A Wilson line of charge q is

$$W_q(C) = \exp\left(iq \int_C a\right). \tag{6.9}$$

Here

$$q \in \mathbb{Z} \tag{6.10}$$

is the $U(1)$ charge carried by the Wilson line.

Thus, in Abelian Chern-Simons theory, an anyon is not introduced as an ordinary local particle field. It is introduced as a charged line operator coupled to the gauge connection. Physically, the curve C is the path traced out by the particle in spacetime.

> An Abelian anyon in $U(1)_m$ Chern-Simons theory is a charged Wilson-line excitation. Its worldline C is assigned the operator
>
> $$W_q(C) = \exp\left(iq \int_C a\right).$$

The anyon is Abelian because braiding two such excitations gives a phase,

$$|\psi\rangle \longmapsto e^{i\theta}|\psi\rangle, \tag{6.11}$$

rather than a non-commuting matrix acting on a multi-dimensional fusion space. Therefore, all braiding data are scalar phases.

6.3 Coupling an Anyon Current to Chern-Simons Theory

To make the anyon appear dynamically, we couple the gauge field to a conserved current j^μ. The coupling term is

$$\Delta S = \int d^3x \, a_\mu j^\mu. \tag{6.12}$$

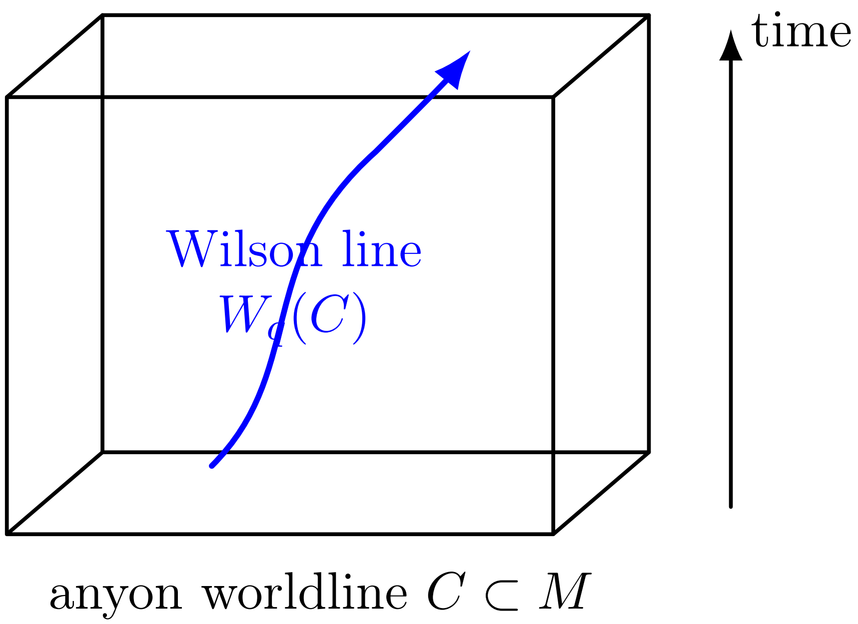


Figure 6.1: In Chern-Simons theory, an Abelian anyon is represented by a Wilson line along its spacetime worldline C.

The full action becomes

$$S[a;j] = \frac{m}{4\pi}\int_M a \wedge da + \int d^3x\, a_\mu j^\mu. \tag{6.13}$$

Gauge invariance requires the current to be conserved:

$$\partial_\mu j^\mu = 0. \tag{6.14}$$

Tong introduces precisely this kind of coupling when discussing quasi-holes and quasi-particles in Abelian Chern-Simons theory [4]. The conservation law means that anyon worldlines cannot simply start or stop in the bulk. They must either form closed loops, end on a boundary, or appear through particle-antiparticle creation and annihilation processes.

On a torus without boundary, the allowed histories are therefore closed worldline configurations in 2 + 1-dimensional spacetime:

$$\partial_\mu j^\mu = 0 \quad \Longrightarrow \quad \text{closed anyon worldlines in the bulk.} \tag{6.15}$$

6.4 Charge-Flux Attachment and Anyonic Excitations

The defining physical feature of Chern-Simons matter coupling is charge-flux attachment. Varying the action in Eq. (6.13) gives the schematic equation of motion

$$\frac{m}{2\pi}\epsilon^{\mu\nu\rho}\partial_\nu a_\rho + j^\mu = 0, \tag{6.16}$$

up to sign convention. Equivalently,

$$f_{\mu\nu} \propto \frac{1}{m}\epsilon_{\mu\nu\rho} j^\rho. \tag{6.17}$$

In Tong's conventions, the corresponding equation is

$$\frac{e^2}{2\pi\hbar} f_{\mu\nu} = \frac{1}{m}\epsilon_{\mu\nu\rho} j^\rho. \tag{6.18}$$

For a static point charge at the origin, this gives

$$\frac{1}{2\pi} f_{12} = \frac{1}{m}\delta^2(x), \tag{6.19}$$

after setting $e = \hbar = 1$. Integrating over a small disk D surrounding the anyon,

$$\frac{1}{2\pi}\int_D f_{12}\, d^2x = \frac{1}{m}. \tag{6.20}$$

Therefore,

$$\Phi = \int_D f = \frac{2\pi}{m}. \tag{6.21}$$

A unit Chern-Simons charge therefore carries magnetic flux $2\pi/m$.

Chern-Simons coupling attaches flux to charge. A unit anyon carries flux

$$\Phi = \frac{2\pi}{m}.$$

This charge-flux composite is the physical origin of Abelian anyonic statistics.

6.5 Braiding Phase from Flux Attachment

Consider a charge q anyon moving around a charge q' anyon. From the flux-attachment result, the q' anyon carries flux

$$\Phi_{q'} = \frac{2\pi q'}{m}. \tag{6.22}$$

A charge q particle moving around this flux acquires an Aharonov-Bohm phase

$$\exp\left(iq\Phi_{q'}\right). \tag{6.23}$$

Substituting the flux gives

$$M_{q,q'} = \exp\left(\frac{2\pi i q q'}{m}\right). \tag{6.24}$$

This is the mutual braiding phase of two Abelian anyons in $U(1)_m$ Chern-Simons theory.

For identical anyons $q = q'$, the full monodromy phase is

$$M_{q,q} = \exp\left(\frac{2\pi i q^2}{m}\right). \tag{6.25}$$

The exchange phase, also called the topological spin, is half of the full braid:

$$\theta_q = \exp\left(\frac{\pi i q^2}{m}\right). \tag{6.26}$$

Simon describes Abelian anyons using the charge-flux composite picture and relates their statistical angle to torus degeneracy and topological quantum information [5].

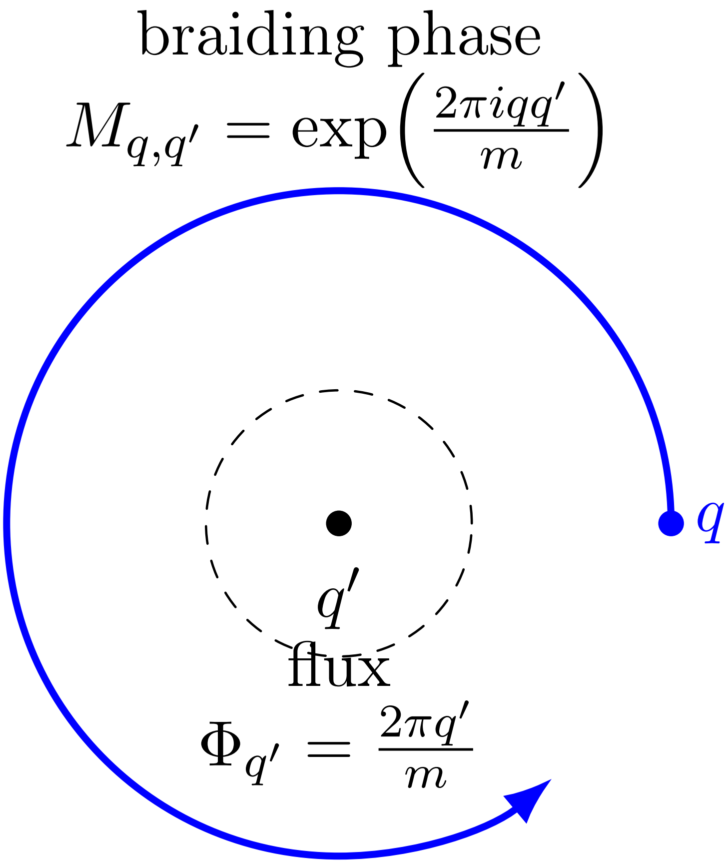


Figure 6.2: A charge q anyon moving around a charge q' anyon acquires an Aharonov-Bohm phase determined by the flux attached to q'.

6.6 Fusion Rules: Abelian Charge Addition

The fusion rules follow from charge addition. Wilson lines multiply as

$$W_q(C)W_{q'}(C) = W_{q+q'}(C). \tag{6.27}$$

In $U(1)_m$ Chern-Simons theory, charges are identified modulo m. Therefore, the distinct anyon sectors are

$$q = 0, 1, \dots, m-1. \tag{6.28}$$

The anyon fusion group is

$$\mathcal{A} = \mathbb{Z}_m. \tag{6.29}$$

The fusion rule is

$$q \times q' = (q + q') \mod m. \tag{6.30}$$

The vacuum sector is

$$0, \tag{6.31}$$

and the anti-anyon of q is

$$\bar{q} = m - q. \tag{6.32}$$

Every fusion product has exactly one output, which is why the theory is Abelian.

The Abelian anyon sectors of $U(1)_m$ Chern-Simons theory form the fusion group

$$\mathcal{A} = \mathbb{Z}_m, \qquad q \times q' = (q + q') \mod m.$$

6.7 Anyon Motion on the Torus

Now we use the topology of the spatial torus

$$T^2 = S^1 \times S^1. \tag{6.33}$$

It has two independent non-contractible cycles,

$$\gamma_1, \qquad \gamma_2. \tag{6.34}$$

A Wilson line can wrap around either cycle. The corresponding charge-q torus Wilson loops are

$$W_1^{(q)} = \exp\left(iq \oint_{\gamma_1} a\right), \tag{6.35}$$

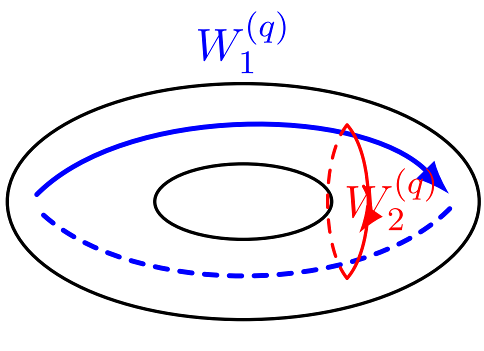
and

$$W_2^{(q)} = \exp\left(iq \oint_{\gamma_2} a\right). \tag{6.36}$$

Physically, this corresponds to a topological process:

1. create an anyon-antianyon pair from the vacuum;
2. move one anyon around a non-contractible cycle of the torus;
3. bring it back and annihilate it with its partner.

No local excitation remains after annihilation, but the process can still act nontrivially on the global ground-state Hilbert space. Simon describes this as a key feature of anyons on a torus: an anyon can wind around a nontrivial cycle and return to annihilate, changing the state in the topologically degenerate ground-state space [5].



anyon motion around non-contractible torus cycles

Figure 6.3: On the torus, Abelian anyons can move around the two non-contractible cycles. These processes are represented by Wilson-loop operators $W_1^{(q)}$ and $W_2^{(q)}$.

6.8 Wilson-Loop Algebra and Anyon Dynamics on T^2

From the canonical quantization of Chern-Simons theory on a torus, the fundamental Wilson loops obey

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1. \tag{6.37}$$

Tong derives this from the canonical commutation relation of the torus holonomies and identifies it with the same algebra that appears when considering anyons on a torus [4].

For charges q and q', the generalized relation is

$$W_1^{(q)} W_2^{(q')} = \exp\left(\frac{2\pi i q q'}{m}\right) W_2^{(q')} W_1^{(q)}. \tag{6.38}$$

This is the torus version of Abelian braiding. A loop around γ_1 and a loop around γ_2 intersect once on the torus. In $2+1$-dimensional spacetime, their histories link once. The linking produces the Abelian phase

$$\exp\left(\frac{2\pi i q q'}{m}\right). \tag{6.39}$$

Thus, the non-commutativity of Wilson loops is not caused by local dynamics. It is purely topological.

6.9 Representation on the Torus Hilbert Space

The torus Hilbert space has dimension

$$\dim \mathcal{H}_{T^2} = m. \tag{6.40}$$

Choose a basis

$$|n\rangle, \qquad n = 0, 1, \ldots, m-1. \tag{6.41}$$

The two fundamental Wilson loops can be represented as clock and shift operators:

$$W_1 |n\rangle = e^{2\pi i n/m} |n\rangle, \tag{6.42}$$

and

$$W_2 |n\rangle = |n+1 \mod m\rangle. \tag{6.43}$$

Now compute

$$W_1 W_2 |n\rangle = W_1 |n+1\rangle \tag{6.44}$$

$$= e^{2\pi i (n+1)/m} |n+1\rangle, \tag{6.45}$$

while

$$W_2 W_1 |n\rangle = W_2 \left(e^{2\pi i n/m} |n\rangle\right) \tag{6.46}$$

$$= e^{2\pi i n/m} |n+1\rangle. \tag{6.47}$$

Therefore,

$$W_1 W_2 |n\rangle = e^{2\pi i/m} W_2 W_1 |n\rangle, \tag{6.48}$$

and hence

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1. \tag{6.49}$$

This explicitly realizes the Wilson-loop algebra on the m-dimensional torus Hilbert space.

Anyon motion around the two cycles of the torus acts on the ground-state Hilbert space through clock and shift operators. This is the quantum-mechanical representation of the Wilson-loop algebra.

6.10 Emergent Behaviour and Dynamics in the TQFT

In pure Chern-Simons theory without sources, the local equation of motion is

$$f_{12} = 0. \tag{6.50}$$

Therefore, there are no local propagating bulk modes. When anyon sources are inserted, they appear as Wilson lines or conserved current insertions. Their motion is encoded by the topology of their worldlines, not by the length, speed, or precise shape of the paths.

The meaningful dynamical processes are:

$$\text{creation of anyon-antianyon pairs,} \tag{6.51}$$

$$\text{braiding of worldlines,} \tag{6.52}$$

$$\text{transport around non-contractible torus cycles,} \tag{6.53}$$

and

$$\text{annihilation.} \tag{6.54}$$

These operations act on the finite-dimensional topological Hilbert space. Their amplitudes depend only on topological data such as linking number, winding number, and intersection number.

6.11 Conclusion

We have shown that Abelian anyons in $U(1)_m$ Chern-Simons theory on

$$M = \mathbb{R}_t \times T^2 \tag{6.55}$$

are Wilson-line excitations labelled by

$$q \in \mathbb{Z}_m. \tag{6.56}$$

Their fusion rule is

$$q \times q' = (q + q') \mod m, \tag{6.57}$$

their mutual braiding phase is

$$M_{q,q'} = \exp\left(\frac{2\pi i q q'}{m}\right), \tag{6.58}$$

and their topological spin is

$$\theta_q = \exp\left(\frac{\pi i q^2}{m}\right). \tag{6.59}$$

On the torus, their Wilson-loop operators obey

$$W_1^{(q)} W_2^{(q')} = \exp\left(\frac{2\pi i q q'}{m}\right) W_2^{(q')} W_1^{(q)}. \tag{6.60}$$

This algebra acts on an m-dimensional torus Hilbert space:

$$\dim \mathcal{H}_{T^2} = m. \tag{6.61}$$

Therefore,

> Abelian anyons in $2+1$-dimensional Chern-Simons theory on a torus emerge as topological Wilson-line excitations. Their fusion, braiding, and motion around non-contractible cycles are encoded by Wilson-loop algebra acting on a finite-dimensional torus Hilbert space.

This prepares the next part of the report, where we will study the Toric Code as a $2+1$-dimensional continuum TQFT and later compare its Abelian anyon structure with the Chern-Simons theory developed here.

Chapter 7

Toric Code as a $2 + 1$D Continuum TQFT

In the previous chapters, Chern-Simons theory was studied as a continuum 2+1-dimensional topological quantum field theory. We now turn to the Toric Code. The Toric Code is not microscopically a continuum quantum field theory. It is a discrete lattice Hamiltonian model. However, its low-energy and long-distance topological sector behaves as a $2 + 1$-dimensional continuum TQFT. The aim of this chapter is to prove this statement carefully.

The main point is that the continuum TQFT description does not arise by treating each microscopic lattice qubit as a smooth field. Instead, it arises after local microscopic details are ignored and only the universal topological data remain. These universal data are the anyon types, fusion rules, braiding phases, string-operator algebra, and topology-dependent ground-state degeneracy. Simon describes the Toric Code as a topologically ordered phase whose long-wavelength physics is described by a TQFT, after introducing it as a Hamiltonian built from vertex/star and plaquette operators [5, Chs. 25–26]. Kitaev originally introduced the model as an exactly solved Hamiltonian model supporting anyonic excitations and fault-tolerant quantum computation [2].

The Toric Code is a microscopic lattice Hamiltonian model whose low-energy topological sector defines a $2 + 1$-dimensional TQFT. The continuum theory is obtained by extracting the universal topological data, not by directly identifying lattice qubits with smooth continuum fields.

7.1 Microscopic Starting Point: The Toric Code Lattice Model

We start with a two-dimensional square lattice Λ embedded in a spatial surface Σ. When time evolution is included, the corresponding spacetime is

$$M = \mathbb{R}_t \times \Sigma. \tag{7.1}$$

Thus the Toric Code has two spatial dimensions and one time dimension. This is the same dimensional setting in which anyons and Chern-Simons theory naturally appear.

A qubit is placed on each edge e of the lattice. If $E(\Lambda)$ denotes the set of lattice edges, the microscopic Hilbert space is

$$\mathcal{H}_\Lambda = \bigotimes_{e\in E(\Lambda)} \mathbb{C}^2. \tag{7.2}$$

On every edge we have Pauli operators

$$\sigma_e^x, \qquad \sigma_e^z. \tag{7.3}$$

The Toric Code Hamiltonian is

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p, \tag{7.4}$$

where s labels vertices or stars, and p labels plaquettes. In the convention used in this report,

$$A_s = \prod_{e\in \mathrm{star}(s)} \sigma_e^x, \tag{7.5}$$

and

$$B_p = \prod_{e\in\partial p} \sigma_e^z. \tag{7.6}$$

Here star(s) is the set of edges adjacent to the vertex s, and ∂p is the boundary of the plaquette p. Some references interchange the roles of σ^x and σ^z. This is only a convention and does not change the topological phase.

The star and plaquette construction is the standard microscopic definition of the Toric Code. Simon introduces the model using vertex and plaquette stabilizers, and Kitaev writes the original exactly solved Hamiltonian in terms of local star and plaquette terms [5, Ch. 25] [2].

7.2 Proof that the Hamiltonian Terms Commute

The first important property of the Toric Code is exact solvability. To prove this, we must show that all stabilizer terms commute:

$$[A_s, A_{s'}] = 0, \qquad [B_p, B_{p'}] = 0, \qquad [A_s, B_p] = 0. \tag{7.7}$$

First, each A_s is a product of σ^x operators. Therefore two star operators commute:

$$A_s A_{s'} = A_{s'} A_s, \tag{7.8}$$

and hence

$$[A_s, A_{s'}] = 0. \tag{7.9}$$

Second, each B_p is a product of σ^z operators. Therefore two plaquette operators commute:

$$B_p B_{p'} = B_{p'} B_p, \tag{7.10}$$

and hence

$$[B_p, B_{p'}] = 0. \tag{7.11}$$

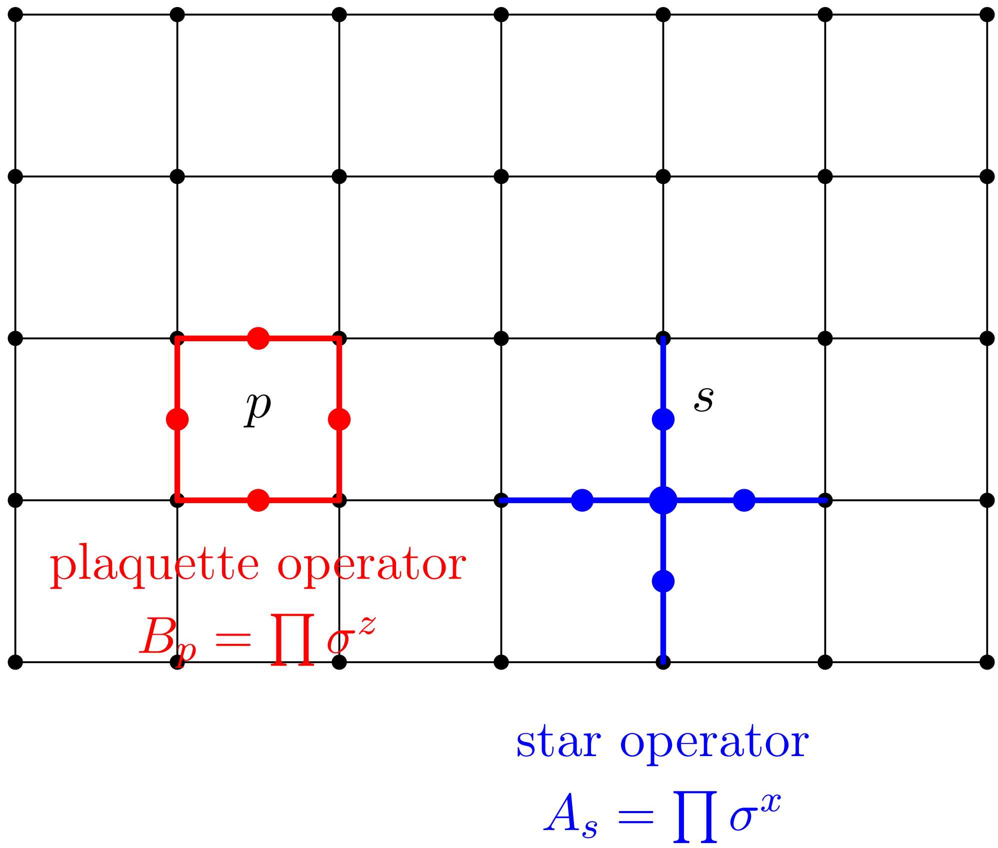


Figure 7.1: The Toric Code is a lattice Hamiltonian model with qubits on edges. The star operator A_s acts on the four edges adjacent to a vertex, while the plaquette operator B_p acts on the four edges around a plaquette.

The nontrivial case is the commutator between a star and a plaquette. On a single shared edge,

$$\sigma_e^x \sigma_e^z = -\sigma_e^z \sigma_e^x. \tag{7.12}$$

So one shared edge produces a minus sign. But on a square lattice, a star and a plaquette share either zero edges or two edges. If they share zero edges, all operators act on different qubits and commute. If they share two edges, the two minus signs cancel:

$$(-1)(-1) = +1. \tag{7.13}$$

Therefore,

$$A_s B_p = B_p A_s, \tag{7.14}$$

and hence

$$[A_s, B_p] = 0. \tag{7.15}$$

Thus all terms in the Hamiltonian commute:

$$\boxed{[A_s, A_{s'}] = [B_p, B_{p'}] = [A_s, B_p] = 0.} \tag{7.16}$$

This commuting-projector structure is central to the exact solvability of the Toric Code and is part of the standard star/plaquette construction discussed by Simon and Kitaev [5, Ch. 25] [2].

7.3 Ground-State Space as a Simultaneous Stabilizer Eigenspace

Since all stabilizers commute, the Hamiltonian can be diagonalized by choosing simultaneous eigenstates of all A_s and B_p. Moreover,

$$A_s^2 = 1, \qquad B_p^2 = 1, \tag{7.17}$$

so their eigenvalues are

$$A_s = \pm 1, \qquad B_p = \pm 1. \tag{7.18}$$

For

$$J_e > 0, \qquad J_m > 0, \tag{7.19}$$

the energy is minimized when each term contributes negatively. Therefore the ground state satisfies

$$A_s|\psi\rangle = |\psi\rangle, \qquad B_p|\psi\rangle = |\psi\rangle \qquad \forall s, p. \tag{7.20}$$

Hence the ground-state space is

$$\mathcal{H}_{\mathrm{GS}} = \{|\psi\rangle : A_s|\psi\rangle = |\psi\rangle,\ B_p|\psi\rangle = |\psi\rangle\ \forall s, p\}. \tag{7.21}$$

Equivalently, the ground-state projector is

$$P_{\mathrm{GS}} = \prod_s \frac{1 + A_s}{2} \prod_p \frac{1 + B_p}{2}. \tag{7.22}$$

Thus the Toric Code ground state is a constraint-satisfying stabilizer space. Simon develops the Toric Code code space in precisely this stabilizer language, while Kitaev's original construction identifies the protected subspace and the anyonic excitations of this exactly solved Hamiltonian [5, Ch. 25] [2].

7.4 Physical Meaning: Local Gauge Invariance and Local Flatness

The stabilizer constraints have a useful gauge-theoretic interpretation. The Toric Code can be viewed as a discrete $\mathbb{Z}_2$ gauge theory. Each edge qubit represents a discrete gauge degree of freedom,

$$g_e \in \mathbb{Z}_2. \tag{7.23}$$

The star operator

$$A_s = \prod_{e \in \mathrm{star}(s)} \sigma_e^x \tag{7.24}$$

acts as a local $\mathbb{Z}_2$ gauge transformation at vertex s. Therefore,

$$A_s|\psi\rangle = |\psi\rangle \tag{7.25}$$

means that physical ground states are invariant under local $\mathbb{Z}_2$ gauge transformations.

The plaquette operator

$$B_p = \prod_{e \in \partial p} \sigma_e^z \tag{7.26}$$

measures the $\mathbb{Z}_2$ flux through the plaquette p. Therefore,

$$B_p|\psi\rangle = |\psi\rangle \tag{7.27}$$

means that there is no local $\mathbb{Z}_2$ flux in the ground state.

Thus the two ground-state constraints have the interpretation

$$A_s = +1 \quad \Longleftrightarrow \quad \text{local gauge invariance,} \tag{7.28}$$

and

$$B_p = +1 \quad \Longleftrightarrow \quad \text{local flatness.} \tag{7.29}$$

The plaquette condition is the lattice analogue of the continuum flatness condition

$$F = 0. \tag{7.30}$$

This is why the Toric Code begins to resemble a topological gauge theory in the ground sector: there are no local field-strength excitations, and the remaining physical information is global. Simon discusses the relation between the Toric Code, quantum double models, continuum topological models, and gauge-theoretic descriptions in his later chapters on the Kitaev quantum double construction [5, Chs. 26 and 29].

> The Toric Code ground-state constraints are the lattice version of gauge invariance and flatness:
> $$A_s = +1 \quad \text{and} \quad B_p = +1.$$
> This is the first sign that the low-energy sector behaves like a topological gauge theory.

7.5 Excitations as Violations of Stabilizer Constraints

Excitations appear when the local stabilizer constraints are violated. If

$$A_s|\psi\rangle = -|\psi\rangle, \tag{7.31}$$

then the vertex s carries an electric excitation,

$$e. \tag{7.32}$$

If

$$B_p|\psi\rangle = -|\psi\rangle, \tag{7.33}$$

then the plaquette p carries a magnetic excitation,

$$m. \tag{7.34}$$

Their composite excitation is

$$\varepsilon = e \times m. \tag{7.35}$$

Therefore the Toric Code has four anyon sectors:

$$1, \qquad e, \qquad m, \qquad \varepsilon = e \times m. \tag{7.36}$$

Simon discusses these vertex and plaquette defects as the excitations of the Toric Code phase, and Kitaev identifies the anyonic excitations of the exactly solved model [5, Ch. 26] [2].

7.6 Fusion Rules

The Toric Code is a $\mathbb{Z}_2$ gauge theory. Therefore two identical charges annihilate:

$$e \times e = 1, \tag{7.37}$$

and two identical magnetic fluxes also annihilate:

$$m \times m = 1. \tag{7.38}$$

The composite of an electric and a magnetic excitation is

$$e \times m = \varepsilon. \tag{7.39}$$

Now consider fusing ε with itself:

$$\begin{align} \varepsilon \times \varepsilon &= (e \times m) \times (e \times m) \tag{7.40} \\ &= (e \times e) \times (m \times m) \tag{7.41} \\ &= 1 \times 1 \tag{7.42} \\ &= 1. \tag{7.43} \end{align}$$

Thus the fusion rules are

$$\boxed{e \times e = 1, \qquad m \times m = 1, \qquad e \times m = \varepsilon, \qquad \varepsilon \times \varepsilon = 1.} \tag{7.44}$$

The fusion algebra is therefore

$$\mathcal{A}_{\mathrm{TC}} = \mathbb{Z}_2 \times \mathbb{Z}_2. \tag{7.45}$$

Every fusion product gives a unique output, so the anyons are Abelian. Simon and Kitaev both discuss the Abelian anyonic content of the Toric Code and its statistical properties [5, Ch. 26] [2].

7.7 String Operators Create and Move Anyons

The Toric Code anyons are created and transported by string operators. With the convention used above, an e-string operator along a path γ is

$$W_e(\gamma) = \prod_{e\in\gamma} \sigma_e^z. \tag{7.46}$$

This operator anticommutes with star operators at the endpoints of γ. Therefore, an open e-string creates an e-anyon pair at its endpoints.

Similarly, an m-string operator along a dual path γ^* is

$$W_m(\gamma^*) = \prod_{e\in\gamma^*} \sigma_e^x. \tag{7.47}$$

This operator anticommutes with plaquette operators at the endpoints of the dual path. Therefore, an open m-string creates an m-anyon pair at its endpoints.

If a string is closed and contractible, it is locally generated by stabilizers and does not change the topological sector. But if a string is closed and non-contractible, it may act nontrivially on the ground-state Hilbert space. This is the bridge from the microscopic lattice model to TQFT: local excitations live at string endpoints, while closed non-contractible strings act globally. Simon discusses the string/ribbon-operator viewpoint in the Toric Code and quantum double settings [5, Chs. 25–26 and 29].

7.8 Braiding from String-Operator Algebra

The mutual braiding phase of the Toric Code is encoded in the algebra of e- and m-string operators. Suppose an e-string and an m-string cross once. At the crossing edge, the corresponding Pauli operators are

$$\sigma_e^z \qquad \text{and} \qquad \sigma_e^x. \tag{7.48}$$

They satisfy

$$\sigma_e^z \sigma_e^x = -\sigma_e^x \sigma_e^z. \tag{7.49}$$

Therefore, for one crossing,

$$W_e(\gamma)W_m(\gamma^*) = -W_m(\gamma^*)W_e(\gamma). \tag{7.50}$$

More generally, if the two paths cross $I(\gamma, \gamma^*)$ times, then

$$W_e(\gamma)W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)} W_m(\gamma^*)W_e(\gamma). \tag{7.51}$$

For one full braid of e around m, the worldline linking number is one. Therefore the mutual braiding phase is

$$M_{e,m} = -1. \tag{7.52}$$

This phase is purely topological. It does not depend on distance, speed, or the precise local shape of the path. It depends only on crossing number or linking number. Simon explicitly discusses the braiding of vertex and plaquette defects in the Toric Code, and Kitaev identifies the corresponding anyonic statistics in the exactly solved model [5, Ch. 26] [2].

7.9 Why the Toric Code is $2+1$D

The Toric Code lives on a two-dimensional spatial lattice:

$$\Sigma = \text{two-dimensional spatial surface}. \tag{7.53}$$

When time is included, the spacetime becomes

$$M = \mathbb{R}_t \times \Sigma. \tag{7.54}$$

Thus anyon motion is described by worldlines in a three-dimensional spacetime. This is exactly the setting in which braiding is meaningful.

Braiding is special to two spatial dimensions because particle worldlines in $2+1$-dimensional spacetime can form nontrivial braids. This is the same reason that Chern-Simons theory is naturally a $2+1$-dimensional TQFT. Simon discusses paths and braid groups in $2+1$ dimensions and later presents the Toric Code as a topological phase described by TQFT data [5, Chs. 3 and 26].

7.10 Why the Toric Code is Topological

The Toric Code is topological because its important low-energy data are insensitive to local geometry. The following quantities do not depend on microscopic distances, angles, or shapes:

$$\text{anyon types,} \tag{7.55}$$

$$\text{fusion rules,} \tag{7.56}$$

$$\text{braiding phases,} \tag{7.57}$$

$$\text{topological spins,} \tag{7.58}$$

$$\text{non-contractible string-operator algebra,} \tag{7.59}$$

and

$$\text{ground-state degeneracy.} \tag{7.60}$$

For example, if a string operator is deformed locally without moving its endpoints or changing its crossing data, its topological action on the ground-state sector is unchanged. In the ground-state space, different representatives of the same homology class act in the same way up to stabilizers. Thus the low-energy observables depend on topology and not local geometry.

This is exactly the sense in which the Toric Code is a topologically ordered phase. Simon describes the Toric Code as a phase of matter and a TQFT, and discusses the robustness of its topological properties under local perturbations [5, Chs. 26–27].

7.11 Why the Continuum Description Appears Only in the Low-Energy Limit

The Toric Code is microscopically discrete. Its Hilbert space is

$$\mathcal{H}_\Lambda = \bigotimes_{e \in E(\Lambda)} \mathbb{C}^2, \tag{7.61}$$

and the lattice spacing is finite. Therefore, one should not say that each qubit becomes a smooth field. That would be incorrect.

The correct statement is that the continuum TQFT appears after taking the long-distance topological limit:

$$a_{\text{lattice}} \to 0, \tag{7.62}$$

while keeping only universal topological data. The low-energy sector ignores

$$\text{lattice spacing,} \qquad \text{microscopic edge positions,} \qquad \text{local path deformations.} \tag{7.63}$$

It preserves

$$\text{anyon labels,} \qquad \text{fusion,} \qquad \text{braiding,} \tag{7.64}$$

$$\text{loop algebra,} \qquad \text{ground-state degeneracy.} \tag{7.65}$$

Thus, at the level of its emergent topological phase, the Toric Code can be described as a continuum TQFT. The corresponding continuum theory will later be identified with the $N = 2$ BF theory, equivalently a doubled Abelian Chern-Simons theory with

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{7.66}$$

This is the continuum topological field-theory description that reproduces the universal Toric Code data [5, Chs. 26 and 29] [3].

The continuum limit of the Toric Code keeps the topological sector, not the microscopic qubits. The emergent TQFT remembers anyons, fusion, braiding, loop algebra, and ground-state degeneracy.

7.12 TQFT-Like Hilbert-Space Assignment

A 2+1-dimensional TQFT assigns a finite-dimensional Hilbert space to a two-dimensional spatial surface:

$$\Sigma \longmapsto \mathcal{H}(\Sigma). \tag{7.67}$$

For the Toric Code, the corresponding low-energy topological Hilbert space is

$$\Sigma \longmapsto \mathcal{H}_{\mathrm{TC}}(\Sigma), \tag{7.68}$$

where

$$\mathcal{H}_{\mathrm{TC}}(\Sigma) = \{|\psi\rangle : A_s|\psi\rangle = |\psi\rangle,\ B_p|\psi\rangle = |\psi\rangle\ \forall s, p\}. \tag{7.69}$$

The dimension of this space depends on the topology of Σ, not on local geometric details.

For a genus-g surface, the $\mathbb{Z}_2$ Toric Code has

$$\dim \mathcal{H}_{\mathrm{TC}}(\Sigma_g) = 2^{2g}. \tag{7.70}$$

For the torus,

$$g = 1, \tag{7.71}$$

so

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 2^2 = 4. \tag{7.72}$$

This shows that the Toric Code ground-state Hilbert space behaves like a TQFT Hilbert-space assignment. Simon discusses the dependence of the Toric Code ground-state structure on topology and gives the general $\mathbb{Z}_N$ degeneracy pattern in his discussion of different topologies and quantum double models [5, Chs. 25 and 29].

7.13 Toric Code as a Topological Quantum Memory

The Toric Code also behaves as a topological quantum memory. Local errors create local anyon pairs:

$$1 \longrightarrow e \times e, \tag{7.73}$$

or

$$1 \longrightarrow m \times m. \tag{7.74}$$

These defects can be detected by local stabilizer measurements.

A logical error, however, requires a string to wrap nontrivially around the spatial surface:

$$\text{non-contractible loop} \quad \Longrightarrow \quad \text{logical operator}. \tag{7.75}$$

Thus the stored quantum information is not localized at a single site. It is stored globally in the topology of the surface. Simon emphasizes the Toric Code as a quantum memory and explains how the surface code is a practical planar descendant of the Toric Code, where local checks detect defects and logical errors are extended strings connecting boundaries or wrapping nontrivially around the surface [5, Chs. 24–25]. This quantum-memory interpretation supports the TQFT interpretation because logical information is protected by topology rather than local microscopic geometry.

7.14 Conclusion

We have shown that the Toric Code begins as a discrete stabilizer Hamiltonian model but has a low-energy topological sector with the structure of a 2 + 1-dimensional continuum TQFT.

First, it is defined on a two-dimensional spatial lattice Σ, so including time gives

$$M = \mathbb{R}_t \times \Sigma. \tag{7.76}$$

Second, its Hamiltonian is a sum of mutually commuting local stabilizers:

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p. \tag{7.77}$$

Third, its ground-state space is the simultaneous +1 eigenspace of all star and plaquette operators:

$$A_s|\psi\rangle = |\psi\rangle, \qquad B_p|\psi\rangle = |\psi\rangle. \tag{7.78}$$

Fourth, violations of these stabilizers create Abelian anyons:

$$1, \qquad e, \qquad m, \qquad \varepsilon = e \times m. \tag{7.79}$$

Their fusion algebra is

$$\mathbb{Z}_2 \times \mathbb{Z}_2. \tag{7.80}$$

Their mutual braiding is encoded by the string-operator algebra

$$W_e(\gamma)W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)}W_m(\gamma^*)W_e(\gamma). \tag{7.81}$$

For one crossing,

$$M_{e,m} = -1. \tag{7.82}$$

Finally, the ground-state Hilbert space depends on the topology of the spatial surface:

$$\dim \mathcal{H}_{\mathrm{TC}}(\Sigma_g) = 2^{2g}. \tag{7.83}$$

Therefore,

> The Toric Code realizes a $2+1$-dimensional continuum TQFT in its low-energy, long-distance topological sector. Its universal data are Abelian anyons, $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion, topological string-operator algebra, nontrivial braiding, and topology-dependent ground-state Hilbert spaces.

This prepares the next chapter, where the Toric Code will be placed explicitly on a torus and the four-dimensional ground-state Hilbert space will be derived from non-contractible string operators.

Chapter 8

Toric Code as a $2 + 1$D Continuum TQFT on a Torus

In the previous chapter, we showed that the Toric Code is not microscopically a continuum quantum field theory. It is a discrete lattice Hamiltonian model. However, its low-energy and long-distance topological sector behaves as a $2+1$-dimensional continuum TQFT. In this chapter, we specialize that statement to the case where the spatial surface is a torus,

$$\Sigma = T^2. \tag{8.1}$$

Thus the spacetime is

$$M = \mathbb{R}_t \times T^2. \tag{8.2}$$

The purpose of this chapter is to prove the torus-specific TQFT structure of the Toric Code. The central result is

$$\boxed{\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4.} \tag{8.3}$$

Physically, this means that the Toric Code on a torus stores two logical qubits:

$$4 = 2^2. \tag{8.4}$$

Mathematically, this fourfold degeneracy appears because the torus has two independent non-contractible cycles,

$$\gamma_1, \qquad \gamma_2. \tag{8.5}$$

The corresponding non-contractible string operators cannot be written as products of local stabilizers. Therefore, they act nontrivially on the ground-state Hilbert space. Simon discusses the Toric Code on different topologies, including the dependence of the code space on topology, and later treats the torus ground-state degeneracy in the quantum double description [5, Chs. 25 and 29]. Kitaev's original model also identifies the protected topological ground-state structure and anyonic excitations of the Toric Code [2].

The Toric Code on a torus is a local stabilizer Hamiltonian model whose ground-state sector is controlled by global topology. The two non-contractible cycles of T^2 generate four topological ground states.

8.1 Defining the Toric Code Lattice on a Torus

To define the Toric Code on a torus, take an $n \times n$ square lattice with periodic boundary conditions in both spatial directions. Periodic boundary conditions mean that the left and right sides are identified, and the top and bottom sides are identified. The resulting spatial surface is

$$T^2 = S^1 \times S^1. \tag{8.6}$$

For an $n \times n$ periodic square lattice, the number of vertices is

$$V = n^2, \tag{8.7}$$

the number of plaquettes is

$$P = n^2, \tag{8.8}$$

and the number of edges is

$$E = 2n^2. \tag{8.9}$$

There are n^2 horizontal edges and n^2 vertical edges, so

$$E = n^2 + n^2 = 2n^2. \tag{8.10}$$

As before, one qubit is placed on every edge of the lattice. Therefore, the microscopic Hilbert space is

$$\mathcal{H}_n = \bigotimes_{e \in E_n} \mathbb{C}^2, \tag{8.11}$$

where E_n is the set of edges of the periodic $n \times n$ lattice. Since there are $E = 2n^2$ edge qubits,

$$\dim \mathcal{H}_n = 2^E = 2^{2n^2}. \tag{8.12}$$

This counting is the microscopic starting point for the torus degeneracy proof. Simon constructs the Toric Code by placing qubits on edges and studying the vertex and plaquette stabilizers on different topologies [5, Ch. 25].

8.2 The Toric Code Hamiltonian on the Torus

The local Hamiltonian has the same form as in Chapter 7:

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p. \tag{8.13}$$

With our convention,

$$A_s = \prod_{e \in \mathrm{star}(s)} \sigma_e^x, \tag{8.14}$$

and

$$B_p = \prod_{e \in \partial p} \sigma_e^z. \tag{8.15}$$

The torus topology changes the global structure of the model, but it does not change the local definition of the Hamiltonian. Locally, every vertex still has a star operator and every plaquette still has a plaquette operator.

The stabilizers commute:

$$[A_s, A_{s'}] = 0, \qquad [B_p, B_{p'}] = 0, \qquad [A_s, B_p] = 0. \tag{8.16}$$

The proof is the same as before. Star operators are products of σ^x operators, so they commute with other star operators. Plaquette operators are products of σ^z operators, so they commute with other plaquette operators. A star and plaquette share either zero edges or two edges. On one shared edge,

$$\sigma_e^x \sigma_e^z = -\sigma_e^z \sigma_e^x. \tag{8.17}$$

If the star and plaquette share two edges, the two minus signs cancel:

$$(-1)^2 = +1. \tag{8.18}$$

Therefore,

$$[A_s, B_p] = 0. \tag{8.19}$$

This commuting-projector structure is the reason the Toric Code is exactly solvable, both on the plane and on the torus [5, Ch. 25] [2].

8.3 Ground-State Constraints on the Torus

Because all stabilizers commute, the ground state is the simultaneous $+1$ eigenspace of all star and plaquette operators:

$$A_s|\psi\rangle = |\psi\rangle \qquad \forall s, \tag{8.20}$$

and

$$B_p|\psi\rangle = |\psi\rangle \qquad \forall p. \tag{8.21}$$

Therefore, the Toric Code ground-state space on the torus is

$$\mathcal{H}_{\mathrm{TC}}(T^2) = \{|\psi\rangle : A_s|\psi\rangle = |\psi\rangle,\ B_p|\psi\rangle = |\psi\rangle\ \forall s, p\}. \tag{8.22}$$

Equivalently, the ground-state projector is

$$P_{\mathrm{GS}} = \prod_s \frac{1 + A_s}{2} \prod_p \frac{1 + B_p}{2}. \tag{8.23}$$

The important point is that the torus ground state is not obtained by specifying local field values. It is obtained by imposing all local stabilizer constraints and then counting the global degrees of freedom that remain. Simon develops this code-space construction using vertex and plaquette constraints and later relates the code space to topological quantum field theory data [5, Chs. 25 and 29].

8.4 Counting Independent Stabilizer Constraints

We now prove the fourfold ground-state degeneracy. The periodic $n \times n$ lattice has

$$V = n^2 \tag{8.24}$$

star operators and

$$P = n^2 \tag{8.25}$$

plaquette operators. Naively, one might think that there are

$$V + P = 2n^2 \tag{8.26}$$

independent stabilizer constraints. However, on a torus, there are two global relations.

First, consider the product of all star operators:

$$\prod_s A_s. \tag{8.27}$$

Every edge belongs to exactly two stars. Therefore, each σ_e^x appears exactly twice in the product. Since

$$(\sigma_e^x)^2 = 1, \tag{8.28}$$

we get

$$\prod_s A_s = 1. \tag{8.29}$$

Thus one star constraint is redundant.

Second, consider the product of all plaquette operators:

$$\prod_p B_p. \tag{8.30}$$

On a torus, every edge belongs to exactly two plaquettes. Therefore, each σ_e^z appears exactly twice. Since

$$(\sigma_e^z)^2 = 1, \tag{8.31}$$

we get

$$\prod_p B_p = 1. \tag{8.32}$$

Thus one plaquette constraint is redundant.

Therefore, the number of independent stabilizer constraints is

$$(V + P) - 2. \tag{8.33}$$

Since

$$V = P = n^2, \tag{8.34}$$

we obtain

$$(V + P) - 2 = 2n^2 - 2. \tag{8.35}$$

This stabilizer-counting argument is the standard way to derive the toric-code degeneracy on a torus [5, Ch. 25].

8.5 Ground-State Degeneracy on the Torus

The microscopic Hilbert space has $E = 2n^2$ qubits, so

$$\dim \mathcal{H}_{\text{micro}} = 2^{2n^2}. \tag{8.36}$$

Each independent stabilizer constraint cuts the Hilbert-space dimension by a factor of 2. Since there are

$$2n^2 - 2 \tag{8.37}$$

independent constraints, the ground-state dimension is

$$\begin{aligned} \dim \mathcal{H}_{\text{TC}}(T^2) &= 2^{2n^2-(2n^2-2)} && (8.38) \\ &= 2^2 && (8.39) \\ &= 4. && (8.40) \end{aligned}$$

Thus,

$$\boxed{\dim \mathcal{H}_{\text{TC}}(T^2) = 4.} \tag{8.41}$$

This is the toric-code torus ground-state degeneracy. Simon discusses the Toric Code on different topologies and the ground-state degeneracy on the torus, while Kitaev's original model identifies the protected topological subspace of the exactly solved Hamiltonian [5, Chs. 25 and 29] [2].

8.6 Meaning of the Four Ground States

The four ground states are not four locally distinguishable patterns. Locally, they all satisfy

$$A_s = +1, \qquad B_p = +1. \tag{8.42}$$

They differ only by global $\mathbb{Z}_2$ holonomies around the two non-contractible cycles of the torus.

The torus has fundamental group

$$\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}. \tag{8.43}$$

For a $\mathbb{Z}_2$ gauge theory, each independent cycle can carry a $\mathbb{Z}_2$ holonomy:

$$+1 \qquad \text{or} \qquad -1. \tag{8.44}$$

Since there are two independent cycles, the number of global sectors is

$$2 \times 2 = 4. \tag{8.45}$$

Equivalently,

$$\begin{aligned} \text{Hom}\left(\pi_1(T^2), \mathbb{Z}_2\right) &= \text{Hom}\left(\mathbb{Z} \times \mathbb{Z}, \mathbb{Z}_2\right) && (8.46) \\ &\cong \mathbb{Z}_2 \times \mathbb{Z}_2. && (8.47) \end{aligned}$$

Therefore,

$$\left|\text{Hom}\left(\pi_1(T^2), \mathbb{Z}_2\right)\right| = 4. \tag{8.48}$$

So the four ground states can be interpreted as four flat $\mathbb{Z}_2$ gauge sectors on the torus. This interpretation matches the view of the Toric Code as a discrete gauge theory and connects naturally to the quantum double and continuum topological descriptions discussed by Simon [5, Ch. 29].

8.7 Non-Contractible String Operators on the Torus

The torus has two independent non-contractible cycles:

$$\gamma_1, \qquad \gamma_2. \tag{8.49}$$

Because of this, the Toric Code has non-contractible string operators. With our convention, the e-string operator along a path γ is

$$W_e(\gamma) = \prod_{e \in \gamma} \sigma_e^z, \tag{8.50}$$

and the m-string operator along a dual path γ^* is

$$W_m(\gamma^*) = \prod_{e \in \gamma^*} \sigma_e^x. \tag{8.51}$$

If γ is closed and non-contractible, then $W_e(\gamma)$ has no endpoints. Therefore, it creates no local excitation. It preserves the ground-state space:

$$W_e(\gamma) : \mathcal{H}_{\mathrm{TC}}(T^2) \longrightarrow \mathcal{H}_{\mathrm{TC}}(T^2). \tag{8.52}$$

Similarly,

$$W_m(\gamma^*) : \mathcal{H}_{\mathrm{TC}}(T^2) \longrightarrow \mathcal{H}_{\mathrm{TC}}(T^2). \tag{8.53}$$

These operators commute with the Hamiltonian because they create no stabilizer violations when the paths are closed:

$$[W_e(\gamma), H_{\mathrm{TC}}] = 0, \qquad [W_m(\gamma^*), H_{\mathrm{TC}}] = 0. \tag{8.54}$$

However, if the closed loop is non-contractible, it need not act as the identity on the ground-state space. This is the key topological feature of the torus. Simon discusses nonlocal string/ribbon operators and their role in the topology-dependent code space [5, Chs. 25 and 29].

8.8 Contractible Versus Non-Contractible Loops

We now prove why topology matters. If a closed string is contractible, then it bounds a disk-like region. In that case, the string operator can be written as a product of stabilizers inside the region. Therefore, on the ground-state space, a contractible loop acts trivially:

$$W_{\text{contractible}}|\psi\rangle = |\psi\rangle. \tag{8.55}$$

A non-contractible loop is different. It does not bound a disk on the torus. Therefore, it cannot be written as a product of local stabilizers. It may commute with the Hamiltonian, but still act nontrivially within the ground-state subspace.

Thus,

$$\boxed{\text{contractible loops are stabilizer-equivalent to the identity, but non-contractible loops are logical opera}} \tag{8.56}$$

This is the Toric Code version of a TQFT observable: the operator depends only on the homology class of the loop, not on its precise local shape. Simon discusses this topological dependence of the code space and the role of nonlocal string/ribbon operators in the Toric Code and quantum double models [5, Chs. 25 and 29].

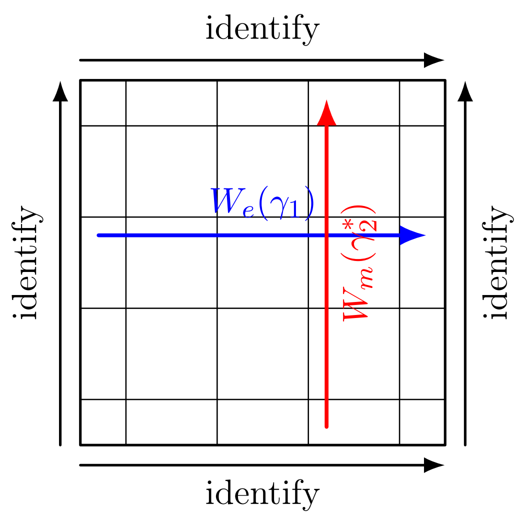


Figure 8.1: On the torus, non-contractible string operators wrap around the two independent cycles. They create no local excitations, but can act nontrivially on the ground-state Hilbert space.

8.9 Logical Loop Algebra on the Torus

Choose two non-contractible cycles on the torus:

$$\gamma_1, \qquad \gamma_2. \tag{8.57}$$

We can define four basic non-contractible string operators:

$$W_e(\gamma_1), \qquad W_e(\gamma_2), \tag{8.58}$$

and

$$W_m(\gamma_1^*), \qquad W_m(\gamma_2^*). \tag{8.59}$$

String operators of the same type commute:

$$[W_e(\gamma_i), W_e(\gamma_j)] = 0, \tag{8.60}$$

and

$$[W_m(\gamma_i^*), W_m(\gamma_j^*)] = 0. \tag{8.61}$$

However, an e-string and an m-string anticommute if they intersect once:

$$W_e(\gamma_i)W_m(\gamma_j^*) = (-1)^{I(\gamma_i,\gamma_j^*)}W_m(\gamma_j^*)W_e(\gamma_i), \tag{8.62}$$

where $I(\gamma_i, \gamma_j^*)$ is the intersection number. For one intersection,

$$I(\gamma_i, \gamma_j^*) = 1, \tag{8.63}$$

so

$$W_e(\gamma_i)W_m(\gamma_j^*) = -W_m(\gamma_j^*)W_e(\gamma_i). \tag{8.64}$$

This is the Toric Code analogue of the Wilson-loop algebra in Chern-Simons theory on a torus. In both cases, nonlocal loop operators act on a finite-dimensional topological Hilbert space, and their algebra is controlled by intersection or linking data rather than local geometry. Simon discusses braiding of vertex and plaquette defects and the ribbon-operator algebra in the Toric Code and quantum double setting [5, Chs. 26 and 29].

8.10 Two Logical Qubits

The four ground states can be understood as two logical qubits. Choose a commuting pair of e-loop operators:

$$\overline{Z}_1 = W_e(\gamma_1), \tag{8.65}$$

and

$$\overline{Z}_2 = W_e(\gamma_2). \tag{8.66}$$

Their eigenvalues are

$$z_1, z_2 \in \{+1, -1\}. \tag{8.67}$$

Therefore a ground state can be labelled as

$$|z_1, z_2\rangle. \tag{8.68}$$

There are four possibilities:

$$|+,+\rangle, \qquad |+,-\rangle, \qquad |-,+\rangle, \qquad |-,-\rangle. \tag{8.69}$$

The corresponding m-loop operators act as logical X-type operators, flipping these $\mathbb{Z}_2$ eigenvalues. Thus the torus ground-state space is

$$\mathcal{H}_{\mathrm{TC}}(T^2) \cong \mathbb{C}^2 \otimes \mathbb{C}^2. \tag{8.70}$$

Hence,

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{8.71}$$

This is why the Toric Code on a torus stores two logical qubits. Simon discusses the quantum-memory interpretation of the Toric Code, and relates the torus topology to topological sectors and nonlocal operators [5, Chs. 24–25].

8.11 Toric Code as a TQFT Hilbert-Space Assignment on T^2

A 2 + 1-dimensional TQFT assigns a vector space to a two-dimensional spatial surface:

$$\Sigma \longmapsto \mathcal{H}(\Sigma). \tag{8.72}$$

For the Toric Code, after taking the low-energy topological sector, we have

$$T^2 \longmapsto \mathcal{H}_{\mathrm{TC}}(T^2). \tag{8.73}$$

We proved that

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{8.74}$$

Therefore, the Toric Code on a torus satisfies the TQFT-style assignment

$$\boxed{T^2 \longmapsto \mathbb{C}^4.} \tag{8.75}$$

This is not the full microscopic Hilbert space of all edge qubits. It is the low-energy topological Hilbert space after imposing the stabilizer constraints. Simon presents the axiomatic TQFT idea of assigning state spaces to spatial manifolds and later identifies the Toric Code as a topological phase whose low-energy sector has this kind of topology-dependent Hilbert space [5, Chs. 7, 25, and 26].

8.12 Continuum Interpretation: Flat $\mathbb{Z}_2$ Gauge Sectors

The Toric Code on a torus can also be interpreted as a discrete flat $\mathbb{Z}_2$ gauge theory. The plaquette constraint

$$B_p = +1 \tag{8.76}$$

means that there is no local $\mathbb{Z}_2$ flux. This is local flatness. But on a torus, flat does not mean globally trivial. There can still be nontrivial holonomies around the two non-contractible cycles

$$\gamma_1, \qquad \gamma_2. \tag{8.77}$$

Thus, just as Chern-Simons theory on a torus has flat connections with nontrivial global Wilson loops, the Toric Code has flat $\mathbb{Z}_2$ gauge configurations with nontrivial global string operators. The four global sectors are

$$(+,+), \qquad (+,-), \qquad (-,+), \qquad (-,-). \tag{8.78}$$

Equivalently,

$$\mathcal{H}_{\mathrm{TC}}(T^2) \cong \mathbb{C}\left[\mathrm{Hom}\left(\pi_1(T^2), \mathbb{Z}_2\right)\right] \cong \mathbb{C}^4. \tag{8.79}$$

This is the continuum topological interpretation of the torus ground-state degeneracy. It prepares the later identification of the long-distance Toric Code phase with $N = 2$ BF theory, equivalently doubled Abelian Chern-Simons theory. Simon discusses the relation between the Toric Code, quantum double models, continuum descriptions, and gauge theory [5, Ch. 29]. The K-matrix description of Abelian topological orders gives the continuum Chern-Simons language used later in the comparison [3].

8.13 General Genus Formula

The torus is the genus-one closed surface. More generally, let Σ_g be a closed surface of genus g. Such a surface has $2g$ independent non-contractible cycles. For the $\mathbb{Z}_2$ Toric Code, each independent cycle can carry a $\mathbb{Z}_2$ holonomy. Therefore,

$$\dim \mathcal{H}_{\mathrm{TC}}(\Sigma_g) = 2^{2g}. \tag{8.80}$$

For the torus,

$$g = 1, \tag{8.81}$$

so

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 2^2 = 4. \tag{8.82}$$

For a $\mathbb{Z}_N$ Toric Code, the analogous formula is

$$\dim \mathcal{H}_{\mathbb{Z}_N}(\Sigma_g) = N^{2g}. \tag{8.83}$$

For $N = 2$, this becomes Eq. (8.80). Simon discusses the $\mathbb{Z}_N$ generalization and the topology-dependent code-space dimension in the Toric Code and quantum double framework [5, Chs. 25 and 29].

8.14 Conclusion

We have proved that the Toric Code on a torus realizes a $2+1$-dimensional continuum TQFT in its low-energy topological sector.

First, the spatial surface is a torus:

$$\Sigma = T^2. \tag{8.84}$$

Adding time gives the $2+1$-dimensional spacetime

$$M = \mathbb{R}_t \times T^2. \tag{8.85}$$

Second, the microscopic theory is a local stabilizer Hamiltonian:

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p. \tag{8.86}$$

The ground states satisfy

$$A_s|\psi\rangle = |\psi\rangle, \qquad B_p|\psi\rangle = |\psi\rangle. \tag{8.87}$$

Third, an $n \times n$ toric-code lattice on T^2 has $2n^2$ edge qubits and $2n^2 - 2$ independent stabilizer constraints. Therefore,

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 2^{2n^2-(2n^2-2)} = 4. \tag{8.88}$$

Fourth, the four states are global topological sectors, not local patterns. They are generated and detected by non-contractible string operators wrapping around the two independent cycles of the torus:

$$\gamma_1, \qquad \gamma_2. \tag{8.89}$$

These operators obey an algebra controlled by intersection number:

$$W_e(\gamma_i)W_m(\gamma_j^*) = (-1)^{I(\gamma_i,\gamma_j^*)} W_m(\gamma_j^*)W_e(\gamma_i). \tag{8.90}$$

Thus, the Toric Code assigns

$$T^2 \longmapsto \mathbb{C}^4 \tag{8.91}$$

in its low-energy topological sector.

> The Toric Code on a torus is a $2+1$-dimensional TQFT in its low-energy sector because all local stabilizer constraints remove local degrees of freedom, while the non-contractible cycles of T^2 leave four global topological ground states acted on by non-contractible string operators.

This prepares the next chapter, where we study Abelian anyons in the Toric Code on a torus and compare their motion, fusion, and braiding with the Abelian anyons of Chern-Simons theory.

Chapter 9

Abelian Anyons in a $2+1$D Toric Code TQFT on a Torus

In the previous chapter, we placed the Toric Code on a torus and proved that its low-energy ground-state Hilbert space is four-dimensional:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{9.1}$$

We also saw that the four ground states are not locally distinguishable. They are global topological sectors produced by the two non-contractible cycles of the torus.

In this chapter, we study the Abelian anyons of the Toric Code on the torus. The aim is to understand how the excitations e, m, and $\varepsilon = e \times m$ are created, moved, braided, fused, and represented by string operators acting on the torus ground-state space. This is the Toric Code analogue of the Wilson-loop analysis performed earlier for Abelian Chern-Simons theory on a torus.

The central statement is:

> In the Toric Code TQFT on a torus, Abelian anyons appear as endpoints of open string operators, while closed non-contractible string operators act as topological operators on the four-dimensional ground-state Hilbert space. Their fusion and braiding data are those of the $\mathbb{Z}_2$ Abelian anyon theory.

This chapter is important for the final comparison with Chern-Simons theory. In Chern-Simons theory, the topological observables are Wilson loops. In the Toric Code, the corresponding observables are non-contractible string operators. Simon describes the Toric Code anyons, their statistical phases, and the torus/topological-memory interpretation, while Kitaev's original construction identifies these excitations as anyons in an exactly solved model [5, Chs. 25–26] [2].

9.1 Toric Code Anyon Sectors on the Torus

The Toric Code Hamiltonian is

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p, \tag{9.2}$$

where

$$A_s = \prod_{e \in \mathrm{star}(s)} \sigma_e^x, \qquad B_p = \prod_{e \in \partial p} \sigma_e^z. \tag{9.3}$$

The ground-state sector is defined by

$$A_s|\psi\rangle = |\psi\rangle, \qquad B_p|\psi\rangle = |\psi\rangle \qquad \forall s, p. \tag{9.4}$$

An excitation is produced when one of these local stabilizer constraints is violated.

If

$$A_s|\psi\rangle = -|\psi\rangle, \tag{9.5}$$

then vertex s carries an electric excitation:

$$e. \tag{9.6}$$

If

$$B_p|\psi\rangle = -|\psi\rangle, \tag{9.7}$$

then plaquette p carries a magnetic excitation:

$$m. \tag{9.8}$$

Their composite is

$$\varepsilon = e \times m. \tag{9.9}$$

Therefore, the Toric Code has four anyon sectors:

$$\boxed{1, \qquad e, \qquad m, \qquad \varepsilon = e \times m.} \tag{9.10}$$

Here 1 denotes the vacuum sector. Simon discusses these vertex and plaquette defects as the elementary anyons of the Toric Code phase, and Kitaev identifies the same excitations in the original exactly solved model [5, Ch. 26] [2].

9.2 Open String Operators and Pair Creation

The Toric Code anyons are created by open string operators. With the convention used in this report, an e-string operator along a direct-lattice path γ is

$$W_e(\gamma) = \prod_{e\in\gamma} \sigma_e^z. \tag{9.11}$$

This operator anticommutes with the star operators at the endpoints of γ. Therefore, if the path γ is open, $W_e(\gamma)$ creates two e anyons at its endpoints:

$$W_e(\gamma) : 1 \longrightarrow e \times e. \tag{9.12}$$

Similarly, an m-string operator along a dual-lattice path γ^* is

$$W_m(\gamma^*) = \prod_{e\in\gamma^*} \sigma_e^x. \tag{9.13}$$

This operator anticommutes with the plaquette operators at the endpoints of the dual path. Therefore, if γ^* is open, $W_m(\gamma^*)$ creates two m anyons:

$$W_m(\gamma^*) : 1 \longrightarrow m \times m. \tag{9.14}$$

This proves that anyons in the Toric Code are not created individually in the bulk. They are created in pairs, because an open string has two endpoints. This is the lattice analogue of the current-conservation condition in a continuum topological field theory: anyon worldlines cannot simply start or stop in the bulk. They must form closed histories or appear through creation and annihilation of particle-antiparticle pairs. Simon discusses this pair-creation/string-operator picture in the Toric Code and quantum double models [5, Chs. 25–26 and 29].

9.3 Closed Strings, Non-Contractible Strings, and the Torus

If a string operator is closed, then it has no endpoints. Therefore it does not create local stabilizer violations. In particular, a closed e-string or m-string commutes with the Hamiltonian:

$$[W_e(\gamma), H_{\mathrm{TC}}] = 0, \qquad [W_m(\gamma^*), H_{\mathrm{TC}}] = 0, \tag{9.15}$$

provided the paths are closed.

However, there is an important distinction between contractible and non-contractible closed strings. If a closed loop is contractible, then it bounds a disk-like region. Such a loop can be written as a product of local stabilizers inside the region, and hence it acts trivially on the ground-state sector:

$$W_{\mathrm{contractible}}|\psi\rangle = |\psi\rangle. \tag{9.16}$$

If a loop is non-contractible, it does not bound a disk on the torus. Therefore it cannot be reduced to a product of local stabilizers. It can act nontrivially on the ground-state Hilbert space:

$$W_{\mathrm{non-contractible}} : \mathcal{H}_{\mathrm{TC}}(T^2) \longrightarrow \mathcal{H}_{\mathrm{TC}}(T^2). \tag{9.17}$$

The torus has two independent non-contractible cycles,

$$\gamma_1, \qquad \gamma_2. \tag{9.18}$$

Therefore, the Toric Code on a torus has non-contractible e- and m-string operators wrapping around these cycles. These are the Toric Code analogues of Wilson loops in Chern-Simons theory. Simon discusses the role of topology and nonlocal string/ribbon operators in the Toric Code and in the quantum double model [5, Chs. 25 and 29].

9.4 Anyon Motion Around a Non-Contractible Cycle

A topological anyon process on the torus can be described in three steps:

1. create an anyon pair from the vacuum;
2. move one anyon around a non-contractible cycle of the torus;
3. bring it back and annihilate it with its partner.

After annihilation, there may be no local excitation left. Nevertheless, the process can act nontrivially on the ground-state Hilbert space.

For example, create an e-anyon pair, move one e around γ_1, and annihilate it with the other e. The resulting operation is a non-contractible e-loop:

$$W_e(\gamma_1). \tag{9.19}$$

Similarly, moving an m anyon around a non-contractible dual cycle gives

$$W_m(\gamma_i^*). \tag{9.20}$$

Thus anyon motion around the torus is not ordinary local dynamics. It is topological dynamics: the result depends only on which non-contractible cycle the anyon winds around. Simon emphasizes this topological-memory interpretation: pair creation, winding around a nontrivial cycle, and annihilation can change the topological ground-state sector without leaving a local excitation [5, Chs. 4 and 25–26].

9.5 Fusion Rules on the Torus

The fusion rules of the Toric Code are unchanged by placing the model on a torus. The torus changes the global ground-state structure, but the local anyon types and their fusion rules remain the same.

Because the theory is $\mathbb{Z}_2$-valued, two identical e anyons annihilate:

$$e \times e = 1. \tag{9.21}$$

Similarly, two identical m anyons annihilate:

$$m \times m = 1. \tag{9.22}$$

The composite of an e anyon and an m anyon is

$$e \times m = \varepsilon. \tag{9.23}$$

Finally,

$$\begin{aligned} \varepsilon \times \varepsilon &= (e \times m) \times (e \times m) && (9.24) \\ &= (e \times e) \times (m \times m) && (9.25) \\ &= 1 \times 1 && (9.26) \\ &= 1. && (9.27) \end{aligned}$$

Therefore,

$$\boxed{e \times e = 1, \qquad m \times m = 1, \qquad e \times m = \varepsilon, \qquad \varepsilon \times \varepsilon = 1.} \tag{9.28}$$

The fusion algebra is

$$\mathcal{A}_{\mathrm{TC}} = \mathbb{Z}_2 \times \mathbb{Z}_2. \tag{9.29}$$

Every fusion product has a unique output, so all anyons are Abelian. Simon and Kitaev both discuss these Abelian anyonic sectors and their fusion/statistical properties in the Toric Code [5, Ch. 26] [2].

9.6 Braiding from String-Operator Algebra

The mutual braiding of e and m is encoded in the algebra of their string operators. Suppose an e-string and an m-string cross once. At the crossing edge, the Pauli operators are

$$\sigma_e^z \qquad \text{and} \qquad \sigma_e^x. \tag{9.30}$$

They satisfy the anticommutation relation

$$\sigma_e^z \sigma_e^x = -\sigma_e^x \sigma_e^z. \tag{9.31}$$

Therefore, for one crossing,

$$W_e(\gamma) W_m(\gamma^*) = -W_m(\gamma^*) W_e(\gamma). \tag{9.32}$$

If the paths cross $I(\gamma, \gamma^*)$ times, then

$$W_e(\gamma) W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)} W_m(\gamma^*) W_e(\gamma). \tag{9.33}$$

A full braid of e around m corresponds to one linking of their worldlines in $2 + 1$-dimensional spacetime. Therefore the mutual braiding phase is

$$\boxed{M_{e,m} = -1.} \tag{9.34}$$

This phase does not depend on the local geometry of the paths. It depends only on their topological crossing or linking number. This is why the Toric Code anyons define a topological anyon theory. Simon explicitly discusses the braiding of vertex and plaquette defects, and Kitaev identifies the anyonic statistics of the model [5, Ch. 26] [2].

9.7 Topological Spins

The topological spin θ_a of an anyon a is the phase obtained by rotating the anyon by 2π, equivalently by exchanging two identical anyons in an Abelian theory. In the Toric Code,

$$\theta_1 = 1. \tag{9.35}$$

The e and m anyons are bosonic:

$$\theta_e = 1, \qquad \theta_m = 1. \tag{9.36}$$

However, their composite

$$\varepsilon = e \times m \tag{9.37}$$

is fermionic:

$$\theta_\varepsilon = -1. \tag{9.38}$$

This can be understood from the fact that e and m have nontrivial mutual braiding:

$$M_{e,m} = -1. \tag{9.39}$$

The composite $\varepsilon = e \times m$ carries both electric and magnetic character. Its exchange statistics therefore include the mutual e-around-m phase, producing

$$\theta_\varepsilon = \theta_e \theta_m M_{e,m} = (1)(1)(-1) = -1. \tag{9.40}$$

Thus,

$$\boxed{\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1.} \tag{9.41}$$

Simon presents the Toric Code S- and T-matrix data and discusses the statistical properties of the vertex and plaquette defects [5, Ch. 26].

9.8 S- and T-Matrices of the Toric Code

The anyon data can be summarized using the modular S- and T-matrices. In the ordered basis

$$(1, e, m, \varepsilon), \tag{9.42}$$

the T-matrix records the topological spins:

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{9.43}$$

The S-matrix records mutual braiding phases. For the Toric Code,

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{9.44}$$

The -1 entries encode the nontrivial mutual braiding between e and m. This topological data is exactly the data that the later continuum Chern-Simons/BF description must reproduce. Simon gives the Toric Code modular data in his discussion of the Toric Code as a topological phase/TQFT [5, Ch. 26]. Wen's K-matrix description provides the continuum Abelian Chern-Simons framework used to reproduce such Abelian topological data [3].

9.9 Action on the Torus Ground-State Hilbert Space

Now we describe how anyon motion acts on the four-dimensional torus Hilbert space. Choose two independent non-contractible cycles of the torus,

$$\gamma_1, \qquad \gamma_2. \tag{9.45}$$

Let η_1^* and η_2^* be dual cycles chosen so that

$$I(\gamma_i, \eta_j^*) = \delta_{ij}. \tag{9.46}$$

Define the logical loop operators

$$\overline{Z}_i = W_e(\gamma_i), \qquad \overline{X}_i = W_m(\eta_i^*), \qquad i = 1, 2. \tag{9.47}$$

Then the loop algebra becomes

$$\overline{Z}_i \overline{X}_j = (-1)^{\delta_{ij}} \overline{X}_j \overline{Z}_i, \tag{9.48}$$

with all other same-type operators commuting.

Choose a basis of the ground-state space labelled by the eigenvalues of the commuting $\overline{Z}_1$ and $\overline{Z}_2$ operators:

$$\overline{Z}_1 |z_1, z_2\rangle = z_1 |z_1, z_2\rangle, \tag{9.49}$$

$$\overline{Z}_2 |z_1, z_2\rangle = z_2 |z_1, z_2\rangle, \tag{9.50}$$

where

$$z_1, z_2 \in \{+1, -1\}. \tag{9.51}$$

There are four basis states:

$$|+,+\rangle, \qquad |+,-\rangle, \qquad |-,+\rangle, \qquad |-,-\rangle. \tag{9.52}$$

The $\overline{X}_i$ operators flip the corresponding $\mathbb{Z}_2$ eigenvalue:

$$\overline{X}_1 |z_1, z_2\rangle = |-z_1, z_2\rangle, \tag{9.53}$$

and

$$\overline{X}_2 |z_1, z_2\rangle = |z_1, -z_2\rangle. \tag{9.54}$$

Thus the torus ground-state Hilbert space is

$$\mathcal{H}_{\mathrm{TC}}(T^2) \cong \mathbb{C}^2 \otimes \mathbb{C}^2. \tag{9.55}$$

This is the algebraic meaning of the fourfold degeneracy:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{9.56}$$

Simon discusses this quantum-memory interpretation of the Toric Code, where non-contractible string operators act as logical operators on the encoded topological Hilbert space [5, Chs. 24–25].

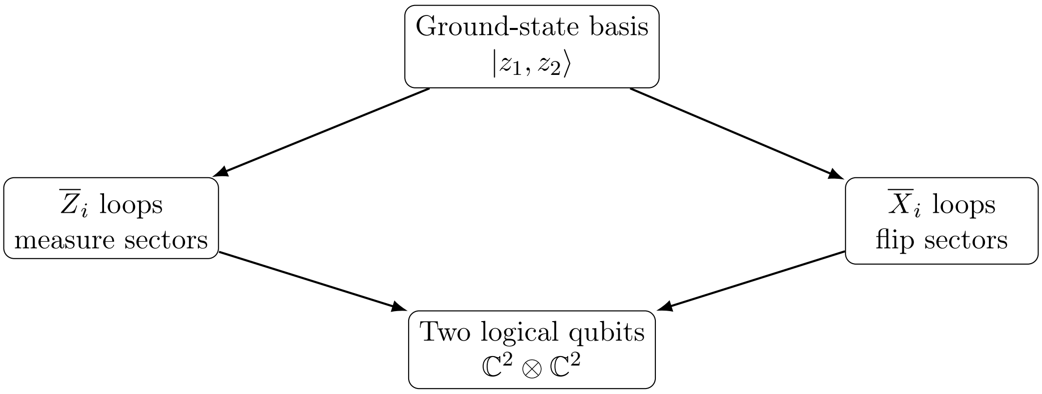


Figure 9.1: Non-contractible anyon string operators act as logical operators on the torus ground-state Hilbert space. The four ground states form two logical qubits.

9.10 Worldline Picture in $2 + 1$ Dimensions

In a $2 + 1$-dimensional spacetime

$$M = \mathbb{R}_t \times T^2, \tag{9.57}$$

anyons trace out worldlines. The topological processes of interest are:

$$\text{pair creation,} \tag{9.58}$$

$$\text{transport around a non-contractible cycle,} \tag{9.59}$$

$$\text{braiding,} \tag{9.60}$$

and

$$\text{annihilation.} \tag{9.61}$$

These histories are insensitive to local geometric details. A worldline can be deformed smoothly without changing the resulting topological operation, as long as its linking and winding data are unchanged.

This is why the Toric Code anyons on a torus define a $2 + 1$-dimensional TQFT. The physical data are not local trajectories, speeds, or distances. The physical data are:

$$\text{anyon labels,} \qquad \text{fusion,} \qquad \text{braiding,} \tag{9.62}$$

$$\text{topological spin}, \qquad \text{winding around torus cycles}, \qquad \text{action on } \mathcal{H}_{\mathrm{TC}}(T^2). \tag{9.63}$$

This is the same kind of topological information that appears in continuum Chern-Simons theory through Wilson lines and Wilson loops. Simon describes the Toric Code as a topological phase/TQFT whose low-energy physics is governed by such topological data, while Kitaev's construction provides the exactly solved microscopic model realizing these anyons [5, Chs. 26–29] [2].

9.11 Relation to the Continuum Chern-Simons/BF Description

The Toric Code anyon data on a torus are reproduced by a doubled Abelian Chern-Simons theory, equivalently a mutual BF theory. In K-matrix form, the relevant continuum theory is described by

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{9.64}$$

The number of anyon sectors is

$$|\det K| = 4, \tag{9.65}$$

matching the four Toric Code sectors

$$1, \quad e, \quad m, \quad \varepsilon. \tag{9.66}$$

The same continuum K-matrix description also reproduces the mutual -1 braiding phase between e and m, and the fourfold torus ground-state degeneracy. This is why the Toric Code and the doubled Abelian Chern-Simons/BF theory describe the same Abelian topological phase at low energies. Wen's K-matrix framework is the standard continuum description for Abelian topological orders, including the relation between K-matrix data, anyons, braiding, and ground-state degeneracy [3].

This relation does not mean that the microscopic Toric Code Hamiltonian is literally equal to the continuum Chern-Simons action. Rather, it means that their universal topological data agree:

$$\text{anyon sectors}, \qquad \text{fusion rules}, \qquad \text{braiding phases}, \tag{9.67}$$

$$\text{topological spins}, \qquad \text{loop/string operator algebra}, \qquad \text{torus ground-state degeneracy}. \tag{9.68}$$

9.12 Conclusion

We have shown that the Abelian anyons of the Toric Code on a torus are described by the four sectors

$$1, \qquad e, \qquad m, \qquad \varepsilon = e \times m. \tag{9.69}$$

They are created and transported by string operators:

$$W_e(\gamma) = \prod_{e\in\gamma} \sigma_e^z, \qquad W_m(\gamma^*) = \prod_{e\in\gamma^*} \sigma_e^x. \tag{9.70}$$

Their fusion rules are

$$e \times e = 1, \qquad m \times m = 1, \qquad e \times m = \varepsilon, \qquad \varepsilon \times \varepsilon = 1, \tag{9.71}$$

so the fusion algebra is

$$\mathbb{Z}_2 \times \mathbb{Z}_2. \tag{9.72}$$

Their mutual braiding is

$$M_{e,m} = -1, \tag{9.73}$$

and their topological spins are

$$\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1. \tag{9.74}$$

On the torus, non-contractible string operators act on the four-dimensional ground-state Hilbert space:

$$\mathcal{H}_{\mathrm{TC}}(T^2) \cong \mathbb{C}^4. \tag{9.75}$$

Their algebra is controlled by intersection number:

$$W_e(\gamma_i) W_m(\gamma_j^*) = (-1)^{I(\gamma_i, \gamma_j^*)} W_m(\gamma_j^*) W_e(\gamma_i). \tag{9.76}$$

Therefore,

> Abelian anyons in the $2+1$-dimensional Toric Code TQFT on a torus are topological string excitations. Their fusion, braiding, topological spin, and action on the torus Hilbert space are completely determined by $\mathbb{Z}_2 \times \mathbb{Z}_2$ topological data.

This completes the Toric Code side of the anyon analysis and prepares the next chapter, where the Chern-Simons and Toric Code TQFT models will be compared directly.

Chapter 10

Comparison of the Two Models of TQFT

In the previous chapters, two $2+1$-dimensional topological descriptions were developed. The first is the continuum Chern-Simons description of a two-dimensional quantum Hall system placed on a torus. The second is the low-energy topological description of the Toric Code on a torus. Both theories are naturally studied on a spacetime of the form

$$M = \mathbb{R}_t \times T^2, \tag{10.1}$$

where T^2 is the two-dimensional spatial torus.

The purpose of this chapter is to compare these two TQFT models side by side. The comparison is not a claim that a microscopic quantum Hall Hamiltonian is the same object as a microscopic Toric Code Hamiltonian. Rather, the comparison is between their low-energy topological field-theory descriptions. In the quantum Hall case, the effective topological field theory is Chern-Simons theory. In the Toric Code case, the effective topological field theory is a $\mathbb{Z}_2$ topological gauge theory, equivalently a $N=2$ BF theory or doubled Abelian Chern-Simons theory in the continuum limit.

A crucial distinction must be made at the beginning. A generic quantum Hall system described by a single-component $U(1)_m$ Chern-Simons theory is not automatically equivalent to the Toric Code. The quantum Hall system motivates the Chern-Simons TQFT framework, but the specific continuum Chern-Simons theory that matches the Toric Code is the non-chiral doubled Abelian Chern-Simons theory, equivalently the $N=2$ BF theory, with

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

Thus, this chapter compares the general Chern-Simons language of Abelian topological order with the particular doubled Chern-Simons/BF theory that reproduces the Toric Code topological data.

The central idea is:

> Two TQFTs should be compared by their topological data: Hilbert spaces on closed surfaces, loop operators, anyon sectors, fusion rules, braiding phases, topological spins, and ground-state degeneracy. They should not be compared by directly equating their microscopic Hamiltonians.

Chern-Simons theory as a TQFT was developed by Witten and appears naturally as the effective topological field theory of quantum Hall systems [1, 4]. The Toric Code was introduced by Kitaev as an exactly solved lattice model with anyonic excitations and topological protection [2]. Simon discusses both the quantum Hall/Chern-Simons viewpoint and the Toric Code as a topological phase whose long-wavelength sector is described by TQFT data [5]. Wen's K-matrix formalism provides the Abelian Chern-Simons language for comparing Abelian topological orders [3].

10.1 The Two Models Being Compared

The two models compared in this chapter are:

$$\text{Model A: Chern Simons Theory system on } T^2 \quad \longrightarrow \quad \text{Abelian Chern-Simons TQFT,} \tag{10.2}$$

and

$$\text{Model B: Toric Code on } T^2 \quad \longrightarrow \quad \text{low-energy } \mathbb{Z}_2 \text{ BF/doubled CS TQFT.} \tag{10.3}$$

For the CST on a Torus side, the prototype Abelian Chern-Simons action is

$$S_{\mathrm{QH}}[a] = \frac{m}{4\pi} \int_M a \wedge da, \qquad M = \mathbb{R}_t \times T^2, \tag{10.4}$$

where a is a compact $U(1)$ gauge field and $m \in \mathbb{Z}$ is the Chern-Simons level. This is the effective topological theory for a Laughlin-type Abelian quantum Hall state, after non-topological microscopic details are ignored [3, 4].

For the Toric Code side, the continuum topological theory is the $N = 2$ BF theory,

$$S_{\mathrm{BF}}[a, b] = \frac{2}{2\pi} \int_M b \wedge da, \tag{10.5}$$

which can also be written as a doubled Abelian Chern-Simons theory

$$S_{\mathrm{TC}}[a^1, a^2] = \frac{1}{4\pi} \int_M K_{IJ} a^I \wedge da^J, \qquad K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{10.6}$$

This continuum theory reproduces the Toric Code anyon content, braiding, fusion, and fourfold torus ground-state degeneracy [3, 5].

Model A: Chern-Simons TQFT on T^2	**Model B: Toric Code TQFT on T^2**
A two-dimensional quantum Hall fluid is placed on a spatial torus T^2. The low-energy topological dynamics are described by Abelian Chern-Simons theory [3, 4].	A two-dimensional Toric Code lattice is placed on a torus T^2. Its low-energy topological sector is described by a $\mathbb{Z}_2$ BF/doubled Chern-Simons TQFT [2, 3, 5].
The fundamental continuum field is a compact $U(1)$ gauge connection a.	The continuum description uses emergent gauge fields a, b, or equivalently two Abelian gauge fields a^1, a^2 with $K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$.
The action is typically $$S_{\text{QH}} = \frac{m}{4\pi} \int a \wedge da.$$	The topological action is $$S_{\text{BF}} = \frac{2}{2\pi} \int b \wedge da,$$ or equivalently the doubled Chern-Simons action in Eq. (10.6).
It is a continuum TQFT from the start at the effective-field-theory level.	It is not microscopically continuum; the continuum TQFT appears only after taking the low-energy topological limit of the lattice model.

10.2 Important Note: Structural Mapping Versus Exact Equality

The comparison in this chapter is a structural mapping between two topological descriptions. A generic Abelian quantum Hall Chern-Simons theory $U(1)_m$ is not automatically the same TQFT as the Toric Code. For example, the simple $U(1)_m$ Laughlin theory is chiral, while the Toric Code is non-chiral and is described by a doubled theory. Therefore, the correct statement is not

$$U(1)_m \text{ Chern-Simons} = \text{Toric Code} \quad \text{for arbitrary } m. \tag{10.7}$$

The correct statement is:

$$\boxed{\text{The Toric Code is matched by the doubled Abelian Chern-Simons/BF theory with } K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.} \tag{10.8}$$

Thus, the quantum Hall/Chern-Simons model provides the continuum TQFT language, while the Toric Code selects a particular non-chiral Abelian topological order within that language. This distinction is standard in Abelian topological-order comparisons: theories are identified by their K-matrix data, anyon sectors, braiding, topological spins, and ground-state degeneracy [3, 5].

10.3 Mapping of Spacetime and Hilbert-Space Assignments

Both models are studied on the same spacetime form:

$$M = \mathbb{R}_t \times T^2. \tag{10.9}$$

A 2 + 1-dimensional TQFT assigns a finite-dimensional Hilbert space to the spatial torus:

$$T^2 \longmapsto \mathcal{H}(T^2). \tag{10.10}$$

For the $U(1)_m$ Chern-Simons theory, quantization on the torus gives an m-dimensional Hilbert space:

$$\dim \mathcal{H}_{U(1)_m}(T^2) = m. \tag{10.11}$$

This was derived earlier from the noncommuting Wilson-loop algebra on the torus and is discussed in Tong's treatment of Chern-Simons theory on T^2 [4].

For the Toric Code, the low-energy Hilbert space on the torus is four-dimensional:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{10.12}$$

This follows from the stabilizer counting proof:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 2^{2n^2-(2n^2-2)} = 4. \tag{10.13}$$

Simon discusses this topology-dependent Toric Code degeneracy, and Kitaev's model provides the underlying lattice Hamiltonian realization [2,5].

Chern-Simons side	**Toric Code side**
Spatial surface: $$\Sigma = T^2.$$	Spatial surface: $$\Sigma = T^2.$$
Spacetime: $$M = \mathbb{R}_t \times T^2.$$	Spacetime: $$M = \mathbb{R}_t \times T^2.$$
TQFT assignment: $$T^2 \mapsto \mathcal{H}_{U(1)_m}(T^2).$$	TQFT assignment: $$T^2 \mapsto \mathcal{H}_{\mathrm{TC}}(T^2).$$
Torus degeneracy: $$\dim \mathcal{H}_{U(1)_m}(T^2) = m.$$	Torus degeneracy: $$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4.$$
The degeneracy comes from quantizing global holonomies of the Chern-Simons gauge field.	The degeneracy comes from two independent non-contractible $\mathbb{Z}_2$ string sectors.

For the Toric Code continuum K-matrix theory in Eq. (10.6), the determinant is

$$|\det K| = \left|\det \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}\right| = 4. \tag{10.14}$$

This agrees with the Toric Code result:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = |\det K| = 4. \tag{10.15}$$

Thus the doubled Chern-Simons/BF continuum theory reproduces the Toric Code torus Hilbert-space assignment [3, 5].

10.4 Mapping of Observables: Wilson Loops and String Operators

The most important operator-level comparison is between Wilson loops in Chern-Simons theory and string operators in the Toric Code.

In Abelian Chern-Simons theory, a charge-q Wilson loop along a closed curve γ is

$$W_q(\gamma) = \exp\left(iq \oint_\gamma a\right). \tag{10.16}$$

On the torus, the two basic Wilson loops wrap around the two non-contractible cycles:

$$W_1, \qquad W_2. \tag{10.17}$$

In the Toric Code, the corresponding topological operators are non-contractible string operators:

$$W_e(\gamma_i) = \prod_{e \in \gamma_i} \sigma_e^z, \tag{10.18}$$

and

$$W_m(\gamma_i^*) = \prod_{e \in \gamma_i^*} \sigma_e^x. \tag{10.19}$$

These operators also wrap around non-contractible cycles of the torus and act on the finite-dimensional ground-state Hilbert space.

Chern-Simons Wilson-loop observable	**Toric Code string observable**
A Wilson loop is defined by $$W_q(\gamma) = \exp\left(iq \oint_\gamma a\right).$$	An e-string loop is $$W_e(\gamma) = \prod_{e\in\gamma} \sigma_e^z.$$
Wilson loops wrap around non-contractible cycles of T^2.	String operators wrap around non-contractible cycles of the toric-code lattice on T^2.
The loop measures or changes global holonomy sectors.	The string operator measures or changes global $\mathbb{Z}_2$ topological sectors.
The algebra of Wilson loops is controlled by intersection/linking data.	The algebra of string operators is controlled by intersection number of direct and dual paths.

This is the first direct dictionary:

$$\boxed{\text{Chern-Simons Wilson loops} \quad \longleftrightarrow \quad \text{Toric Code non-contractible string operators.}} \tag{10.20}$$

Witten and Tong emphasize Wilson loops as the natural observables of Chern-Simons theory, while Simon and Kitaev describe the corresponding nonlocal string/ribbon operators in the Toric Code [1, 2, 4, 5].

10.5 Mapping of Anyon Excitations

In Chern-Simons theory, anyons are represented by Wilson lines. A Wilson line along a spacetime curve C is

$$W_q(C) = \exp\left(iq \int_C a\right). \tag{10.21}$$

This represents the worldline of a quasiparticle with topological charge q.

In the Toric Code, anyons are endpoints of open string operators. An open e-string creates a pair of e anyons:

$$W_e(\gamma) : 1 \longrightarrow e \times e. \tag{10.22}$$

An open m-string creates a pair of m anyons:

$$W_m(\gamma^*) : 1 \longrightarrow m \times m. \tag{10.23}$$

Chern-Simons anyons	**Toric Code anyons**
An anyon is represented by a Wilson line: $$W_q(C) = \exp\left(iq \int_C a\right).$$	An anyon is represented as the end-point of an open string operator: $$W_e(\gamma) = \prod \sigma^z, \qquad W_m(\gamma^*) = \prod \sigma^x.$$
Anyon labels in $U(1)_m$: $$q = 0, 1, \ldots, m-1.$$	Anyon labels in the Toric Code: $$1, \quad e, \quad m, \quad \varepsilon = e \times m.$$
Fusion in $U(1)_m$: $$q \times q' = (q + q') \mod m.$$	Fusion in the Toric Code: $$e^2 = m^2 = \varepsilon^2 = 1, \qquad e \times m = \varepsilon.$$
The anyon group is $$\mathbb{Z}_m.$$	The anyon group is $$\mathbb{Z}_2 \times \mathbb{Z}_2.$$

For the exact Toric Code continuum K-matrix theory, the anyon sectors are equivalence classes

$$\ell \in \mathbb{Z}^2 / K\mathbb{Z}^2. \tag{10.24}$$

For

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \tag{10.25}$$

there are

$$|\det K| = 4 \tag{10.26}$$

anyon sectors, matching

$$1, \quad e, \quad m, \quad \varepsilon. \tag{10.27}$$

This is the precise continuum Abelian Chern-Simons/BF realization of the Toric Code anyon content [3, 5].

10.6 Mapping of Braiding and Loop Algebra

In Abelian $U(1)_m$ Chern-Simons theory, the mutual braiding phase of charges q and q' is

$$M_{q,q'} = \exp\left(\frac{2\pi i q q'}{m}\right). \tag{10.28}$$

On the torus, the fundamental Wilson loops satisfy

$$W_1 W_2 = e^{2\pi i/m} W_2 W_1. \tag{10.29}$$

This algebra arises because the two non-contractible cycles of the torus intersect once [4].

In the Toric Code, an e-string and an m-string anticommute when they cross once:

$$W_e(\gamma)W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)}W_m(\gamma^*)W_e(\gamma). \tag{10.30}$$

For one crossing,

$$I(\gamma, \gamma^*) = 1, \tag{10.31}$$

so

$$W_e(\gamma)W_m(\gamma^*) = -W_m(\gamma^*)W_e(\gamma). \tag{10.32}$$

This gives the mutual braiding phase

$$M_{e,m} = -1. \tag{10.33}$$

Chern-Simons braiding	**Toric Code braiding**
Mutual braiding in $U(1)_m$: $$M_{q,q'} = \exp\left(\frac{2\pi i q q'}{m}\right).$$	Mutual braiding in the Toric Code: $$M_{e,m} = -1.$$
Wilson-loop algebra on T^2: $$W_1W_2 = e^{2\pi i/m}W_2W_1.$$	String-operator algebra on T^2: $$W_eW_m = (-1)^I W_mW_e.$$
The phase is determined by intersection or linking number.	The sign is determined by direct-dual string intersection number.

For the Toric Code K-matrix, the general Abelian Chern-Simons braiding formula is

$$M_{\ell,\ell'} = \exp\left(2\pi i\, \ell^T K^{-1}\ell'\right). \tag{10.34}$$

Taking the two fundamental anyons to be represented by

$$\ell_e = (1, 0), \qquad \ell_m = (0, 1), \tag{10.35}$$

and using

$$K^{-1} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}, \tag{10.36}$$

we find

$$\ell_e^T K^{-1}\ell_m = \frac{1}{2}. \tag{10.37}$$

Therefore,

$$M_{e,m} = \exp(2\pi i \cdot 1/2) = -1. \tag{10.38}$$

Thus the continuum doubled Chern-Simons/BF theory reproduces the Toric Code mutual braiding phase [3].

10.7 Mapping of Topological Spins

Topological spin is another important part of the topological data.

For $U(1)_m$ Chern-Simons theory, the topological spin of an anyon labelled by q is

$$\theta_q = \exp\left(\frac{\pi i q^2}{m}\right). \tag{10.39}$$

For the Toric Code,

$$\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1. \tag{10.40}$$

In K-matrix language, the topological spin is

$$\theta_\ell = \exp\left(\pi i\, \ell^T K^{-1} \ell\right). \tag{10.41}$$

Using

$$K^{-1} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}, \tag{10.42}$$

we get

$$\theta_e = 1, \qquad \theta_m = 1, \tag{10.43}$$

while for

$$\ell_\varepsilon = \ell_e + \ell_m = (1, 1), \tag{10.44}$$

we get

$$\ell_\varepsilon^T K^{-1} \ell_\varepsilon = 1, \tag{10.45}$$

and therefore

$$\theta_\varepsilon = e^{\pi i} = -1. \tag{10.46}$$

So the doubled continuum theory reproduces the Toric Code spin data. Simon gives the Toric Code S- and T-matrix data, and Wen's Abelian K-matrix formalism supplies the continuum formulas [3, 5].

10.8 Complete Topological Data Table

The comparison can now be summarized by topological data.

Chern-Simons model on T^2	**Toric Code model on T^2**
Continuum action: $$S = \frac{m}{4\pi} \int a \wedge da.$$	Continuum low-energy action: $$S = \frac{2}{2\pi} \int b \wedge da.$$ Equivalently, $$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$
Physical origin: two-dimensional quantum Hall fluid with topological response.	Physical origin: lattice stabilizer code; continuum description emerges in the low-energy topological sector.
Basic observables: Wilson loops $$W_q(\gamma) = \exp\left(iq \oint_\gamma a\right).$$	Basic observables: string loops $$W_e(\gamma), \qquad W_m(\gamma^*).$$
Anyon labels for $U(1)_m$: $$q \in \mathbb{Z}_m.$$	Anyon labels: $$1,\ e,\ m,\ \varepsilon.$$
Fusion: $$q \times q' = (q + q') \mod m.$$	Fusion: $$e^2 = m^2 = \varepsilon^2 = 1, \quad e \times m = \varepsilon.$$
Braiding: $$M_{q,q'} = \exp\left(\frac{2\pi i q q'}{m}\right).$$	Braiding: $$M_{e,m} = -1.$$
Torus Hilbert-space dimension: $$\dim \mathcal{H}_{U(1)_m}(T^2) = m.$$	Torus Hilbert-space dimension: $$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4.$$
Chiral Abelian topological order for the simple Laughlin $U(1)_m$ case.	Non-chiral doubled Abelian topological order.

This table shows that the quantum Hall/Chern-Simons model and the Toric Code model have the same structural TQFT ingredients: finite-dimensional torus Hilbert spaces, topological loop operators, anyon sectors, fusion rules, and braiding phases. However, the precise numerical data differ unless the continuum Chern-Simons theory is chosen to be the doubled $K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$ theory corresponding to the Toric Code.

10.9 Diagrammatic Dictionary Between the Two TQFTs

The following diagram summarizes the mapping of concepts.

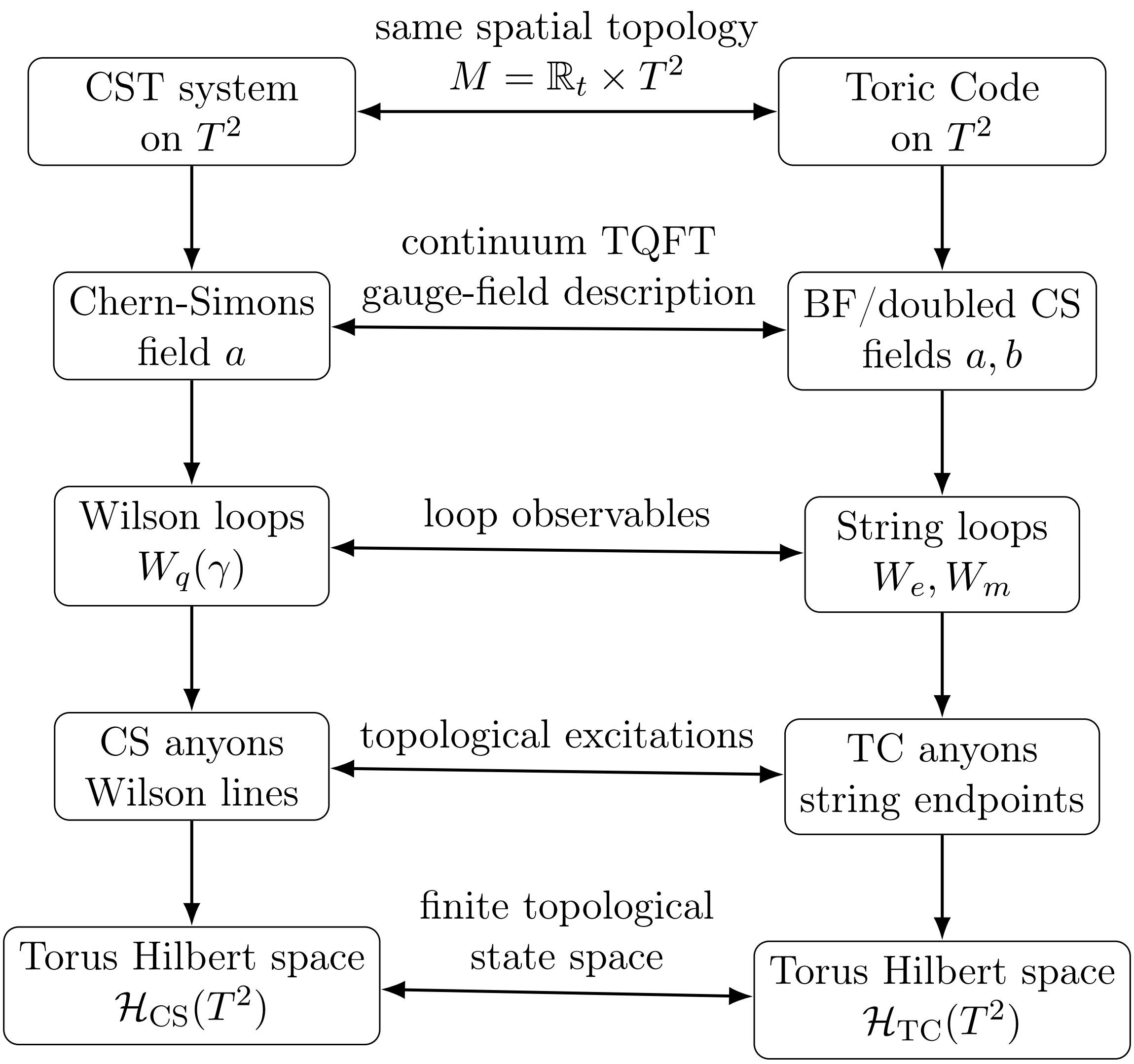


Figure 10.1: Dictionary between the quantum Hall/Chern-Simons TQFT on a torus and the Toric Code TQFT on a torus. The correspondence is between topological structures: gauge fields, loop operators, anyons, and finite-dimensional torus Hilbert spaces.

10.10 Proof of the Mapping

We now state the comparison as a proof.

First, both theories are $2+1$-dimensional because their spacetime is

$$M = \mathbb{R}_t \times T^2. \tag{10.47}$$

The quantum Hall system is a two-dimensional electron fluid evolving in time, while the Toric Code is a two-dimensional lattice system evolving in time. In both cases, anyon worldlines live in three-dimensional spacetime. This is why braiding is possible in both models [4, 5].

Second, both theories have topological loop observables. In Chern-Simons theory these are Wilson loops,

$$W_q(\gamma) = \exp\left(iq\oint_\gamma a\right), \tag{10.48}$$

while in the Toric Code they are non-contractible string operators,

$$W_e(\gamma) = \prod \sigma^z, \qquad W_m(\gamma^*) = \prod \sigma^x. \tag{10.49}$$

In both theories, the important loops are non-contractible loops wrapping around γ_1 and γ_2 on T^2. These loops act on the topological Hilbert space rather than creating local bulk waves [1, 2, 4, 5].

Third, both theories have anyons described by line operators. In Chern-Simons theory, anyons are Wilson lines in spacetime. In the Toric Code, anyons are endpoints of open string operators. Their histories are worldlines in $2+1$-dimensional spacetime.

Fourth, both theories encode braiding through the algebra of line or loop operators. For $U(1)_m$ Chern-Simons theory,

$$W_1W_2 = e^{2\pi i/m}W_2W_1. \tag{10.50}$$

For the Toric Code,

$$W_eW_m = -W_mW_e \tag{10.51}$$

when the two loops intersect once. In both cases, the phase is determined by topological intersection or linking number.

Fifth, both theories assign a finite-dimensional Hilbert space to the torus. The $U(1)_m$ theory gives

$$\dim \mathcal{H}_{U(1)_m}(T^2) = m, \tag{10.52}$$

while the Toric Code gives

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{10.53}$$

For the doubled Chern-Simons theory with

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \tag{10.54}$$

we have

$$|\det K| = 4, \tag{10.55}$$

so the continuum theory reproduces the Toric Code torus degeneracy.

Therefore,

$$\boxed{\text{The Toric Code is not generically equal to a single } U(1)_m \text{ quantum Hall TQFT,}} \tag{10.56}$$

but

$$\boxed{\text{the Toric Code is matched by the doubled Abelian Chern-Simons/BF TQFT with } K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.} \tag{10.57}$$

10.11 Conclusion

The comparison shows that both models fit the same general $2+1$-dimensional TQFT pattern. A quantum Hall system on a torus is naturally described by Chern-Simons theory, whose topological observables are Wilson loops and Wilson lines. The Toric Code on a torus is a microscopic lattice model, but its low-energy sector is described by a continuum $\mathbb{Z}_2$ BF/doubled Chern-Simons theory, whose topological observables are non-contractible string operators and anyon worldlines.

The dictionary is:

Chern-Simons gauge field	$\longleftrightarrow$	emergent Toric Code gauge fields
Wilson loop	$\longleftrightarrow$	non-contractible string operator
Wilson line anyon	$\longleftrightarrow$	string-endpoint anyon
CS loop algebra	$\longleftrightarrow$	TC string algebra
torus Hilbert space	$\longleftrightarrow$	four-dimensional TC ground space

(10.58)

Thus the two models are compared not by microscopic equality, but by the matching of topological data. This prepares the next chapter, where the Toric Code and Chern-Simons descriptions can be brought together into the final derivation of the continuum topological field theory from the discrete stabilizer model.

Chapter 11

Derivation of Chern-Simons Theory from the Toric Code: Proof

The previous chapter compared two $2 + 1$-dimensional TQFT descriptions side by side. We now give the central proof of the report: in the low-energy topological sector, the Toric Code is described by a continuum $2 + 1$-dimensional topological gauge theory, and this continuum theory can be written as a doubled Abelian Chern-Simons theory. More precisely, the Toric Code does not become the single chiral $U(1)_m$ Chern-Simons theory of a Laughlin quantum Hall state. Instead, it becomes the non-chiral $N = 2$ BF theory, equivalently the doubled Abelian Chern-Simons theory with

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{11.1}$$

Thus the proof is not a microscopic equality

$$H_{\mathrm{TC}} = S_{\mathrm{CS}}. \tag{11.2}$$

This expression is not meaningful because H_{TC} is a lattice Hamiltonian, while S_{CS} is a continuum action. The correct statement is a low-energy topological equivalence:

$$\boxed{\text{Toric Code} \xrightarrow{\text{low-energy topological sector}} \mathbb{Z}_2 \text{ BF theory} \equiv \text{doubled Abelian Chern-Simons theory.}} \tag{11.3}$$

Kitaev's Toric Code gives the microscopic exactly solved lattice realization of the topological phase [2]. Simon discusses the Toric Code as a TQFT/topological phase and its relation to quantum double and gauge-theoretic descriptions [5]. Wen's K-matrix formalism provides the Abelian Chern-Simons description of Abelian topological orders [3]. BF theory provides the continuum topological gauge theory language in which $\mathbb{Z}_N$ gauge theories appear naturally [6, 7].

> The goal of this chapter is to prove that the universal topological sector of the Toric Code is captured by the continuum action
>
> $$S_{\mathrm{BF}} = \frac{2}{2\pi} \int b \wedge da,$$
>
> or equivalently by the doubled Abelian Chern-Simons K-matrix theory
>
> $$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

11.1 What Must Be Proven

To derive the Chern-Simons description from the Toric Code, we must show that the continuum theory reproduces all universal topological data of the lattice model. The relevant data are:

$$\text{ground-state degeneracy on } T^2, \tag{11.4}$$

$$\text{anyon sectors,} \tag{11.5}$$

$$\text{fusion rules,} \tag{11.6}$$

$$\text{braiding phases,} \tag{11.7}$$

$$\text{topological spins,} \tag{11.8}$$

and

$$\text{non-contractible loop or string-operator algebra.} \tag{11.9}$$

This is the correct notion of equivalence between topological phases. Two microscopic models can be very different locally and still describe the same TQFT if they have the same universal topological data [3, 5].

For the Toric Code, the data are:

$$\mathcal{A}_{\mathrm{TC}} = \{1, e, m, \varepsilon = e \times m\}, \tag{11.10}$$

$$e^2 = m^2 = \varepsilon^2 = 1, \qquad e \times m = \varepsilon, \tag{11.11}$$

$$M_{e,m} = -1, \tag{11.12}$$

$$\theta_1 = \theta_e = \theta_m = 1, \qquad \theta_\varepsilon = -1, \tag{11.13}$$

and

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4. \tag{11.14}$$

These are the data that the continuum Chern-Simons/BF theory must reproduce.

11.2 Step 1: Start from the Toric Code Hamiltonian

The microscopic Toric Code Hamiltonian is

$$H_{\mathrm{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p, \tag{11.15}$$

where

$$A_s = \prod_{e \in \mathrm{star}(s)} \sigma_e^x, \qquad B_p = \prod_{e \in \partial p} \sigma_e^z. \tag{11.16}$$

The ground-state subspace is defined by

$$A_s|\psi\rangle = |\psi\rangle, \qquad B_p|\psi\rangle = |\psi\rangle. \tag{11.17}$$

Thus all local star and plaquette violations are absent in the low-energy sector.

The star constraint is interpreted as a local $\mathbb{Z}_2$ gauge invariance condition, while the plaquette constraint is interpreted as a local flatness condition. Therefore, the low-energy Toric Code is a discrete flat $\mathbb{Z}_2$ gauge theory. This gauge-theoretic interpretation of the Toric Code is part of the standard quantum double/topological-order description [2, 5].

The first conclusion is:

$$\boxed{\text{Low-energy Toric Code} = \text{flat } \mathbb{Z}_2 \text{ gauge theory with global holonomies.}} \tag{11.18}$$

11.3 Step 2: Extract the Long-Distance Topological Sector

The Toric Code is defined on a lattice, but a TQFT is a continuum theory. The continuum description is obtained by discarding lattice-scale details while keeping the topological data. Thus we ignore:

$$\text{microscopic edge lengths,} \qquad \text{precise local path shapes,} \qquad \text{local stabilizer fluctuations above the gap} \tag{11.19}$$

We keep:

$$\text{anyon sectors,} \qquad \text{fusion,} \qquad \text{braiding,} \qquad \text{topological spins,} \qquad \text{global loop algebra.} \tag{11.20}$$

The stabilizer gap ensures that the low-energy sector is separated from local excitations. Hence, at energies much smaller than the gap, the effective theory is not sensitive to local microscopic details. It is controlled only by topology. This is exactly the regime in which a continuum TQFT is expected to describe a lattice topological phase [3,5].

Thus,

$$\boxed{\text{lattice stabilizer model} \quad \longrightarrow \quad \text{continuum topological gauge theory.}} \tag{11.21}$$

11.4 Step 3: Identify the Continuum BF Action

The continuum topological gauge theory for a $\mathbb{Z}_N$ gauge theory in $2+1$ dimensions is BF theory:

$$S_{\text{BF}} = \frac{N}{2\pi} \int_M b \wedge da. \tag{11.22}$$

Here a and b are compact Abelian one-form gauge fields. In the Toric Code case,

$$N = 2. \tag{11.23}$$

Therefore,

$$S_{\text{BF}} = \frac{2}{2\pi} \int_M b \wedge da. \tag{11.24}$$

BF theories are metric-independent topological gauge theories, and the $N = 2$ case captures the $\mathbb{Z}_2$ gauge structure of the Toric Code [5–7].

The fields a and b play complementary roles. One field couples naturally to electric-type Wilson lines, while the other couples naturally to magnetic-type Wilson lines. This mirrors the Toric Code, where e and m anyons are created by two different kinds of string operators:

$$W_e(\gamma) = \prod_{e \in \gamma} \sigma_e^z, \qquad W_m(\gamma^*) = \prod_{e \in \gamma^*} \sigma_e^x. \tag{11.25}$$

11.5 Step 4: Show that BF Theory is Topological

The BF action

$$S_{\text{BF}} = \frac{N}{2\pi} \int_M b \wedge da \tag{11.26}$$

contains no spacetime metric. It is written entirely using differential forms and the orientation of the three-dimensional manifold. Therefore,

$$\frac{\delta S_{\mathrm{BF}}}{\delta g_{\mu\nu}} = 0. \tag{11.27}$$

Thus the stress-energy tensor vanishes in the topological sector:

$$T_{\mu\nu} = 0. \tag{11.28}$$

This proves that BF theory is a TQFT.

Next, vary the action. Varying with respect to b gives

$$\delta_b S_{\mathrm{BF}} = \frac{N}{2\pi} \int_M \delta b \wedge da. \tag{11.29}$$

Since δb is arbitrary, the equation of motion is

$$da = 0. \tag{11.30}$$

Varying with respect to a gives

$$\delta_a S_{\mathrm{BF}} = \frac{N}{2\pi} \int_M b \wedge d(\delta a). \tag{11.31}$$

Integrating by parts and ignoring boundary terms on a closed manifold gives

$$\delta_a S_{\mathrm{BF}} = -\frac{N}{2\pi} \int_M db \wedge \delta a. \tag{11.32}$$

Therefore,

$$db = 0. \tag{11.33}$$

The equations of motion are therefore

$$\boxed{da = 0, \qquad db = 0.} \tag{11.34}$$

This is the continuum analogue of the Toric Code ground-state constraints:

$$A_s = +1 \quad \text{and} \quad B_p = +1. \tag{11.35}$$

Both descriptions say that there are no local gauge or flux excitations in the ground sector. Only global flat connections and holonomies remain.

11.6 Step 5: Compactness Reduces the Continuum Theory to $\mathbb{Z}_2$

The coefficient N in

$$S_{\mathrm{BF}} = \frac{N}{2\pi} \int b \wedge da \tag{11.36}$$

is not just a normalization. Because a and b are compact gauge fields, the integer N determines the discrete gauge structure. In the presence of compactness, the allowed topological sectors are classified by $\mathbb{Z}_N$ holonomies. For the Toric Code,

$$N = 2, \tag{11.37}$$

so the global holonomies are $\mathbb{Z}_2$-valued.

Thus, for a non-contractible cycle γ,

$$\exp\left(i\oint_\gamma a\right) \in \{+1,-1\}, \tag{11.38}$$

and similarly for b. This is exactly what happens in the Toric Code: non-contractible string operators have eigenvalues

$$\pm 1. \tag{11.39}$$

Therefore, compact $N=2$ BF theory reproduces the discrete $\mathbb{Z}_2$ holonomy structure of the Toric Code.

11.7 Step 6: Derive Doubled Chern-Simons Theory from BF Theory

Now we show that the BF action is equivalent to a doubled Abelian Chern-Simons action. Start with

$$S_{\text{BF}} = \frac{N}{2\pi}\int b\wedge da. \tag{11.40}$$

Introduce two Abelian gauge fields

$$a^1 = a, \qquad a^2 = b. \tag{11.41}$$

The general Abelian Chern-Simons K-matrix action is

$$S_K = \frac{1}{4\pi}\int_M K_{IJ}a^I\wedge da^J. \tag{11.42}$$

Choose

$$K = \begin{pmatrix} 0 & N \\ N & 0\end{pmatrix}. \tag{11.43}$$

Then

$$S_K = \frac{1}{4\pi}\int \left(Na^1\wedge da^2 + Na^2\wedge da^1\right) \tag{11.44}$$

$$= \frac{N}{4\pi}\int (a\wedge db + b\wedge da)\,. \tag{11.45}$$

Using integration by parts on a closed manifold,

$$\int_M a\wedge db = \int_M b\wedge da \tag{11.46}$$

up to a boundary term. Therefore,

$$S_K = \frac{N}{2\pi}\int_M b\wedge da. \tag{11.47}$$

Hence,

$$\boxed{S_{\text{BF}} = S_K \quad \text{with} \quad K = \begin{pmatrix} 0 & N \\ N & 0\end{pmatrix}.} \tag{11.48}$$

For the Toric Code, $N = 2$, so

$$\boxed{K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.} \tag{11.49}$$

This proves that the continuum $N = 2$ BF theory is a doubled Abelian Chern-Simons theory. This is the Chern-Simons theory that captures the low-energy Toric Code, not a single chiral $U(1)_m$ theory. The use of K-matrix Abelian Chern-Simons theory to classify Abelian topological orders is standard in Wen's framework [3].

11.8 Step 7: Match the Anyon Sectors

In Abelian K-matrix Chern-Simons theory, anyons are labelled by integer vectors

$$\ell \in \mathbb{Z}^2 \tag{11.50}$$

modulo local particles generated by the columns of K:

$$\ell \sim \ell + K\Lambda, \qquad \Lambda \in \mathbb{Z}^2. \tag{11.51}$$

Therefore the anyon group is

$$\mathbb{Z}^2/K\mathbb{Z}^2. \tag{11.52}$$

For

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \tag{11.53}$$

the determinant is

$$|\det K| = 4. \tag{11.54}$$

Thus there are four anyon sectors.

A convenient choice of representatives is

$$1 = (0,0), \qquad e = (1,0), \qquad m = (0,1), \qquad \varepsilon = (1,1). \tag{11.55}$$

This matches the Toric Code anyon content:

$$\boxed{\{1, e, m, \varepsilon\}.} \tag{11.56}$$

Thus the doubled Chern-Simons theory reproduces the four superselection sectors of the Toric Code [2, 3, 5].

11.9 Step 8: Match the Fusion Rules

In Abelian K-matrix theory, fusion corresponds to addition of anyon vectors modulo $K\mathbb{Z}^2$:

$$\ell \times \ell' = \ell + \ell' \mod K\mathbb{Z}^2. \tag{11.57}$$

Using the representatives

$$e = (1,0), \qquad m = (0,1), \qquad \varepsilon = (1,1), \tag{11.58}$$

we find

$$e \times e = (2,0) \sim (0,0) = 1, \tag{11.59}$$

because $(2,0)$ is a local vector in $K\mathbb{Z}^2$. Similarly,

$$m \times m = (0,2) \sim (0,0) = 1. \tag{11.60}$$

Also,

$$e \times m = (1,0) + (0,1) = (1,1) = \varepsilon. \tag{11.61}$$

Finally,

$$\varepsilon \times \varepsilon = (1,1) + (1,1) = (2,2) \sim (0,0) = 1. \tag{11.62}$$

Therefore,

$$\boxed{e^2 = m^2 = \varepsilon^2 = 1, \qquad e \times m = \varepsilon.} \tag{11.63}$$

This is exactly the Toric Code fusion algebra:

$$\mathbb{Z}_2 \times \mathbb{Z}_2. \tag{11.64}$$

11.10 Step 9: Match the Mutual Braiding

In Abelian Chern-Simons K-matrix theory, the mutual braiding phase between anyons ℓ and ℓ' is

$$M_{\ell,\ell'} = \exp\left(2\pi i\, \ell^T K^{-1} \ell'\right). \tag{11.65}$$

For the Toric Code K-matrix,

$$K^{-1} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}. \tag{11.66}$$

Let

$$\ell_e = (1,0), \qquad \ell_m = (0,1). \tag{11.67}$$

Then

$$\ell_e^T K^{-1} \ell_m = (1,0) \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2}. \tag{11.68}$$

Therefore,

$$M_{e,m} = \exp\left(2\pi i \cdot \frac{1}{2}\right) = -1. \tag{11.69}$$

This exactly matches the Toric Code string-operator result:

$$W_e(\gamma) W_m(\gamma^*) = (-1)^{I(\gamma,\gamma^*)} W_m(\gamma^*) W_e(\gamma). \tag{11.70}$$

For one crossing, $I = 1$, so the phase is -1. Hence the Chern-Simons/BF theory reproduces the Toric Code mutual braiding [2, 3, 5].

11.11 Step 10: Match the Topological Spins

In Abelian K-matrix theory, the topological spin of anyon ℓ is

$$\theta_\ell = \exp\left(\pi i\, \ell^T K^{-1} \ell\right). \tag{11.71}$$

For $e = (1, 0)$,

$$\ell_e^T K^{-1} \ell_e = 0, \tag{11.72}$$

so

$$\theta_e = 1. \tag{11.73}$$

For $m = (0, 1)$,

$$\ell_m^T K^{-1} \ell_m = 0, \tag{11.74}$$

so

$$\theta_m = 1. \tag{11.75}$$

For

$$\varepsilon = (1, 1), \tag{11.76}$$

we compute

$$\ell_\varepsilon^T K^{-1} \ell_\varepsilon = (1, 1) \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1. \tag{11.77}$$

Therefore,

$$\theta_\varepsilon = e^{\pi i} = -1. \tag{11.78}$$

Thus,

$$\boxed{\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1.} \tag{11.79}$$

This matches the Toric Code topological spin data discussed earlier [3, 5].

11.12 Step 11: Match the Torus Ground-State Degeneracy

For Abelian Chern-Simons theory with K-matrix K, the ground-state degeneracy on a torus is

$$\dim \mathcal{H}(T^2) = |\det K|. \tag{11.80}$$

For the Toric Code K-matrix,

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \tag{11.81}$$

so

$$\det K = -4. \tag{11.82}$$

Therefore,

$$|\det K| = 4. \tag{11.83}$$

Hence,

$$\boxed{\dim \mathcal{H}_K(T^2) = 4.} \tag{11.84}$$

This matches the Toric Code stabilizer counting result:

$$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 2^{2n^2 - (2n^2 - 2)} = 4. \tag{11.85}$$

Thus the continuum doubled Chern-Simons theory reproduces the torus Hilbert-space assignment of the Toric Code [3,5].

For a genus-g closed surface, the K-matrix theory gives

$$\dim \mathcal{H}(\Sigma_g) = |\det K|^g. \tag{11.86}$$

For the Toric Code K-matrix, this is

$$4^g = 2^{2g}, \tag{11.87}$$

which matches the $\mathbb{Z}_2$ Toric Code degeneracy on genus g surfaces.

11.13 Step 12: Match the Loop Operator Algebra

The Toric Code has non-contractible string operators on the torus:

$$W_e(\gamma_i), \qquad W_m(\gamma_i^*), \qquad i = 1, 2. \tag{11.88}$$

Their algebra is

$$W_e(\gamma_i)W_m(\gamma_j^*) = (-1)^{I(\gamma_i,\gamma_j^*)}W_m(\gamma_j^*)W_e(\gamma_i). \tag{11.89}$$

In the doubled Chern-Simons/BF theory, the corresponding observables are Wilson loops of a and b:

$$U_a(\gamma) = \exp\left(i \oint_\gamma a\right), \qquad U_b(\gamma^*) = \exp\left(i \oint_{\gamma^*} b\right). \tag{11.90}$$

The BF coupling implies that when the two loops have intersection number one, their operators acquire the phase

$$\exp\left(\frac{2\pi i}{N}\right). \tag{11.91}$$

For $N = 2$,

$$\exp\left(\frac{2\pi i}{2}\right) = -1. \tag{11.92}$$

Therefore,

$$U_a(\gamma)U_b(\gamma^*) = -U_b(\gamma^*)U_a(\gamma) \tag{11.93}$$

for one intersection. This is exactly the Toric Code string-operator algebra. Hence,

$$\boxed{\text{TC string algebra} = \text{BF/Chern-Simons Wilson-loop algebra at } N = 2.} \tag{11.94}$$

11.14 Summary Table of the Proof

The following table summarizes the derivation.

Toric Code low-energy sector	**Continuum BF/doubled Chern-Simons theory**
Ground-state constraints: $$A_s = +1, \qquad B_p = +1.$$	Flatness equations: $$da = 0, \qquad db = 0.$$
Discrete $\mathbb{Z}_2$ gauge structure.	Compact BF theory at level $N = 2$: $$S_{\mathrm{BF}} = \frac{2}{2\pi}\int b \wedge da.$$
Anyon sectors: $$1,\ e,\ m,\ \varepsilon.$$	Anyon vectors: $$(0,0),\ (1,0),\ (0,1),\ (1,1).$$
Fusion: $$e^2 = m^2 = \varepsilon^2 = 1, \quad e \times m = \varepsilon.$$	Fusion by vector addition modulo $K\mathbb{Z}^2$.
Mutual braiding: $$M_{e,m} = -1.$$	$$M_{\ell,\ell'} = \exp\bigl(2\pi i\, \ell^T K^{-1} \ell'\bigr), \quad M_{e,m} = -1.$$
Topological spins: $$\theta_e = \theta_m = 1, \quad \theta_\varepsilon = -1.$$	$$\theta_\ell = \exp\bigl(\pi i\, \ell^T K^{-1} \ell\bigr).$$
Torus degeneracy: $$\dim \mathcal{H}_{\mathrm{TC}}(T^2) = 4.$$	$$\dim \mathcal{H}_K(T^2) = \lvert \det K \rvert = 4.$$
Non-contractible string operators.	Wilson loops of a and b.

11.15 Final Proof Statement

We can now state the derivation as a theorem.

The low-energy topological sector of the Toric Code is equivalent to the $N = 2$ BF theory

$$S_{\mathrm{BF}} = \frac{2}{2\pi} \int b \wedge da.$$

This BF theory is equivalently the doubled Abelian Chern-Simons theory

$$S_K = \frac{1}{4\pi} \int K_{IJ} a^I \wedge da^J, \qquad K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

The equivalence is proven by matching anyon sectors, fusion rules, braiding phases, topological spins, loop-operator algebra, and torus ground-state degeneracy.

In symbolic form,

$$\boxed{\text{Toric Code} \quad \xrightarrow{\text{low energy}} \quad \mathbb{Z}_2 \text{ BF theory} \quad \equiv \quad \text{doubled Abelian Chern-Simons theory.}} \tag{11.95}$$

This completes the derivation of the continuum Chern-Simons topological field theory from the Toric Code at the level of universal low-energy topological data. The next optional step is to discuss how this derivation may be modified or enriched when one embeds the Toric Code into continuous-variable systems such as GKP-type codes.

Chapter 12

Deriving Chern-Simons Theory from the Toric Code with GKP: Extra Proof Sketch

This chapter is an additional proof sketch. It is not needed for the main derivation in Chapter 11, where the Toric Code was already shown to flow in its low-energy topological sector to $N = 2$ BF theory, equivalently doubled Abelian Chern-Simons theory. The purpose of this chapter is to explain how the same idea can be viewed through a continuous-variable or GKP-type embedding.

The Gottesman-Kitaev-Preskill (GKP) code encodes a qubit into the Hilbert space of a harmonic oscillator using a lattice of position and momentum displacements [8]. This makes it conceptually useful for connecting a discrete stabilizer code to continuum variables. In the present report, the GKP discussion should be understood as an optional bridge:

$$\begin{gathered}\text{discrete Toric Code qubits}\\ \Downarrow\\ \text{continuous oscillator variables with GKP constraints}\\ \Downarrow\\ \text{continuum topological gauge theory}\end{gathered} \tag{12.1}$$

The important note is:

> The GKP construction does not change the topological order of the Toric Code. It gives a continuous-variable representation of the same encoded qubit and stabilizer structure. The low-energy topological field theory remains the $N = 2$ BF theory, equivalently the doubled Abelian Chern-Simons theory with
>
> $$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

12.1 Step 1: Recall the GKP Encoding

A GKP code encodes a qubit into a single bosonic mode with canonical quadratures

$$[\hat{q}, \hat{p}] = i. \tag{12.2}$$

Instead of using a finite two-dimensional microscopic Hilbert space directly, the GKP code uses a continuous-variable oscillator Hilbert space and imposes periodic stabilizer constraints in phase space. The ideal square-lattice GKP stabilizers can be written schematically as displacement operators

$$S_q = \exp\big(2i\sqrt{\pi}\,\hat{q}\big), \qquad S_p = \exp\big(-2i\sqrt{\pi}\,\hat{p}\big). \tag{12.3}$$

The logical Pauli operators can be represented as smaller displacements:

$$\overline{Z} = \exp\big(i\sqrt{\pi}\,\hat{q}\big), \qquad \overline{X} = \exp\big(-i\sqrt{\pi}\,\hat{p}\big). \tag{12.4}$$

These obey the qubit Pauli anticommutation relation

$$\overline{X}\,\overline{Z} = -\overline{Z}\,\overline{X}. \tag{12.5}$$

Thus, a GKP oscillator can behave as an encoded qubit at low energies or inside the protected code subspace. This is why it can be used as a continuous-variable representation of the qubits that appear on the edges of the Toric Code [8].

12.2 Step 2: Replace Toric Code Edge Qubits by GKP Encoded Qubits

In the Toric Code, a qubit lives on each edge e of a two-dimensional lattice:

$$\mathcal{H}_\Lambda = \bigotimes_{e\in E(\Lambda)} \mathbb{C}^2. \tag{12.6}$$

The star and plaquette stabilizers are

$$A_s = \prod_{e\in \mathrm{star}(s)} \sigma_e^x, \qquad B_p = \prod_{e\in\partial p} \sigma_e^z. \tag{12.7}$$

In a GKP embedding, each edge qubit is represented by an oscillator whose encoded logical Pauli operators are

$$\sigma_e^x \longrightarrow \overline{X}_e, \qquad \sigma_e^z \longrightarrow \overline{Z}_e. \tag{12.8}$$

Therefore the Toric Code stabilizers become encoded stabilizers:

$$A_s^{\mathrm{GKP}} = \prod_{e\in \mathrm{star}(s)} \overline{X}_e, \tag{12.9}$$

and

$$B_p^{\mathrm{GKP}} = \prod_{e\in\partial p} \overline{Z}_e. \tag{12.10}$$

Inside the ideal GKP code subspace, these operators obey the same algebra as the original Pauli stabilizers. Therefore, the GKP embedding preserves the Toric Code stabilizer structure:

$$[A_s^{\mathrm{GKP}}, A_{s'}^{\mathrm{GKP}}] = 0, \qquad [B_p^{\mathrm{GKP}}, B_{p'}^{\mathrm{GKP}}] = 0, \qquad [A_s^{\mathrm{GKP}}, B_p^{\mathrm{GKP}}] = 0. \tag{12.11}$$

The same two-minus-sign cancellation occurs whenever a star and plaquette overlap on two edges. Hence the GKP representation does not alter the topological stabilizer algebra of the Toric Code.

12.3 Step 3: The Low-Energy Subspace Still Gives a $\mathbb{Z}_2$ Gauge Theory

The GKP embedding introduces continuous oscillator variables, but the encoded logical information remains qubit-like. In the low-energy code subspace, the relevant eigenvalues are still discrete:

$$A_s^{\text{GKP}} = +1, \qquad B_p^{\text{GKP}} = +1. \tag{12.12}$$

Thus the same physical interpretation remains:

$$A_s^{\text{GKP}} = +1 \quad \Longleftrightarrow \quad \text{local } \mathbb{Z}_2 \text{ gauge invariance,} \tag{12.13}$$

and

$$B_p^{\text{GKP}} = +1 \quad \Longleftrightarrow \quad \text{local } \mathbb{Z}_2 \text{ flatness.} \tag{12.14}$$

Therefore the GKP-embedded Toric Code still has the same low-energy gauge-theoretic content:

$$\boxed{\text{GKP-embedded Toric Code} \quad \xrightarrow{\text{low energy}} \quad \text{flat } \mathbb{Z}_2 \text{ gauge theory.}} \tag{12.15}$$

This is the same intermediate step used in Chapter 11. The GKP representation changes the microscopic Hilbert space, but it does not change the emergent topological gauge theory.

12.4 Step 4: String Operators Become Products of Displacements

In the ordinary Toric Code, the anyon string operators are

$$W_e(\gamma) = \prod_{e\in\gamma} \sigma_e^z, \qquad W_m(\gamma^*) = \prod_{e\in\gamma^*} \sigma_e^x. \tag{12.16}$$

In the GKP representation, these become products of logical displacement operators:

$$W_e^{\text{GKP}}(\gamma) = \prod_{e\in\gamma} \overline{Z}_e, \tag{12.17}$$

and

$$W_m^{\text{GKP}}(\gamma^*) = \prod_{e\in\gamma^*} \overline{X}_e. \tag{12.18}$$

Because

$$\overline{X}_e\overline{Z}_e = -\overline{Z}_e\overline{X}_e, \tag{12.19}$$

the same string-operator algebra is recovered:

$$W_e^{\text{GKP}}(\gamma)W_m^{\text{GKP}}(\gamma^*) = (-1)^{I(\gamma,\gamma^*)}W_m^{\text{GKP}}(\gamma^*)W_e^{\text{GKP}}(\gamma). \tag{12.20}$$

For one crossing,

$$I(\gamma,\gamma^*) = 1, \tag{12.21}$$

so

$$W_e^{\text{GKP}}W_m^{\text{GKP}} = -W_m^{\text{GKP}}W_e^{\text{GKP}}. \tag{12.22}$$

Thus the GKP-embedded model preserves the Toric Code mutual braiding phase

$$M_{e,m} = -1. \tag{12.23}$$

12.5 Step 5: Continuum Limit Gives the Same BF Theory

Because the low-energy GKP-embedded model has the same $\mathbb{Z}_2$ gauge constraints and the same string-operator algebra as the Toric Code, its continuum topological field theory is also the $N = 2$ BF theory:

$$S_{\mathrm{BF}} = \frac{2}{2\pi} \int_M b \wedge da. \tag{12.24}$$

The equations of motion are

$$da = 0, \qquad db = 0, \tag{12.25}$$

which express local flatness. The compactness and level $N = 2$ encode the residual $\mathbb{Z}_2$ topological sectors. BF theory is a metric-independent TQFT and is equivalent to a doubled Abelian Chern-Simons theory [3, 6, 7].

As in Chapter 11, define

$$a^1 = a, \qquad a^2 = b. \tag{12.26}$$

Then

$$S_{\mathrm{BF}} = \frac{1}{4\pi} \int_M K_{IJ} a^I \wedge da^J, \tag{12.27}$$

with

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{12.28}$$

Therefore,

$$\boxed{\text{GKP-embedded Toric Code} \xrightarrow{\text{low-energy continuum limit}} \text{doubled Abelian Chern-Simons theory.}} \tag{12.29}$$

12.6 Step 6: Topological Data are Unchanged

The GKP embedding preserves the Toric Code topological data. The anyon sectors remain

$$1, \qquad e, \qquad m, \qquad \varepsilon = e \times m. \tag{12.30}$$

The fusion rules remain

$$e^2 = m^2 = \varepsilon^2 = 1, \qquad e \times m = \varepsilon. \tag{12.31}$$

The braiding remains

$$M_{e,m} = -1. \tag{12.32}$$

The topological spins remain

$$\theta_1 = 1, \qquad \theta_e = 1, \qquad \theta_m = 1, \qquad \theta_\varepsilon = -1. \tag{12.33}$$

On the torus, the ground-state degeneracy remains

$$\dim \mathcal{H}(T^2) = 4. \tag{12.34}$$

Equivalently, the continuum K-matrix theory gives

$$|\det K| = \left| \det \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \right| = 4. \tag{12.35}$$

Thus the GKP embedding does not produce a new topological phase. It gives a continuous-variable microscopic route to the same Toric Code topological phase.

12.7 Summary Table

GKP/Toric Code description	**Continuum TQFT description**
Each edge qubit is encoded into an oscillator using GKP stabilizers.	The low-energy degrees of freedom are described by compact topological gauge fields.
Logical Pauli operators: $$\overline{X}, \qquad \overline{Z}.$$	Wilson operators of the continuum gauge fields: $$\exp\left(i\oint a\right), \qquad \exp\left(i\oint b\right).$$
Encoded star and plaquette constraints: $$A_s^{\mathrm{GKP}} = +1, \qquad B_p^{\mathrm{GKP}} = +1.$$	Flatness equations: $$da = 0, \qquad db = 0.$$
String algebra: $$W_e W_m = (-1)^I W_m W_e.$$	BF/Chern-Simons loop algebra: $$U_a U_b = \exp(2\pi i I/2) U_b U_a.$$
Toric Code topological order: $$1, e, m, \varepsilon.$$	Doubled Chern-Simons theory: $$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$
Torus degeneracy: $$\dim \mathcal{H}(T^2) = 4.$$	$$\lvert \det K \rvert = 4.$$

12.8 Conclusion

The GKP construction provides an optional continuous-variable route from the Toric Code to continuum topological field theory. By encoding each Toric Code edge qubit into an oscillator, the microscopic Hilbert space becomes continuous-variable, but the protected logical Pauli algebra remains the same. Consequently, the star and plaquette stabilizers, string operators, fusion rules, braiding phases, and torus degeneracy are unchanged.

Therefore,

$$\boxed{\begin{array}{c}\text{Toric Code with GKP encoding}\\ \xrightarrow{\text{low energy}}\\ N=2\text{ BF theory}\\ \equiv\\ \text{doubled Abelian Chern–Simons theory}\end{array}} \tag{12.36}$$

This chapter should be read as a proof sketch rather than a new independent derivation. The main derivation remains the topological-data matching proof of Chapter 11. The GKP viewpoint is useful because it shows how a discrete stabilizer model can be embedded into continuous-variable degrees of freedom while preserving the same emergent $2+1$-dimensional TQFT.

Chapter 13

Conclusion

This report has shown how Chern-Simons theory can be understood as the continuum topological field-theory description of the Toric Code in its low-energy sector. We began by introducing quantum field theory, topological quantum field theory, and the special role of 2+1-dimensional continuum TQFTs, where anyonic braiding and non-contractible loop observables naturally arise. We then studied Chern-Simons theory on a torus, identifying Wilson loops, finite-dimensional torus Hilbert spaces, Abelian anyons, fusion rules, braiding phases, and topological spins. On the Toric Code side, we showed that the microscopic lattice Hamiltonian is not itself a continuum field theory, but that its gapped low-energy sector behaves as a 2+1-dimensional TQFT: local stabilizer constraints remove local degrees of freedom, while non-contractible string operators on the torus generate a four-dimensional topological ground-state space. By comparing the two descriptions, we found that the Toric Code is captured not by a single chiral $U(1)_m$ Chern-Simons theory, but by the non-chiral $N = 2$ BF theory, equivalently the doubled Abelian Chern-Simons theory with $K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$. This identification was verified by matching the universal topological data: four anyon sectors $1, e, m, \varepsilon$, $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion, mutual braiding phase $M_{e,m} = -1$, topological spins $\theta_e = \theta_m = 1$, $\theta_\varepsilon = -1$, the non-contractible loop algebra, and the torus ground-state degeneracy $\dim \mathcal{H}(T^2) = 4$. Therefore, the derivation is a proof of low-energy topological equivalence: the Toric Code flows, after microscopic lattice details are discarded, to the same universal TQFT described by $N = 2$ BF/doubled Chern-Simons theory. The optional GKP discussion further showed that embedding the Toric Code qubits into continuous-variable oscillator degrees of freedom does not change the topological order, but provides another route for understanding how a discrete stabilizer model can give rise to a continuum topological gauge theory. Overall, the report demonstrates that the bridge from the Toric Code to Chern-Simons theory is built not through microscopic equality, but through the precise matching of topological Hilbert spaces, anyon data, and loop-operator algebra.

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